Mind It! v1.69 — QC Report (builds 121–172)

Date: 12 October 2026 · **Version:** 1.69 (unchanged) · **Android versionCode:** 121 · **Source:** git tag `checkpoint-build121`

Where the testing happened: everything in this report was run on the build machine (Linux sandbox): the real Windows laptop app (Electron, the same app shell as `Mind It!.exe`) under a virtual screen, with the tablet and phone pages opened in Chromium at real sizes and touch emulation. **No physical Windows PC, Galaxy Tab A9+, phone or TV was used.** The live acceptance test is **PENDING USER LIVE TEST** (section J).

Overall readiness: Ready for your live acceptance test — not yet cleared for unrestricted live use. Every automated check, audit, stress test and the 60-minute soak passed in the sandbox, and the rollback was tested; what's missing is a real game on your real laptop, tablet, phones and TV (section J). Keep build 119 until that passes.

Build 172 addendum (5 November 2026)

What you asked for (5 Nov): "yes do the toast fix too". This was the one finding of the build 171 audit left for you to decide: with the Audience window open, the small pop-up notes covered the top of the timer buttons.

Same version: v1.69. The Galaxy Tab and Mind It! TV apps are versionCode 172, and `APP_BUILD=172`. The change is one new module (`app172.js`); no older module was edited. No rule, score, timer or wager changed. Not a 5th build: no Mac app, and the guides are build 171's.

A · What's new

- **Pop-up notes never sit on a control.** Build 106 puts notes beside the host panel, at the top of the left column. With the Audience window open, build 171 put the timer card there, so a two-line note covered the top of Pause Timer / +30 Sec / End Timer for its 4.5 s on 1366×768, 1366×700 and 1280×800 laptops. The time-up note did this as the next team's clock started.
- Notes now move down past the controls they would cover, clearing a small card such as the timer card as a whole, to the first free place.
- The check runs right after build 152's own placement of the notes, in the same frame and before anything is drawn, so a note appears straight in its place.
- Nothing moves where nothing is covered, and nothing moves when no free place fits on the screen.

B · Checks

- `fix172` **23/23** on the build-172 page (14/23 on the build-171 page, where the 9 misses are the overlaps it was written for):
  - 1366×768, 1366×700, 1280×800 and 1920×1080 with the window open: the note covers no control and starts under the timer card.
  - Three notes at once cover no control and stay on screen.
  - The answer and wrong-answer screens.
  - The board, and notes going back up after a question.
  - The time-up note.
  - The window closed and the Galaxy Tab: unchanged.
- `fix171` 44/44 on the build-172 page.
- **Full run:** browser 144 suites **4,409/4,409**, laptop app 80 suites **1,445/1,445** (the build-172 Android apps inside the laptop app); `fix170` passed on its second try (three "no page errors" checks on the phone screens with empty error lists, a timing flake of the test) and again on its own 75/75.
- **Pictures:** BEFORE (build 171) and AFTER (build 172) from the real laptop app in the same states (`design-plans/ui171/after172-compare/`). Only the note moved; the other differences are the usual run-to-run wobble of the captures (the right column a few pixels higher or lower, clocks, random team badges).
- **Reproducible:** Galaxy Tab APK ec3c9a93… and Mind It! TV APK d9793f71…, versionCode 172, signed with the original key; the tablet APK was built from a fresh copy of the repository and carries the tested page (8685254d…).
- **Windows EXE** 329dcd14… (106,686,152 bytes) carries the page, `main.js` and both build-172 Android apps; its 4 parts join to the same file (Wine joiner 4/4). No Mac app this build.

C · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or TV.

Build 171 addendum (4 November 2026)

What you asked for (4 Nov): install the UI skills (the ui-skills.com list plus Emil Kowalski's apple-design) and audit the app's interface in two phases. Phase 1 compared the running app with mockups and waited for your approval. You approved: "ok do them all". In the same message you added: "when a team crosses the time and doesnt answer, make it autopass without negatives".

Same version: v1.69. The Galaxy Tab and Mind It! TV apps are versionCode 171, and `APP_BUILD=171`. The changes are a new module (`app171.js`) and a `BUILD 171` block in `style.css`; no older module was edited. Scores, wagers and bonuses are unchanged. Time's up now passes by itself. Not a 5th build, so there is no Mac app this time. The guides that describe the time-up rule were rebuilt.

A · What's new

- **P1 · The laptop with the Audience window open (build 154's wide layout), question and answer screens.** The action card (host-only answer, who can still answer, CORRECT / WRONG ANSWER, PASS, …) starts at the top of the right column, with More controls under it. The timer card and the "Showing in the Audience View Window" note sit in the left column. Before, they sat above the action card: at 1366×768, Pass and Pause timer were cut off and four more buttons needed scrolling. The card also jumped from the left column (question) to the right (scored). This is a `style.css` change only.
- **P2 · The Undo pop-up never covers a host control.** Build 152 kept it above fixed action bars. It now tries the bottom (centre, left half, right half), then the strip under the top bar, then the first free height. On touch screens, the Undo button and Mastermind It!'s "✏️ Change a wager" toggle are 44 px tall (they were 40 and 42).
- **P3 · The Audience window's "Full screen" button** rests out of sight 10 s after the mouse stops, and whenever the window is full screen. It comes back when the mouse moves, like the Mind It! TV page's own button (`app117`).
- **T · Time's up passes by itself.** Wherever the ⏰ Time's up card used to offer WRONG ANSWER / PASS / Dismiss, PASS is now pressed automatically, 2 s after the clock runs out:
  - no points are lost;
  - the next team gets 30 s;
  - it counts as a pass, so the pass-on bonus applies;

- the TV says "TIME'S UP → … 'S TURN", and the host sees a note.
- When the last team runs out of time, the question ends as No one answered.
- Unchanged: Blind It! / Rewind It! squares and Triple Gamble (no card there before either; the team keeps them) and Classic answering.
- In Phone Buzzer Mode it passes wherever the card used to appear.
- In those 2 s, CORRECT / WRONG ANSWER (the laptop, its keys, the Galaxy Tab remote) still score the team whose time ran out, as they did while the card showed, and then nothing passes. Passing at the very moment the clock ran out would have handed a late tap to the next team.
- Older test suites keep the card through a test-only switch (`__tzTestNoAutoPass` in `tests/support/mockuser.js`).

B · Checks

- `fix171` **44/44** (browser, the real page):
  - **P1**, 1366×768 and 1366×700 with the window open: all 8 action buttons on screen (the last ends at y = 626); the card at x = 690 on the question, answer and scored screens; window closed and the Galaxy Tab unchanged.
  - **P2**: no overlap after scoring, after a wrong answer, on Mastermind It! marking or on the Galaxy Tab; Undo undoes; Undo and Change a wager 44 px on touch.
  - **P3**: the button rests, comes back on a mouse move, and rests again.
  - **T**, the pass itself: the 2 s, then the pass to Team Chai with no points lost and 30 s; the TV banner "TIME'S UP → TEAM CHAI'S TURN · 30 SEC"; the host's note; the buzzer, then the pass sound 2.0 s later; Team Chai's right answer +938 = 750 + 188 pass-on bonus.
  - **T**, inside the 2 s: WRONG ANSWER, CORRECT and the Galaxy Tab remote's WRONG ANSWER score Desert Foxes as before, and nothing passes afterwards.
  - **T**, the other cases: the last team gives "No one answered"; Rewind It!, Triple Gamble and Classic answering are unchanged; Phone Buzzer Mode passes where its card appeared; the older suites' switch still shows the card.
- **Full run, 223 suites:** browser 143 suites **4,383/4,383**; laptop app 80 suites **1,445/1,445**, with the build-171 apps inside the laptop app.
  - Four suites passed on their second try: `fix128t` , `fix135` , `desk` , `fix130a` (`fix130a` during a page-server restart). All four passed again on the final page.
- **Re-run on the final page** (955a43b3…; the page changed at 11:04 Dubai for the 2 s): the 16 laptop-app suites that ran before then, plus `fix130a` , **290/290**; 11 browser suites (`fix171` – `fix164` , `fix128t` , `fix135`) **500/500**.
- **Random full games** (`game171.py` , seed 1171): 4 games (3 teams on the laptop, 6 on the Galaxy Tab sideways with Blind It!, 9 upright with Rewind It! and Online Mode, 12 on a 1920 laptop), **0 failures**.
  - 43 clocks were made to run out; every one checked clean: no score moved, the pass recorded as a time-out, the next team's 30 s. 6 were by the last team; on 13 a verdict was pressed inside the 2 s.
  - The layout checker found the same 213 flags, in the same places, on the build-170 and build-171 pages with the same seed.
- **Stress phases and the 60-minute soak** (the real laptop app, the automatic pass on, a fresh one for each): idle 2/2, network 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6.
  - The **soak scored 15/15**: 60 minutes of real-time quiz with the tablet and 10 phones, 70 questions, 3 whole games to the winner, 309 tablet actions, **0 false disconnects**, no script errors. The laptop server's memory went from 199 to 212 MB.
  - The idle phase first scored 1/2. It leaves a question up for 5 minutes with the clock running, and the new rule passed the question round the teams and closed it. Its clock is now paused, as a host stepping away would, and it scored 2/2.
- **Pictures:** real BEFORE (build 170) and AFTER (build 171) captures in the same states, viewports and themes. Against the approved mockups, every P1/P2 screen has the same arrangement, a few pixels apart (`design-plans/ui171/after-compare/`).
- **Reproducible:** Galaxy Tab APK 6b24c9f8… and Mind It! TV APK f4796aba…, versionCode 171, signed with the original key; the tablet APK was built from a fresh copy of the repository and carries the tested page (955a43b3…).
- **Windows EXE** 5ff9114c… (106,688,294 bytes) carries the page, `main.js` and both build-171 Android apps. Its 4 parts join to the same file (Wine joiner 4/4). No Mac app this build: build 170's stays the latest, and the next is due with build 175.
- **Guides:** the Complete Idiot's Guide (22 pages), Quick Start (4), Quizmaster Cheat Sheet (2) and Players' Guide (2) describe the automatic pass. Every page was measured and fits. The Cheat Sheet also says End Timer only stops the clock.

C · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or TV, and no Wi-Fi was involved. The Galaxy Tab was emulated in Chromium at 1280×800 with touch.

Please try one question on a real night: let the clock run out and watch the turn pass 2 s later. Then press WRONG ANSWER just as the clock hits zero, and check that the team whose time ran out loses the points.

One thing not changed, for you to decide: with the Audience window open, the small pop-up notes overlap the top 4–5 px of Pause Timer / +30 Sec / End Timer for 4.5 s. They have done since build 154, and it wasn't part of what you approved. They could start under the timer card instead.

Build 170 addendum (3 November 2026)

What you asked for (3 Nov): from the build 170 ideas, "we only do the below" — 🛟 the Galaxy Tab as backup host, 📲 a picture from the phones, 🎯 who can still win, 🚪 a TV lobby — then "🎯 Who can still win? is rejected, Go Ahead with the remaining 3". A before/after page showed all three first.

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 170, `APP_BUILD=170` . New module `app170.js` and style rules; nothing in an older module was edited. The laptop app's server (`main.js`) and the tablet's server (`MindItHost.java` , `MainActivity.java`) gained the backup routes. No rule, score, timer or wager changed. A 5th build: the Mac app, the guides (builds 169 and 170) and the stress phases with the 60-minute soak are all in.

A · What's new

- 🛟 **The Galaxy Tab as backup host.** While the Galaxy Tab app is the laptop's remote, the laptop sends it a copy of the quiz once a second (only when something changed; a heartbeat otherwise): the whole game, the timer, the sealed Mastermind It! wagers and Final answers as the server holds them, and the profiles the quiz uses. Pictures and sounds travel once. The tablet keeps the copy in its own files. The laptop's top bar says "🛟 Backup: Galaxy Tab · up to date"; the tablet says "🛟 Backup ready · N s ago". When the laptop has been silent for about 10 seconds, the tablet offers "🛟 Take over on this tablet" (or "Keep waiting for the laptop"). Taking over opens the quiz on the tablet's own server on the same question with the timer paused, the next host epoch and the same game id. Team phones and TV browsers look for the tablet on their own and move there with their team. The tablet tells the laptop (when it can reach it) that it took over; the laptop then retires the quiz for good ("The Galaxy Tab took over this quiz at …", a Safety Backup kept) and sends late phones to the tablet. A laptop app started again later doesn't offer that quiz for Resume.
- 🚪 **TV lobby.** Before Begin game, the TV shows the show's name, a welcome message, a countdown to the start time (or "The quiz starts soon"), each team as its phone joins, "N of M teams have joined", and the join QR code. Setup → Audience View & Presentation: the message (80 characters) and an optional start time. Squeezed when many long names would reach the QR code.

- 📲 **Every team's night as a picture.** When the game ends, each team's phone draws a 1080×1920 picture: place, score, how far from 1st (or ahead of 2nd), the show and the date, and its best moment of the night (a Rewind It! spin, a Mastermind It! wager, its best answer, a streak). Save picture downloads it; Share appears only where the phone allows sharing files from the page. Each phone gets only its own team's moment.
- **Fixed (found by the build 170 checks):** a laptop app stopped suddenly soon after Begin game could open again on Setup without offering the quiz, when the only quick save that survived was from Setup (the browser's storage in the laptop app writes to disk seconds — under load, much longer — behind the game). The complete copy on disk (build 140) is now offered when it is newer than a Setup quick save. The laptop also waits for that copy before asking whether the Galaxy Tab took the quiz over.

B · Checks

- `fix170` **75/75** (browser): the TV lobby (welcome, countdown, start time, teams popping up, "N of M teams have joined", 4 and 12 teams with long names kept clear of the QR code, the switch off), Setup's message and start time, the phone's picture at three phone sizes (place, score, gap, moment; Save; Share hidden where the phone can't share files; nothing before the end), the backup screens (the laptop's sign, the tablet's "Backup ready" both ways round, the take-over offer and its sizes, the banner after taking over, a phone moving, the laptop retiring), and the crash-save fix (six cases, then a quiz really reopened from its complete copy).
- `fix170b` **34/34** (the real laptop app and the tablet's server built from the APK's Java source): A · the laptop killed mid-question — the tablet holds the copy and says so, offers to take over about 10 s later, takes over with the same scores, question and paused timer (31 s left on both), the same game with the next host epoch; both team phones and a TV browser move to the tablet with their teams and no codes; the quiz carries on; the laptop app started again doesn't offer the quiz and sends a late phone to the tablet. B · the laptop page frozen during Mastermind It! wagers — the sealed wagers carried over exactly, the laptop app hears the tablet and retires the quiz, phones go straight to the tablet, a wrong key is refused.
- `fix170c` **22/22**: C1 · the picture's data on both servers (only once the game has ended, only the team's own moment, none for a phone without a team); C2 · a question with a picture on the Rewind It! wheel (4×): the picture reaches the tablet once, apart from the game; after the laptop dies the tablet takes over with the 4× and the same picture, and a right answer pays 4×; C3 · a laptop page that can't hear the tablet asks it, finds out it took over, retires the quiz and sends phones there.
- **Full run: 222 suites** (on the page before the crash-save fix): browser 142 suites **4,323/4,323**; laptop app 80 suites 1,438/1,443. The misses: `fix170c` C3 started the tablet's server before the laptop app, so the server took the laptop's port — a mistake in the test, fixed (22/22); `fix142d` / `fix143d` ran before the build-170 tablet app existed (17/17 and 13/13 once it was in the laptop app); `fix170b` passed on its second try — its first found the crash-save gap above, now fixed.
- **Re-run on the final page** (66ad1746…): `fix170` 75/75; laptop app and browser: `fix170b`, `fix170c`, `fix121`, `fix130m`, `fix130x`, `fix140sd`, `fix140s`, `fix151`, `fix151u` **267/267**; `fix142d`, `fix143d` **30/30** with the build-170 apps inside the laptop app; 23 browser suites (`fix170`, `fix169`, `fix168`, `fix168p`, `fix167`, `fix166`, `fix165`, `fix164`, `fix163`, `fix155`, `fix155h`, `fix154`, `fix154p`, `fix153l`, `fix152tab`, `fix151c`, `fix141`, `fix140s`, `qc2`, `lighter111`, `fix134`, `fix135`, `fix138p` — the newest builds, the layout and size checks, and every suite that resumes a saved game) **1,231/1,231**.
- **Random full games** (`game168.py`, seed 1170): 4 games (9 teams on the laptop, 3 on the Galaxy Tab sideways with Rewind It!, 12 upright with Blind It! and Online Mode, 6 on a 1920 laptop with Rewind It! over 3 rounds and a Wild Hot Potato), **0 failures**; every Rewind It! score matched the rules. The layout checker found no new kind of flag. It counted more flags than build 169's run, so the first game was played again with the same seed on the build-169 page: 330 flags on both, from the laptop's More controls opened on a short screen (it scrolls there; checked).
- **Stress phases and the 60-minute soak** (the real laptop app, a fresh one for each, on the final page): idle 2/2, network 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6; the **soak 15/15** — 60 minutes of real-time quiz with the tablet and 10 phones, 71 questions, 3 whole games to the winner, 308 tablet actions, the window hidden and shown 5 times, **0 false disconnects**, no script errors; the laptop server's memory 201 → 211 MB, the Quizmaster page's JS heap 15 → 19 MB.
- **Reproducible:** Galaxy Tab APK cde4222c… and Mind It! TV APK 7665934b…, versionCode 170, signed with the original key; the tablet APK was built from a fresh copy of the repository and carries the tested page (66ad1746…).
- **Windows EXE** 1c9b7476… (Electron 44.9.3) carries the page, `main.js` and both build-170 Android apps; its 4 parts join to the same file (Wine joiner 4/4). **Mac app** (Apple Silicon, ad-hoc signed) 4c3530db…: the page, `main.js` and the tablet app; 6 parts join to the same zip.
- **Guides** refreshed for builds 169 and 170 (the TV lobby, the backup host with a picture from the real take-over, the named buttons, Now answering, "Open the steps ›", No loss, the phones' picture): the Complete Idiot's Guide 22 pages, the Quick Start 4, the Quizmaster Cheat Sheet 2 and the Players' Guide 2 — every page measured to fit (the two one-pagers' new lines were shortened until they did).

C · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or TV, and no Wi-Fi was involved: the laptop app ran on the build machine under a virtual screen, the Galaxy Tab's server ran from the same Java source as the APK, and the tablet, phones and TVs were browser pages at their real sizes. Please rehearse a take-over before relying on it: start a quiz on the laptop with the Galaxy Tab app as its remote and a phone or two joined, then close the laptop's lid or switch off its Wi-Fi mid-question and take over on the tablet.

Build 169 addendum (3 November 2026)

What you asked for (3 Nov): "any improvements for the host journey on laptop or tablet?" — a full night was played through your host buttons on the laptop (1366×768) and the Galaxy Tab both ways (1280×800, 800×1280), and a before/after page proposed four changes and three small ones. Then: "do them all".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 169, `APP_BUILD=169`. New module `app169.js` and style rules; nothing in an older module was edited. No rule, score, timer, wager, save or sync changed (the remote's state gets one read-only field, the answering team). Not a 5th build and not a scoring change: no Mac app and no stress phases or soak this time (both due at 170). The guides are build 168's.

A · What's new

- **H1 · The team's name on CORRECT and WRONG ANSWER** — on its own line above the words, on the laptop, the Galaxy Tab both ways, the tablet as a remote and the phone remote. It follows the turn after a wrong answer, a Rewind It! / Blind It! square's team and a passed 🥔 Hot Potato; Classic answering (scored per team) shows none. The buttons' words, the C / W keys and what each button does are unchanged (the name is drawn on top).
- **H2 · Galaxy Tab upright: Now answering** — the question screen opens with the answering team in large type, the question, and every team's score (the answering team lit, teams that already tried dimmed). Undo & resets moved below the other cards.
- **H3 · Laptop board panel fits the screen** — Filling the board… Skip and the round's buttons sit under Now picking, then the scores in slimmer rows: the whole panel fits 1366×768; at 1280×720 Skip and Go To The Final are on screen.
- **H4 · Setup** — one gold button (Begin game); "Start" became "Open the steps ›" with a gold outline; the "Add questions to the Question Database" card left Setup in the laptop and tablet apps (the main menu and the tablet home keep the Question Database, which has Import Template; a plain browser keeps the card); the tablet's Setup says "this tablet".
- **Small:** Mastermind It! marking's "0" reads "No loss"; after a round, Save backup file is outlined so Start Round is the one gold button.

B · Checks

- `fix169` **60/60**: every change on the laptop and the Galaxy Tab both ways — the names (after a wrong answer, Rewind It!, a passed Hot Potato, a 52-letter name that fits, Classic answering without names, both remotes), Now answering (the question, 4 and 12 teams, tried teams, gone once answered, not on the laptop or sideways), the board panel order and what's on screen at 1366×768 and 1280×720, Setup in the laptop app / tablet app / a phone browser, "No loss" still meaning wrong with no penalty, the outlined backup button, and the app's sizes (text ≥ 14 px, touch ≥ 44 px, nothing off the side).

- **Full run: 219 suites** (on the page before the two fixes below): laptop app 78 suites **1,389/1,389**; browser 141 suites 4,113/4,116. The misses: `fix155h` found the upright tablet's question screen 41 px taller than the screen (build 155's rule: it fits one screen) — fixed: the question card's own "Question on screen" title and "Now playing" line go when Now answering shows; `fix154` found the laptop board panel reordered while the Audience View window is open (build 154's Pick a clue card leads then) — fixed: the panel keeps build 154's order then; `fix154p`, `dbbtn` and `newf` were written for the old Setup ("Start" gold, the Question Database card) — `fix154p` now clicks "Open the steps ›", and older suites keep the card they test (fix169 checks the new Setup); `fix167` missed a picture download once (70/70 on the re-run).
- **Re-run on the final page** (dc844081…) with the build-169 apps: 25 browser suites (`fix169`, `fix168`, `fix168p`, `fix167`, `fix166`, `fix165`, `fix164`, `fix155`, `fix155e`, `fix155a`, `fix155l`, `fix155h`, `fix154`, `fix154r`, `fix154p`, `fix153l`, `fix153z`, `fix153c`, `fix153p`, `fix153v`, `fix152l`, `fix152tab`, `fix151`, `newf`, `dbbtn`) **1,198/1,198** and 6 laptop-app suites (`fix163d`, `fix161o`, `fix155d`, `fix153d`, `fix142d`, `fix143d`) **71/71**.
- **Random full games** (`game168.py`, seed 1169): 4 games (9 teams on the laptop, 12 on the Galaxy Tab sideways, 3 upright, 6 on a 1920 laptop; Blind It! and Rewind It! with a passed Wild Hot Potato), **0 failures**; the layout checker found no new kind of flag and none on the new panel or the named buttons.
- **Reproducible:** Galaxy Tab APK 8bab6efa… and Mind It! TV APK 4be73ead…, versionCode 169, signed with the original key; the tablet APK carries the tested page (dc844081…).
- **Windows EXE** f74b6043… (Electron 44.9.3) carries the page, `main.js` and both build-169 apps; its 4 parts join to the same file (Wine joiner 4/4).

C · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or TV. Please look at a question on the upright Galaxy Tab (Now answering), a wrong answer passing to the next team (the name on the buttons), and the board on your laptop's screen.

Build 168 addendum (2 November 2026)

What you asked for (2 Nov): "for rewind it, make the negative multiplier for 4x and 5x to 3.5x instead of 3x"; then, from the Wild-wheel ideas, "2,5,9,11,13" — 🛡️ Safety Net, 🃏 Wildcard, 🏹 Robin Hood, 😺 Rascala's Choice, 🥔 Hot Potato — with "The wild wheel is an optional feature so the host can activate it before the quiz", "the wild wheel, will only have the 5 wild choices, the multiplier wont be there", "the wild wheel only comes in default when there are 4 or more rounds and will only happen in the round before mastermind it", "for the options where a team chooses the multiplier, the host will input it in the device", "for robin hood if the last placed team is the one who got it then they get 3x the value but lose 1x", "hot potato if passed, the team who gets the question passed on to, dont get negative points", "if we are playing 4 rounds (3 rounds + 1 mastermind it) then on the third round will have 2 rewind it and 1 wild rewind it, rather than 2 rewind it" and "just reconfirming, wild rewind it only happens when there 3 or more question rounds". And "release build 168 with all the planned features".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 168, `APP_BUILD=168`. New module `app168.js` and style rules; nothing in an older module was edited. Not a 5th build: **no Mac app** (next at 170). The **Complete Idiot's Guide** (20 pages, a new Wild-wheel picture) and the **Quizmaster Cheat Sheet** are rebuilt for the new rules; the Quick Start is build 166's, the Rehearsal Checklist and the Players' Guide build 160's. A scoring change is a **major** change by your rule, so this build had the stress phases and the 60-minute soak.

A · What's new

- **4× and 5× lose 3.5×:** a wrong answer on a Rewind It! square loses 1×, 2×, 3×, **3.5×, 3.5×** the value (rounded to whole dirhams); right still wins value × the spin. One rule feeds the score, the WRONG ANSWER button, the host's hint, both payout tables (laptop and TV), the question line on the TV, Setup's description, the numbers sent to the team's phone and the phone's own text.
- 🎡 **Wild Rewind It!:** with **3 or more question rounds** (Mastermind It! doesn't count) the **last question round** gets **one extra square** — its usual Rewind It! squares stay on the classic wheel (so with 3 rounds + Mastermind It!, round 3 has 2 Rewind It! + 1 Wild Rewind It!). The TV says WILD REWIND IT! and spins a five-slice wheel, one chance in five each, no 1×–5× slices:
  - 🛡️ **Safety Net** 2×, a wrong answer costs nothing;
  - 🃏 **Wildcard** — the team chooses 1×–5×, the host taps it in (or presses 1–5); no question until it's entered;
  - 🏹 **Robin Hood** 2× and +1× to the last-place team (level last places all get it; one Undo takes back both); wrong −1×; the last-place team spinning it: right 3×, no gift, wrong −1×;
  - 😺 **Rascala's Choice** — the last-place team (of the others) chooses 1×–5× for the spinner; the host taps it in;
  - 🥔 **Hot Potato** 3× / −3×, or pass it once to any team: right +3×, wrong costs them nothing.
- **On by itself, can be turned off:** Setup → Rounds → Special Question → 🤠 Classic wheel only / 🎡 + a Wild Rewind It! square (greyed with fewer than 3 question rounds, and says why), and a Wild Rewind It! Off/On row in Prepare tonight's quiz. Same as last time and Rematch keep the choice. Blind It! nights have no Wild square.
- **Everywhere it shows:** the TV (its own wheel, the result with its emoji, the five slices' rules), the host's card, the remote's buttons, the team's phone in Buzzer Mode (it spins the Wild wheel; while a 🃏 / 😺 choice is being entered it shows no multiplier), the saved game, the Night Recap (a passed Hot Potato counts for the team that answered), and Rehearsal bots that make their own choice.

B · Checks

- `fix168` **54/54**: the square counts and rounds (2 / 3 / 4 / 5 question rounds, with and without Mastermind It!, turned off, a round opened again), the classic 1×–5× table on every value, each Wild slice's score through the app's own buttons (Robin Hood's gift, level last places, the last-place rule, Undo; Hot Potato's pass and no loss), the wheel's fairness (50,000 draws, 9,892–10,118 per slice) and landing angles, Setup / Prepare tonight's quiz / Same as last time, the saved record and the Night Recap, the bots, and the app's sizes (text ≥ 14 px, touch controls ≥ 44 px) on the laptop and the Galaxy Tab both ways.
- `fix168p` **23/23** (new): a Wild square spun from the **team's phone** in Buzzer Mode — 🛡️, 🃏 (the host enters 4×: the phone is then told +4× / −3.5×), 🏹, 🥔 (passed: nobody loses), the phone's card.
- Older suites that encoded "at most 3×" were brought to the new rule: `fix148` 64/64, `fix149` 100/100, `fix151` 37/37, `fix151p` 12/12, `fix167` 70/70.
- **Your backup** (private; counts only): 25 players, 3 nights; every past night opens, saved scores and places unchanged, no page errors. With your usual 2 question rounds there's no Wild square; a new game from the same Setup with 3 question rounds puts the Wild square in round 3 beside its 2 Rewind It! squares.
- **Random full games** (`game168.py`, which now also works out every Rewind It! score from your rules by itself and checks where the Wild square sits): **12 games, 0 failures** — seeds 1168 and 2168 (3 to 12 teams, the laptop and the Galaxy Tab both ways, Rewind It! and Blind It!, one 3-round game with Wild switched off) and seed 3168 with every game a Wild game. All 50 Rewind It! scores matched the rules (among them 6 wrong 4× / 5× answers losing 3.5×, and 6 Wild spins: 😺 Rascala's Choice ×3 — each a wrong answer on a chosen 5×, −3.5× — and 🃏 Wildcard ×3, right on a chosen 1×, 4× and 5×; each choice tapped in by the host). Safety Net, Robin Hood and Hot Potato didn't come up in these games; `fix168` and `fix168p` cover each of them. The Wild square appeared only in the last of 3 question rounds, once, never with 2 rounds, on a Blind It! night or with Wild switched off. The crown, the end-of-night order and the saved records were right every time. The layout checker's flags were the same kinds as build 167's, on screens 168 didn't change; on the Wild wheel it flags each slice's emoji against its own name — the square boxes around rotated text, not a real overlap (checked on a 1920 TV picture).
- **Full run: 217 suites** on the build-168 page (b34c7dd0…):
  - browser: 139 suites, 4,177/4,178 — the one miss was `fix167` still expecting the old "lose at most 3×" on a 5× spin; updated and re-run: 70/70. Passed on retry: `fix151a`.
  - laptop app: 78 suites, **1,389/1,389**.
- **Stress phases and the 60-minute soak** (a scoring change is major), each on a fresh laptop app: idle 2/2, network 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6, and the **soak 15/15** — 60 minutes of real-time play with the TV, the tablet and 10 phones: 70 questions, 3 whole games to the winner, 0 false disconnects, no script errors; the laptop's server 200 → 214 MB, the Quizmaster page 14 → 21 MB. (The soak plays 2 question rounds with Blind It!, so it never reaches a Wild square: the Wild wheel is covered by the suites and the random games above.)

- **Reproducible:** Galaxy Tab APK 58dc709a… and Mind It! TV APK 821eb0ca…, versionCode 168, signed with the original key (5f4819e4…); the tablet APK carries the tested page (b34c7dd0…). Windows EXE ce9427d1… (Electron 44.9.3) carries the page, `main.js` and both build-168 apps; its 4 parts join to the same file (Wine joiner 4/4).

C · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, TV or Mi Box. Please try a Rewind It! night with **3 question rounds + Mastermind It!** to see the Wild Rewind It! square in round 3: the Wild wheel on the TV, tapping in a 🃏 / 😺 choice, a 🥔 pass, and a team spinning it from their phone in Buzzer Mode.

Build 167 addendum (2 November 2026)

What you asked for (2 Nov): from the ideas list, "5,6,7,8" — 🤠 Rewind It! highlights in the Night Recap, the champion's belt history, a season finale on the TV with a picture for the chat, and head-to-head records with a nemesis — and "ensure this includes the historic games too". Then: "release the build once all work is done, and from next time onwards dont wait for my request to release".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 167, `APP_BUILD=167`. New module `app167.js` (Q1–Q4) and style rules; nothing in an older module was edited. Not a 5th build: **no Mac app** (next at 170; the release carries none, as build 164). The **Complete Idiot's Guide** is rebuilt for the new features (19 pages, three new pictures); the Quick Start and the Quizmaster Cheat Sheet are build 166's, the Rehearsal Checklist and the Players' Guide build 160's. Stress phases and the soak: not due (no change to how a live game runs, the connections or saving; the next is due at build 170).

A · What's new

- **Q1 · Rewind It! on the Night Recap:** a row between the night's numbers and the highlights — the biggest spin, the luckiest and unluckiest spin (worked out from the night's score journal, so an undone score doesn't count) and who won most from the wheel (or what the wheel gave in all, when one spin already says it). The same on each past night's page in Historic Games. A game saved from 167 on keeps each spin's result (a new field `d` on each saved spin; older builds ignore it). Records saved before 167 kept the spins but not their results: their page shows the biggest spin, who spun most and the average, and says so. The last night played on a device gets its results back from its journal.
- **Q2 · 👑 Champions (Players & History):** every reign from all of History in the order the nights were played (Add Past Game counts on its own date), with the owner's crown rule from build 122 (the whole winning team, or a one-player winner; deleted players don't count; a team with no players by its name): current champions, longest reign, most nights wearing the crown, every reign with "Dethroned by …" or "Didn't defend it". Each player's page: "👑 N nights wearing the crown".
- **Q3 · 🏆 Season finale:** from Players & History → 🏆 Season or the Season standings card at the end of an Online Mode night (tonight counts before End Quiz & save, as on the card). The TV: intro → Grand Rascala → 3rd → 2nd → Season Champion (a missing place is skipped), Next / Back / Space; ✓ Finish, ✕ or Esc puts the TV back. Nothing is kept in the saved game. The picture: 1080×1350, the whole season. The remote leaves the button out (its controls open on the host's screen). Setup's small TV preview keeps showing tonight's walk-on.
- **Q4 · ⚔️ Head to Head:** each player's 😈 nemesis and 🎯 favourite opponent (2+ meetings, the one ahead on the record) on their page; any two players compared in Players & History → ⚔️ Head to Head (record, nights on the same team, every night they met). Build 64's head-to-head now counts merged players as one person and goes by score for a record without places.

B · Checks

- `fix167` **70/70**: the four features from real play (the wheel spun and scored by the app, the Night Recap on the TV, the saved record, a past night's page, old records), the reigns (dethroned, absent, shared wins, a new player joining, Add Past Game's date, a deleted player, a team without players), the current champions against the crown the app gives a new game, nemesis and favourite, merged players, the finale (both entry points, every step, Space / Back / Esc / a tap beside the panel, the picture, nothing saved, the remote, previews), and the app's sizes (text ≥ 14 px, touch controls ≥ 44 px, nothing off the side) on the laptop and the Galaxy Tab both ways.
- **Your backup** (private; counts only): 3 games, 25 players, one merged player. The 3 real Rewind It! spins of the last night were read from its journal; its page got its results back. Champions: 2 reigns, the right current champion. The October–December finale: 10 players, all five steps, joint Grand Rascalas. No page errors.
- **TV sizes:** the Night Recap with the Rewind It! row at 3, 6 and 10 teams and the finale (every step, two seasons) on a 1920 and a 1280 TV and upright 1080×1920 and 800×1280: nothing outside the screen; the recap still shows 1–3 of its highlights beside the new row. (Upright with 10 teams the recap's table is small — the same in build 166.)
- **Random full games** (the build-166 game runner, plus checks of the new screens after each saved game): 4 games (seed 1167) — 3 teams on the laptop at 1366 (Online Mode, Rewind It!), 6 on the Galaxy Tab sideways (Blind It!), 12 upright (Rewind It!), 9 on the laptop at 1920 (Blind It!); Mastermind It! in three; players moving between teams. **0 failures.** Every Rewind It! spin was saved with its result (5 of 5, 3 of 3) and the night's page showed them; after every game the current champions were that night's winners; the crown went to the right team each time (with a new player in game 3); 👑 Champions and ⚔️ Head to Head opened; the finale's table matched the season table. The layout checker's flags were the same kinds as in build 166's games, on screens build 167 doesn't change; two kinds its games never reached — the 12-team Mastermind It! wager screen on the upright Galaxy Tab (the small TV preview is below the visible part of the screen there) and a 🔥 streak badge beside a place label on the Blind It! wager screen — are noted to be looked at in the next build.
- **Full run: 216 suites** on the build-167 page:
  - browser: 138 suites, 4,114/4,121 — the 7 misses were `fix153l` finding the new crown line at 13 px (the app's minimum is 14 px). Fixed (every new panel brought to the app's sizes), re-run on the final page 90/90.
  - laptop app: 78 suites, 1,365/1,367 — `fix142d` and `fix143d` checked that the laptop app carries the tablet app of its own build, run before the build-167 APKs existed (17/17, 13/13 once they were in); `fix115` had no result because the test page server on this machine was answering on the local address only (a setup slip, not the app): 22/22 once served properly. Passed on retry: `fix158d`.
  - re-run on the final page (e65f8ae1…) with the build-167 apps: `fix167`, `fix166`, `fix158`, `fix155`, `fix153l`, `fix153z`, `tabs2`, `recap`, `fix142d`, `fix143d`, `fix115`: **395 + 22 checks, all passed**.
- **Reproducible:** Galaxy Tab APK 06fb27d2… and Mind It! TV APK 0b7055d0…, versionCode 167, signed with the original key; the tablet APK carries the tested page (built twice from a fresh clone: byte-identical). Windows EXE 2df24e00… (Electron 44.9.3) carries the page, `main.js` and both build-167 apps; its 4 parts join to the same file (Wine joiner 4/4).

C · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, TV or Mi Box. Please try: a Rewind It! night to the end and its 📊 Night Recap; Players & History → 👑 Champions and ⚔️ Head to Head on the tablet; and the 🏆 Season finale on the real TV, with 📋 Copy picture for the chat into your group chat.

Build 166 addendum (1 November 2026)

What you asked for (1 Nov, after build 165 went out): the live standings ("player name was going over their score", Galaxy Tab, mid-game); "the top three werent showing before the rascala" (you chose the podium, then The Rascala!); a new player shown as the returning champion; "the players tab in the tablet needs to be optimized"; "make rewind it on by default and blind it the alternate option"; "do a complete check, run around 2-3 games with different number of teams with random permutations"; and the Mac app and guides put off from 165. Then "release the build once all work is done".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 166, `APP_BUILD=166`. New module `app166.js` (N1–N6) and style rules. The 5th build since 160: **Mac app** included; **guides refreshed** (Complete Idiot's Guide, Quick Start, Quizmaster Cheat Sheet; the Rehearsal Checklist and Players' Guide are build 160's).

A · What was wrong, and the fix

- **The end of the night (N1):** final standings → awards → the podium (3rd, 2nd, winner) → 🐱 The Rascala! → Season standings (Online Mode) → End Quiz. The automatic Rascala 9 s after the winner (build 148) would have cut the podium short, so it is off; ▶ Next: The Rascala! shows it.
- **The returning champion (N2):** reproduced on the released build 165. In Online Mode the 🎲 random turn order is drawn *after* the crown is given, and the crown was stored as a place in the team list — so it went to whoever was drawn into the champions' place (in 16 test starts, every Online Mode start crowned the team drawn first, twice the new player; every room start was right). The crown is now worked out after the draw and follows its team. Your rule (build 122: the whole winning team; a one-player winner's team) is unchanged. A game typed in with Add Past Game no longer counts as "last game" because it was entered today.
- **Rewind It! by default (N3):** a new quiz starts with Rewind It!; a game saved part-way keeps the special it started with.
- **The live score column (N4):** found by the randomized games, not by the standings screen checked earlier. On the TV picture's score column with **9+ teams, while every team was level** (0 at the start of a game), the cards fell apart: the name sat on the score and the picture moved to the middle. Also the 👑 CHAMPION badge sat on the champions' name with 10+ teams, and the 🔥 streak badge covered "Now playing" (Gaming theme) or collided with the crown. All fixed; with 4–8 teams the column is unchanged from 165.
- **Player Profiles (N5, N6):** on the web page, Player Profiles opened before the profiles had loaded said "No players yet" until closed — it now fills in. On touch screens, scrolling a panel re-drew the blurred page behind it on every step (15 of 40 frames over 50 ms on a 4× slower CPU with your library; 16–18 ms a frame without the blur): panels now sit on a plain dark backdrop. Upright, panels use the tablet's full width (four players a row instead of three).

B · The complete check: randomized full games

A driver plays whole games through the host's own buttons (every question with random right / wrong / pass / no one, Blind It! wagers, the Rewind It! wheel, a mid-game look at the standings, Mastermind It! wagers and marks, Reveal all, the end of the night with Back now and then, End Quiz & save), checking every score change against its button, the end-of-night order, the saved game (scores, places, special, Online Mode), the returning champion, and the layout of every new screen.

- **8 games** in two runs: 3, 6, 9 and 12 teams; the laptop at 1366 and 1920, the Galaxy Tab sideways and upright; Rewind It! and Blind It!; Online Mode in five; 2–3 rounds plus Mastermind It!; random themes; players moved between games (last night's winners kept together — renamed, with a new player — or split up).
- Second run (after the fixes): **4 games, 0 failures**; the champion right in all three later games, including after Online Mode's draw.
- First run: no failures in the games that ran on the laptop; the two on the tablet played to the end and saved, then the runner lost the page because End Quiz on the tablet returns to the tablet's home (the runner was fixed). This run found the score-column problem.

C · Tests

- Score column: 1,512 screens (14 themes × 4–12 teams × level / scores × champion × streaks, the Galaxy Tab's TV picture both ways and a 1920 TV) plus 288 on the laptop and a 1280 TV: nothing on a name or score except 3 px of line spacing (the same on 165).
- Galaxy Tab panels with your library, both ways: clean. The Rewind It! wheel at 6 and 12 teams on three TV sizes: clean.
- `fix166` 64/64 (end order, crown, Rewind default, score column, Player Profiles, tablet panels); `fix155e` 52/52 and `fix148` 64/64 updated for the new order; `champ` 9/9 and `champ122` 15/15 (your champion rule).
- **Full run: 215 suites**, on the final page (dad1247d…), the laptop app carrying the build-166 tablet and TV apps:
  - browser: 137 suites, 4,025/4,029;
  - laptop app (Electron 44.9.3): 78 suites, 1,358/1,367.
  - **What failed, and why** (all were the tests, not the app; every one re-run alone and passing):
    - `fix153z` , `fix151u` : expected the old order (the podium as the last card). Updated: 63/63, 33/33.
    - `fix141` , `fix128t` , `fix130y` (and the shared tablet hand-over helper): written for Blind It! as the new quiz's special, so their Daily Double opened the Rewind It! wheel. They now choose Blind It!, as their checks are about the Blind It! wager: 50/50, 22/22, 22/22. The older suites keep Blind It! through a test-only switch (like build 148's switch for the automatic Rascala); `fix166` checks the real default.
    - `fix130x` : the same game setting changed the phone's text so a check's 200-character window caught the word "team code" from a lower section; with Blind It! chosen 32/32.
    - `fix116` : a tablet-server stand-in left running by an earlier suite held port 8765, so the laptop app moved to 8766. Stopped: 26/26.
    - `fix139d` (the server limits under load): 23/23 alone.
  - Passed on retry: `v169` , `fix130t` .
- **Reproducible:**
  - The Galaxy Tab APK (18232e19…) and the Mind It! TV APK (706127e2…) are versionCode 166, signed with the original key; the tablet APK carries the tested page and asks for the same 11 permissions as 165 (min SDK 24, target 35).
  - The Windows EXE (4e8ba415…, Electron 44.9.3) carries the same page, `main.js` and both build-166 apps; its 4 parts join to the same file (Wine joiner test 4/4).
  - The Mac app (Apple Silicon zip 432db9a4…, ad-hoc signed) carries the same page, `main.js` and both apps; 6 parts join to the same zip.

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, TV or Mi Box. The Players tab speed was measured on a slowed-down desktop browser with your library, not on the tablet itself. Please try: an Online Mode night with last week's champion (the crown), a game with 9+ teams on the Galaxy Tab (the score column at the start), and the end of the night (podium, then The Rascala!).

Build 165 addendum (1 November 2026)

What you asked for (1 Nov, after build 164 went out): "have a layout check on all the screens, ensure its in perfect condition and if fixes are there then complete it and release the build", then "put off the doc updation and mac build for 166, in 165 put update app on main menu screen".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 165, `APP_BUILD=165` . New module `app165.js` (M1–M5) and style rules (M6, M7). No Mac app and no guide or QC-report changes in 165 (both came with build 166).

A · What was checked

- **A crawler** (`sweep165.py`): from the main menu or the tablet home, every Setup tab, the game board's menus and 30 panels opened directly; every safe button clicked two or three levels deep (nothing that deletes, resets, restores, starts, ends, downloads, imports, syncs or shares). On every new screen: sideways page scroll, text cut, text off screen, text on top of text, low contrast, tap targets under 24 px. Laptop 1366×768 and 1920×1080, Galaxy Tab 1280×800 and 800×1280, a phone as the host (390×844), the web page.
- The game screens at 6 and 12 teams on the laptop, the Galaxy Tab and the tablet remote; phones with 6 teams; the TV page and the Galaxy Tab's start page; all 14 themes on 12 Audience View screens at three sizes; the returning champion's badge; the Audience View audit at 9 sizes.

B · What didn't fit, and the fix

- **Galaxy Tab upright, during play:** the cards you swipe sideways lost their layout ("Now Playing / team / Next question" on one line).
- **Returning champions with 6 teams:** the 👑 Champion tag made the score cards taller and squeezed the board's category names; with 5+ cards in a row the badge is "👑 ×N".
- **Gaming theme, 8+ teams:** the board's last column went under the score column; a long side score touched its "1st" badge.
- **Comic theme, 5–6 teams:** category names and the bottom row of the board were cut.
- **Galaxy Tab home, upright, nothing saved yet:** big gaps; now centred.
- **Laptop main menu in a 1280×650 window with Continue Last Game:** 17 px too tall; a fourth fit step.
- **A phone as the host:** player rows overlapped the next row; long "Pick a Clue" words were cut.
- ⬆ **Update App (your request):** a fifth tile on the laptop's main menu (the build and its update state; opens the updater) and an Update App tile on the Galaxy Tab home.

C · Tests

- Audience View audit on the final page: 405 screens, no issues. `fix165` 41/41. Crawler rechecks: 300 screens each on the laptop, tablet landscape, tablet portrait and phone host; 100 on laptop 1920 and the web page. What was left flagged was the checker (bold words wrapping, a closed fold's contents).
- **Full run** on the final page (a026509b…), the laptop app carrying the build-165 tablet and TV apps: browser 136 suites, 3,975/3,976; laptop app 78 suites, 1,389/1,389. The one miss: `fix154p` checked the main menu's tiles as an exact list and didn't know the fifth tile (fixed; 117/117). Passed on retry: `fix154r`, `fix151a`, `fix163d`, `fix158d`.
- APKs versionCode 165, original key; the tablet APK carries the tested page. Windows EXE (Electron 44.9.3) carries the same page and both apps; 4 parts, Wine joiner 4/4. Released as v1.69-build165; all 12 files downloaded again and checked.

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, TV or Mi Box. After release you reported that the tablet's app update says the app is harmful: that is Google Play Protect warning about a sideloaded app that can install updates (the 165 app asks for nothing new compared with 164).

Build 164 addendum (1 November 2026)

What you asked for (1 Nov, after build 163 went out): "any upgrades especially for online mode / lets make maximum teams 12 and adjust all the visual layouts for that too, so check everything". Online Mode ideas were sent separately. Then "its ok, release the build as soon as its ready".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 164, and `APP_BUILD=164`.

- New module `app164.js` (`build.py` order `…,163,164,6`) and style rules for portrait screens and the standings fit. No server change.
- Code from commit 1d69d66 (tests and guides changed after it, the page didn't).
- Not a 5th build, so **no Mac app**: the Mac stays on build 160 until build 165.
- Guides: the Complete Idiot's Guide and the Quizmaster Cheat Sheet now say 2–12 teams (rebuilt for build 164). The Quick Start, the Rehearsal Checklist and the Players' Guide are build 160's.

A · The limit

- Build 153's limit of 10 (`B153_MAXT`) is 12. Everything that offers or caps the number of teams reads it: Quick Setup's Number Of Teams, + Add Team, 🎲 Random Teams, Add Past Game, an Excel team list, presets and the rehearsal bots' room.
- The laptop app's server and the Galaxy Tab's server already kept 12 teams (checked in their code).

B · Every layout checked at 12 teams

Compared at 12 (and 11) with the same screens at 10, so only what's new at 12 counts:

- **Audience View:** every screen of the Audience View audit (board, questions, wagers, chips, standings, Mastermind It!, winner, awards, recap, intro, how to play…), with long names, at 1920×1080, 1280×720, 1024×768, 1920×1200 and 800×1280.
- **Laptop and Galaxy Tab host:** the same screens plus Setup's Teams and Quick Setup, Random Teams and Connected Devices, at 1366×768 and 1920×1080, and the Galaxy Tab at 1280×800 and 800×1280.
- **Tablet remote:** portrait and landscape.
- **Phones:** on the laptop app, a 390-px phone on team 1, a 360-px phone on team 12 and one not joined, every screen. With the scoreboard opened, all 12 rows fit.
- **Host Cam:** the same place at 10 and 12, covering nothing.
- **Pictures:** standings, Night Recap and champions, at 10, 11 and 12.

What didn't fit, and the fix:

- **Night Recap picture:** the statistics ran into the footer. Rows are 30 px instead of 34 for 11–12 teams; the 10-team picture is pixel-identical to build 163's.
- **Portrait screens (800×1280):**
  - the standings under the podium ran off the bottom;
  - the Mastermind It! answers squeezed the question card to 50 px and hid the answer (now wager cards six to a row from 9);
  - the standings' round title wrapped over "Standings · Round N". This one happened at any number of teams.
- **Standings under the podium in landscape with 12 long names:** a fourth line went below the screen. Found by the new suite, not by the audit, whose names were shorter. Now the screen measures itself once the podium has settled and takes the rows one size smaller at a time, only from 8 rows (11+ teams).

C · Tests

- **Before the full run**, on the release page (9be33f7f…):
  - the 12- and 11-team Audience View audits: 225 screens each, **no issues**;
  - the standard Audience View audit on the podium screens, at all 9 sizes: 63 screens, no issues;
  - 27 suites that touch the TV page, layouts and team counts: all passed in the end. Three needed a second look, all the test set-up:
    - `fix162` compares Setup word for word with build 161, and Quick Setup now offers 11 and 12. The comparison leaves those two out; 21/21.
    - `fix161` compares with build 160's page on port 8817, where I had put build 163 for my own comparisons. Stopped; 15/15.
    - `fix130a` and `fix163d`: "Target crashed". The sandbox's disk was full (268 MB left), and build 163's page crashed the same way. After deleting old local build copies and audit screenshots (13 GB free): 41/41 and 6/6.
- **Full run: 213 suites**, on the final page (9be33f7f…), the laptop app carrying the build-164 tablet and TV apps:

- browser: 135 suites, 3,930/3,931;
- laptop app (Electron 44.9.3): 78 suites, 1,389/1,389.
- The one miss was **`fix158b`**: "with 8 teams only 2 bots fit (10 teams at most)". With 12 teams allowed, 3 bots fit beside 8 teams. The test now follows the limit ("with 10 teams only 2 bots fit"); 19/19 alone.
- **Passed on retry: `fix129t` and `fix138d`.** Both run a Galaxy Tab server and ran at the same moment, as `fix132n` / `fix132t` did in build 163. Also `fix139d`, the slow-TV test, which depends on how fast this sandbox makes TV pictures (0.23 s this run).
- The new suite **`fix164` 20/20**:
  - 11- and 12-team saves;
  - the TV's board, question and Blind It! wager, and every end-of-night screen, at 5 sizes;
  - the recap picture;
  - the portrait fixes.
- Also `fix153c` 34/34 (limits at 12), `fix163d` 6/6, and `fix142d` 17/17 / `fix143d` 13/13 with the build-164 apps inside the laptop app.

- **Reproducible:**
  - The Galaxy Tab APK (c1d8e124…) and the Mind It! TV APK (480128e2…) are versionCode 164 and signed with the original key. The tablet APK carries the tested page.
  - The Windows EXE (97829f5a…, Electron 44.9.3) carries the same page, `main.js` and both build-164 apps. Its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, TV or Mi Box. In particular:

- **a real 12-team night:** every screen was checked with made-up teams here, on simulated devices.
- the Windows EXE on Windows (built and checked here; its joiner tried under Wine).
- No stress phases or soak for this build (it changes the limit and layouts, not the game logic or the connections; the servers already took 12).

Build 163 addendum (30 October 2026)

What you asked for (30 Oct, after build 162 went out): "any upgrades?". Then "1": the TV picture without text for a moment about 2 s after a question opens (seen by `fix161o` in build 162's tests, also with build 161's page). Then "release 163".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 163, and `APP_BUILD=163`.

- New module `app163.js` (`build.py` order `…,162,163,6`). It changes the TV page only. No server change, no change to the tablet's home screen or the TV app's own code.
- Code from commit cc20c65.
- Not a 5th build, so **no Mac app**: the Mac stays on build 160 until build 165.
- The guides are build 160's. Nothing they describe changed.

A · What was wrong

- **Measured on the laptop app**, with a TV in today's Chromium and one in Chromium 105. Each TV recorded itself every 25 ms, inside the TV page: sampling from outside missed it, because the laptop page is busy at that moment.
- **The Audience View on the laptop never lost the question.** The TVs did, every time, on both engines:
  - the question: gone at about 2.25–2.35 s after it opened, back 0.3–0.5 s later (also with the question timer stopped: the laptop still sends a new picture about every 0.5 s);
  - after ✓ Correct: the TV empty for about 1.7 s;
  - after going back to the board: the whole board gone for about 1.5 s;
  - a 25–75 ms blip as the question opened.
- **Cause.** The TV merges each new picture into its page by position (build 130), and two plain boxes count as the same element. When a question opens, the TV plays the theme's moment itself (build 150): one layer in front of the question and one after it, each removed by a timer when the moment ends. When the laptop removed its own copies of those layers, the next picture was merged by position: the TV's layer in front became the question, and the question's own element was dropped. The layer's timer then removed what was now the question, until the next picture put it back.
- In place since build 150 (the theme moments on the TV).

B · The fix

- **T1:** when the TV merges a picture, a moment's layer is only ever matched with a moment's layer. A layer the laptop no longer has at that place is removed; it never becomes part of the page.
- **T2:** the timer that ends a moment removes its element only while that element is still a layer.
- Presentation on the TV only: no timer, score, turn, save, sync or phone change.
- A TV picks the fix up the next time it loads the quiz page (↻ Refresh All Screens, or restarting the TV app).

C · Tests

- **Full run: 212 suites**, on the final page (9ed07f18…), the laptop app carrying the build-163 tablet and TV apps:
  - browser: 134 suites, 3,935/3,935, all passed. `fix132n` passed on retry (below).
  - laptop app (Electron 44.9.3): 78 suites; the 76 with a result passed, 1,310/1,310. `fix115` and `fix132t` had no result (below).
  - The new suites: `fix163` 8/8, `fix163d` 6/6. `fix142d` 17/17 and `fix143d` 13/13 with the build-163 apps inside the laptop app. `fix139d` 23/23 (a new TV picture every 0.20 s here, so the slow TV had pictures to skip).
- **What needed a second look — all the test set-up, not the app:**
  - **`fix115` no result:** it opens the page at this computer's network address (port 8800). The sandbox had restarted before the run and I started that page server for this computer only (127.0.0.1). Started again for every address: alone 22/22. Restarting it took the 8800 page away for about a minute during the run; no other suite was flagged.
  - **`fix132n` (retry) and `fix132t` (no result):** both run a Galaxy Tab server on this computer, and they ran at the same moment (the browser and laptop-app halves of the run go side by side). On its first try `fix132n`'s iPad remote dropped 3 times; `fix132t` timed out waiting for "Approve Transfer". Alone: `fix132t` 57/57, `fix132n` 7/7 three times out of three. Three extra `fix132n` runs while the laptop-app half was still going: 1 passed, 2 couldn't connect at all. I didn't pin down exactly how the two servers got in each other's way (each takes the next free port). `fix132n` had passed first time in the 15 earlier full runs I have. Nothing in the Galaxy Tab's server or the connection code changed in 163.

- **Before the run** (on the same code): the 28 suites that touch the TV page, motion and themes: 717/717. `fix163d` on build 162's page: 1/6 (the blanks in A), on 163: 6/6. `fix163` shows the loss on build 162's page.
- `fix146d` **:** in that earlier run it passed only on retry: the square it opened was one of the game's randomly placed wager squares, so "pauses on the question" found the wager screen. It now clears the round's wager squares when its game starts: 28/28 twice, and first time in the full run.
- **Reproducible:**
  - The Galaxy Tab APK (9c70dc9d…) and the Mind It! TV APK (290a91da…) are versionCode 163 and signed with the original key. The tablet APK carries the tested page.
  - The Windows EXE (831b6e77…, Electron 44.9.3) carries the same page, `main.js` and both build-163 apps. Its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, TV or Mi Box. In particular:

- **a real TV or Mi Box.** The TVs here were Chromium windows (today's and version 105, the Mind It! TV app's minimum Android web engine), on the same computer as the laptop app.
- the Windows EXE on Windows (built and checked here; its joiner tried under Wine).
- No stress phases or soak for this build (only for a major change; build 163 changes how the TV draws, not the game or the connections).

Build 162 addendum (30 October 2026)

What you asked for (30 Oct, after build 161 went out): "any upgrades?". Then "1,2,5":

- 1: Setup on the Galaxy Tab is slow: draw only the section you're looking at.
- 2: All Questions takes 2.8 s: show the first 100 questions at once, the rest as you scroll.
- 5: clean full test runs without retries.

Then "release 162, have a code and layout review too before releasing it".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 162, and `APP_BUILD=162`.

- New module `app162.js` (`build.py` order `…,161,162,6`) and one style rule. No server change, no change to the tablet's home screen.
- Code from commit e49ae92. Later commits change only tests.
- Not a 5th build, so **no Mac app**: the Mac stays on build 160 until build 165.
- The guides are build 160's. Nothing they describe changed.

A · What changed

- **Measuring first.** A Setup redraw on the tablet was 0.6–1.3 s depending on the tab (your library, 1280 × 800 touch screen at 1.5× pixels, processor slowed 4×). Build 161's "about 1.5 s" was measured with the test pages' switch that shows all seven tabs at once, which a real tablet never does.
- **Not built literally.** Setup builds all seven tabs, then about 40 later add-ons rearrange them by finding labels across all of them. Building only the visible tab would mean rewriting Setup itself. Instead the work for the hidden tabs is cut:
  - **L1 · line icons once per element.** One redraw handed the same 584 buttons and headings to the icon pass 3,091 times.
  - **L2 · the question editor** is drawn only on Rounds & Questions (and whenever you jump to a question to fix).
  - **L3 · the phone-remote QR code** is worked out once per address.
  - **L4 · Player Management** is drawn only on Teams.
- **Setup at 4× slower, per tab, build 161 → 162:** Quick Setup 734 → 361 ms · Game 632 → 372 · Teams 950 → 699 · Gameplay 927 → 416 · Rounds 1,098 → 753 · Audio 1,296 → 839 · Advanced 667 → 363. All seven: 6.3 s → 3.8 s.
- **Q1 · the Question Database.** All 400 rows are still made, with everything later builds add to them. Rows 101 and on are hidden until you scroll within about 2,000 px of the end; then the next 50 are shown (about 0.35 s on the slowed tablet). The same for a large category, Recently Added and search results. All Questions at 4×: about 4.8 s → about 1.7 s.

B · Code and layout review (before release)

- **Code review** (a separate reviewer, reading only): no serious problems. Three small points, fixed:
  - each search left the old list's scroll watcher behind; a list that has gone is no longer watched;
  - with the Tab key, questions 101 and on couldn't be reached; reaching one of the last 5 rows shown now shows the next 50;
  - if Setup ever fails to sort its cards into tabs (it then shows one long page), the question editor and Player Management are drawn in full there.
  - Not changed: the browser's own Find (Ctrl+F) only finds questions already shown; the Question Database's search box searches them all.
- **Layout review:** the same made-up library (5,000 questions, 50 players, 100 games) and the same quiz in build 161 and build 162, screenshotted at four sizes (laptop 1440 × 900, tablet landscape 1280 × 800, tablet portrait 800 × 1280, phone 390 × 844). Each of the seven Setup tabs, All Questions (opening, around row 100, scrolled to the end), a search, the category list, the game board and a question: **pixel-identical**, apart from the "Build 162" line and the Christmas theme's random snow.
  - One apparent difference (the turn-order badges in Quick Setup's TV preview, portrait tablet) came from capturing a very tall page in one shot: it appeared in either build at random, and with the preview on screen both builds show it every time.

C · Tests

- **Full run: 210 suites**, on the final page (6fa4f77a…), **all passed**:
  - browser: 133 suites, 3,930/3,930;
  - laptop app (Electron 44.9.3, carrying the build-162 tablet and TV apps): 77 suites, 1,383/1,383.
  - `fix142d` 17/17 and `fix143d` 13/13: the laptop app carries the build-162 Galaxy Tab and Mind It! TV apps.
  - `fix139d` **23/23**, passed for the first time since build 157. This run the laptop made a new TV picture every 0.21 s, faster than the slow TV's 0.4 s, so the TV had pictures to skip: it skipped 22 of 47 and ended on the newest. In builds 157–161 the laptop made one only every ≈0.34 s here, so there was nothing to skip. Nothing in the app changed; the sandbox was quicker this time.
- **Two suites passed only on retry. Both were the test, not the app:**
  - `fix162` (first try 20/21). The check that Setup is word for word build 161's differed on two things that change with timing: the Mind It! logo numbers its colour gradient each time it is drawn (`mhx65` against `mhx66`), and the pre-show checklist is replaced every 3 s (since build 141), its line icons going in on the next frame. The capture caught "⬇ Back Up Now" between the two.
    - Checked: 56 Setup tab draws in each build with the processor slowed 4×, looking for a button left without its icon 0.9 s after a draw and again 3 s later: **none in either build**.

    - Fixed in the test: the icons the next frame would add are put in before the capture, and the logo's counter is left out, the same for both builds. L1's own check (every button has its icon after a redraw) is unchanged.
  - `fix146t` (first try 12/13). The Christmas-theme change on a tablet mid-question: the theme was right, and the message didn't say which other part failed. Once it did (under load, 2 runs of 3), the tablet had gone back to the **wager screen**, with the game unchanged. The cell the test opens is sometimes one of the game's wager cells (they are placed at random), and going back to the wager screen is right. The test expected the question screen every time. It now expects the screen it left, and each step waits for its own result instead of a fixed 0.3–0.8 s.
  - `fix162` **again, under load (1 run of 6):** L1's own check found "⬇ Back Up Now" without its icon two frames after a redraw: the same 3-s checklist replacement. The check now looks again for up to 10 frames, and only an icon still missing then counts.
  - After these fixes: alone, `fix162` 21/21 and `fix146t` 13/13, 3 times out of 3. Under load (three `fix162` and one `fix146t` at once), 3 rounds out of 3: every copy passed. Once more with the opened cell forced to be a wager cell: the tablet left from the wager screen and came back to it, game unchanged, 13/13.
- **Test fixes made before this run** (after the first full run, on an earlier copy of the page):
  - **Phone suites** `fix155d`, `fix151d` **and** `fix114` sometimes had no result under load. The test made new teams and opened the phones at once. The phone's first check could reach the laptop's server before the laptop page had sent it the new teams, so the phone rightly said "Your team code has changed". The tests now wait until the server lists the new teams. Builds 161 and 162 send new teams equally fast. In this run: 16/16, 14/14, 26/26.
  - `fix145` : a fixed 0.15 s pause before checking a 0.43 s animation; it now waits in the page until the card is 0.1 s in. 29/29.
  - `fix161o` (old-engine TV, laptop app) failed about 4 runs in 10. Both TVs had no text at the moment of measuring. That moment is in build 161's page too: for about 2 s after a question opens, the laptop's TV picture carries no text. The test now waits for the text, then compares. 6 of 6 alone, 9/9 in this run.
    - **Not looked into:** whether those 2 s without text can be seen on a real TV. It is not new in 162.
- **Related suites before the run** (fix162, the Setup and Question Database suites, the layout suites): 392/392.
- **Reproducible:**
  - The Galaxy Tab APK (3bcbf62f…) and the Mind It! TV APK (0bc31707…) are versionCode 162 and signed with the original key. The tablet APK carries the tested page.
  - The Windows EXE (902a52b5…, Electron 44.9.3) carries the same page, `main.js` and both build-162 apps. Its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or Mi Box. In particular:

- **the Galaxy Tab's speed on the tablet itself.** Measured with the processor slowed 4×, which is an estimate of a tablet, not the tablet.
- the Windows EXE on Windows (built and checked here; its joiner tried under Wine).
- No stress phases or soak for this build (only for a major change; build 162 changes how Setup and the Question Database draw, not the game or the connections).

Build 161 addendum (30 October 2026)

What you asked for (29 Oct, after build 160 went out): "any upgrades?". Then "1,2,7":

- 1: check older devices (the Galaxy Tab's and the Mi Box's web engine).
- 2: the Galaxy Tab's speed with a library the size of yours.
- 7: a soak that really plays for 60 minutes.

Then "release 161".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 161, and `APP_BUILD=161`.

- New module `app161.js` (`build.py` order `…,160,161,6`); the tablet's home screen (`android/assets/start.html`) gets the warning card. No server change.
- Code from commit eb1465d. Later commits change only tests and scripts.
- Not a 5th build, so **no Mac app**: the Mac stays on build 160 until build 165.
- The guides are build 160's. Nothing they describe changed.

A · What changed

- **E1 · a too-old web engine says so.**
  - The Galaxy Tab and the Mi Box show Mind It! in Android's own web engine (Android System WebView). On a TV box it can be years old.
  - I listed the features the page uses and tried the page in two real old Chromium builds (downloaded Chromium snapshots, run here):
    - **Chromium 105** (Mind It! TV's minimum, `MIN_CHROME=105`): the board, a question, the Blind It! wager and the standings come out at the same text sizes as today's Chrome (54 of 54). No script errors. The only differences are cosmetic: long questions wrap line by line instead of in balanced lines (needs 114), and the Comic theme's Blind It! title draws its outline over the letters (needs 123).
    - **Chromium 100:** no script errors and the host controls work, but the TV picture is the wrong size on all 4 screens tried. The page is laid out in container units (`cqw` / `cqh` , 2,500+ uses) and `:has()` , which arrived in 105. Rewriting that for older engines would mean a second layout.
  - So an engine without those two features now shows a red banner, on the TV page, phones, the laptop page and the tablet's home screen. It names the device ("this TV", "this phone", "this tablet"), its engine version and what to do. On Android that is: update "Android System WebView" (and "Google Chrome") in the Play Store; on an iPhone or iPad, Software Update. "Continue anyway" hides it for that session.
  - The check asks the engine whether it supports the features. It doesn't read the version number, so old iPhones are caught too.
- **G1 · the Galaxy Tab's speed.**
  - Method: your library's size (1,264 questions), a 1280 × 800 touch screen at 1.5× pixels, and the processor slowed 4× (Chrome's "mid-tier mobile", about a Galaxy Tab against the build machine). Each action was timed on builds 160 and 161:

Action (tablet, 4× slower)	Build 160	Build 161
Start the game	1.38 s	0.84 s
Mark an answer right	0.95 s	0.77 s
Mark an answer wrong	0.77 s	0.62 s
Open a question	0.69 s	0.56 s
Question Database · All Questions	3.3 s	2.8 s

Action (tablet, 4× slower)	Build 160	Build 161
Setup redraw	≈1.5 s	≈1.5 s

  - The biggest single cost was asking the window for its size in the middle of a redraw. That makes the browser lay out the whole page there and then, and again when the redraw finishes. The window's size only changes on a resize (turning the tablet, the keyboard), so build 161 remembers it and updates it then. This applies to the tablet layout switch (build 155), the Quizmaster top bar's "Remote View" line, and "is this a phone-sized remote?".
  - The Question Database worked out each question's matching key for every square of the quiz, for each of up to 400 rows: about 60,000 times per draw of All Questions. The key of a text never changes, so it is now remembered.
  - Setup didn't get quicker: its cost is the size of the page. I tried letting the browser skip off-screen Setup panels (`content-visibility`): 40 % faster, but taps then jumped the page by up to 8,000 px. Not used.
  - Using the tablet as the laptop's remote was already quick: about 0.3 s per tap.
- **Soak · the full hour.** `stress110.py soak 60` stopped starting new questions 10 minutes before the end, so it played about 55 minutes. It now plays to the last minute, and reports whether the real-time quiz ran the whole time.

B · Found while building

- **The tablet APK copy into the laptop app** (so the tablet can update from the laptop over the Wi-Fi) was a manual step in earlier builds. It is now part of the EXE script, which checks the EXE carries it.
- **The soak log contains the test's seeded host name.** It is redacted in the committed result.
- I ran the EXE script twice by mistake (a listing command started it again). The second run was replaced by a third clean run. The released EXE, its parts and the joiner test are all from that third run. Nothing had been published.

C · Tests

- **Full run: 209 suites**, on the final page (3384678c…):
  - browser: 132 suites, 3,908/3,909;
  - laptop app (Electron 44.9.3): 77 suites, 1,380/1,383.
  - Retries: `fix140s`, `v169` and `fix155d` passed on retry.
- `fix145b` **34/35 in the full run, 35/35 three times out of three alone.** The check missed was "a redraw mid-way keeps both going". It reads the card's shake animation 200 ms after a wrong answer; under the full run's load the shake had already ended. I record it as timing under load, not as a clean pass.
- `fix142d` **17/17 and** `fix143d` **13/13** once the laptop app carried the build-161 tablet app (the EXE step).
- `fix139d` **22/23 — not passed**, as in builds 157–160. In this sandbox the laptop makes a new picture only every ≈0.34 s, so the slow TV keeps up and has nothing to skip.
- **New suite** `fix161` **15/15** (re-run after the stress test with build 160 served alongside, so both comparisons ran in full):
  - **E1:** today's Chrome shows no warning. With the two features removed (simulated), the host page, a phone, the TV page and the tablet's home screen each show the warning, worded for that device. "Continue anyway" hides it, also after a reload.
  - **Real Chromium 105:** board, question, Blind It! wager and standings at the same text sizes as today's Chrome; no warning, no errors.
  - **Real Chromium 100:** 4 of 4 screens off size; the page and the tablet's home screen both show the warning; no script errors.
  - **G1:** the tablet layout and the phone-sized remote give the same answers as build 160 at every size, also after turning the tablet. The top bar and its ⋯ menu are word for word build 160's. A game redraw asks the window's size 5 times (build 160: 12). The Question Database's quiz places are the same for all 40 questions tried.
- **New suite** `fix161o` **9/9 (laptop app):** the laptop app hosts; the TV page, the tablet remote and a phone all run in real Chromium 105. The TV draws the board at today's text sizes (54 of 54), the tablet's ✓ CORRECT scores, the phone's Blind It! wager of 750 reaches the laptop. No warnings, no script errors.
- **Stress:** all six phases 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6), each on a fresh laptop app, with nothing else running.
- **The soak (** `stress110.py soak 60` **) 15/15:** the real-time quiz ran **60.2 minutes**, the whole hour (build 160: 55).
  - 3 whole games with 8 teams, 71 questions, Final wagers from the phones every time. One sudden-death tiebreaker was played from the tablet and recorded in Historic Games.
  - 301 tablet actions, 85 samples of TV + tablet + 10 phones, window hidden and shown 5 times.
  - 0 false disconnects, no script errors. Laptop server memory went 200 → 212 MB, the Quizmaster page's JS heap 17 → 20 MB.
- **Reproducible:**
  - The Galaxy Tab APK (4b59a15d…) and the Mind It! TV APK (9a5107a7…) are versionCode 161 and signed with the original key. The tablet APK carries the tested page and the new home-screen card.
  - The Windows EXE (4e415b9c…, Electron 44.9.3) carries the same page, `main.js` and both build-161 APKs. Its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or Mi Box. In particular:

- **the old-engine warning on a real old device.** The old engines were real Chromium builds (105 and 100) on Linux, not an old Android WebView on a TV box. Your Mi Box's and Galaxy Tab's engine versions are not known here.
- **the Galaxy Tab's speed on the tablet itself.** Measured with the processor slowed 4×, which is an estimate of a tablet, not the tablet.
- the Windows EXE on Windows (built and checked here; its joiner tried under Wine).

Build 160 addendum (29 October 2026)

What you asked for (29 Oct, after build 159 went out): "any upgrades?". Then "4,5":

- 4: test with big libraries and fix the slow paths.
- 5: a safety net for the theme colours on older Android WebViews.

Then "release 160, update all the guides, do the stress tests".

Same version: v1.69. Galaxy Tab and Mind It! TV are versionCode 160, and `APP_BUILD=160`.

- New module `app160.js` (`build.py` order `…,159,160,6`); also `style.css`. No server change.
- Code from commit e40f2cb. Later commits change only tests, guides and scripts.
- Every 5th build, so the **Mac app** is included this time.

A · What changed

- **S1–S3 · big libraries stay fast.** I built a made-up library: 5,000 questions, 50 players and 100 past games of 120 questions each. Then I timed every main screen and action on builds 159 and 160:

Action	Build 159	Build 160
All Questions	0.58 s	0.15 s
Mastermind It! tab	0.54 s	0.08 s
Setup redraw	0.49 s	0.14 s
Setup → Rounds	0.72 s	0.20 s
Setup → Teams	0.71 s	0.16 s

  Everything else was already under 0.1 s. What's left is the browser's own layout work.

  - **S1:** the 👁 Preview button and the 📎 media mark searched the whole library for each row (5,000 × 5,000). Now there is one lookup table per draw.
  - **S2:** the duplicate check runs once the app is idle, 4 s after start-up, so the first Question Database open doesn't pay for it.
  - **S3:** the scenery clock (build 148) and Setup's live TV preview now wait for the next frame. They no longer force two extra full style and layout passes on every redraw.

- **T1 · the theme colours without colour mixing.** Build 159 mixes theme colours with CSS `color-mix()`. A web view without it drops those rules, and some host-control backgrounds then came out **see-through**.
  - **Correction:** the build-159 addendum said older WebViews would "keep the navy". That was wrong.
  - Build 160 adds plain theme colours for browsers without `color-mix()` (`@supports not`). Browsers that have it are unchanged.

- **B1 · a bot's answer on the TV keeps out of the way.**
  - Build 158 put the banner a fixed 27 % up the picture. With 7 or more teams that covered a two-line question. This was found while taking the guide pictures.
  - The first build-160 fix moved the banner to the card's bottom edge. The new guide picture then showed it covering the ⚡ Speed Bonus countdown. With 2–6 teams it also touched the top of the score cards, by 17 px.
  - The banner now takes the first free place, trying these in order:
    1. the card's bottom edge;
    2. beside the chip;
    3. lifted just above the chip or the scores;
    4. the card's top edge.
  - Free means inside the card's width and clear of the question, the chip, the score cards and "Now playing".

- **Guides — all five PDFs updated for build 160.**
  - **Quick Start:**
    - a new section on practising with bots, and the Question Database shortcuts;
    - ⟳ Sync devices, the back button and Close, and 🏁 End game now;
    - theme colours on the host screens, restoring a backup on the tablet, and the 4 EXE parts.
  - **The Complete Idiot's Guide (14 → 16 pages):**
    - "New in builds 156–160", one-tap add, and practising alone with bots;
    - the laptop's Season tab, the back button;
    - six new build-160 pictures with made-up names.
  - **Cheat sheet, rehearsal checklist and one-pager:** new NEW tags. All fit their pages (checked with `pages2pdf`).
  - **Players' guides:** unchanged, because nothing players see changed.
  - **Names check:** none of the 110 names in your backup appear in the guide sources or the PDF text.
  - **Wording fix:** the back button is labelled with the screen it returns to (for example "← Players & History"), so the guides describe it that way.

B · Found while building

- **fix159 "Sparky passes in character" failed now and then.** After the host marks Quanta wrong, the turn moves on to Sparky by itself. With the test's fast bots, Sparky sometimes passed before the test handed it the turn, and the test then cleared the line already said. This was confirmed from the bots' own speech log. The test now keeps what was said; the app is unchanged. It then passed 4 times out of 4.
- **The EXE script's header** still said "no Mac app" (copied from build 159). Corrected.

C · Tests

- **Full run: 207 suites** — browser 131 suites 3,890/3,892; laptop app (Electron 44.9.3) 76 suites 1,341/1,344.
  - It ran on the page before B1's final placement (the B1 code is the only difference).
  - Retries: `fix159` , `fix151` and `qc2` passed on retry.
- **Re-run alone on the final page (5fc6fd15…):** 194/194.
  - Suites: `fix160` 15, `fix159` 25, `fix158` 26, `fix158b` 19, `fix154r` 60, `v169` 12, `fix158d` 7, `fix155d` 16, `fix151d` 14.
  - In the full run, `fix154r` missed 1 check (a picture link on the TV), `v169` missed 1 (Present On This Screen, 0.2 s allowed), and `fix155d` and `fix151d` finished without a result (phones connecting under load).
  - Alone, all passed. `fix151d` needed a retry, so I record these as timing under load, not as clean passes.
- **`fix142d` 17/17 and `fix143d` 13/13** once the laptop app carried the build-160 tablet app (the EXE step).
- **`fix139d` 22/23 — not passed**, as in builds 157–159. In this sandbox the laptop makes a new picture only every ≈0.37 s, so the slow TV keeps up and has nothing to skip. It still ends on the newest picture.
- **New suite `fix160` 15** (fails on 159), on a made-up library of 5,000 questions, 50 players and 100 games (6.8 MB):
  - **S1:** drawing All Questions (400 rows) asks for the library 414 times, against about 3 times per row in build 159. Every row still gets its 👁 Preview, picture questions their 📎, and Preview opens. Timings: All Questions 0.17 s and the Mastermind It! tab 0.08 s (build 159: 0.54–0.65 s).

- **S3:** a Setup redraw no longer asks for the scenery's animations on the spot, and the preview is drawn by the next frame. Redraw 0.18 s, Rounds 0.15 s, Teams 0.17 s (build 159: 0.49, 0.72 and 0.71 s).
- **S2:** a few seconds after start-up the duplicates are already worked out.
- **B1:** at 4 and 7 teams, with and without the Speed Bonus chip, the banner is never over the question, the chip, Now Playing or a score, and stays inside the picture.
- **T1:** with `color-mix()` removed (simulated), the tablet remote's cards are see-through without the safety net and keep the Christmas colours with it.

- **Stress:** all six phases 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6), each on a fresh laptop app. **The soak (`stress110.py soak 60`) 14/14:** 55 minutes of real-time quiz, 56 minutes by the clock, so not quite a full hour. It played 3 whole games (65 questions, one sudden-death tiebreaker played from the tablet) with 290 tablet actions and 10 phones, and hid and showed the window 5 times. 0 false disconnects and no script errors. Laptop server memory went 195 → 212 MB, the Quizmaster page's JS heap 17 → 20 MB. One browser-only test (`fix160` , about a minute) ran during the idle phase, which passed anyway.
- **Reproducible:**
  - A fresh clone at e40f2cb builds the tested page.
  - The Galaxy Tab APK (741da351…) and the Mind It! TV APK (1b8abdbf…) are versionCode 160 and signed with the original key.
  - The Windows EXE (ff4f4817…, Electron 44.9.3) carries the same page, `main.js` and both build-160 APKs. Its 4 parts join to the same file (Wine joiner test 4/4).
  - The Mac app (Apple Silicon, ad-hoc signed, zip adef3914…) carries the same page, `main.js` and the tablet app. Its 6 parts join to the same zip.

D · Not tried for real

Nothing was run on your laptop, Mac, Galaxy Tab, phones or Mi Box. In particular:

- the speed with a big library on your own laptop (measured here with a made-up 5,000-question library);
- the theme colours in the Galaxy Tab's own WebView — the old-WebView case was simulated in Chromium, not tried on an old Android;
- the bot banner's position on a real TV;
- the Mac app on a real Mac (built and signed here, never opened on macOS).

Build 159 addendum (29 October 2026)

What you asked for (29 Oct, after build 158 went out): "any improvements r features?" — then "2,5" (2: bots you mark yourself; 5: bot personalities) and "In the main menu, the questions database isnt working, and the app is responding slowly" with a screenshot of the main menu (a scrollbar; the dotted line through the bottom row); "when you choose a theme, can the colours also be the same for the host control"; then "release 159".

Same version: v1.69 · versionCode 159 (Galaxy Tab and Mind It! TV) · `APP_BUILD=159` . New module `app159.js` (`build.py` order `…,158,159,6`); also `style.css` . No server change. Files from commit 3a79ce7. No Mac app (every 5th build: 160).

A · What changed

- **P1 · the Question Database froze.** Reproduced with the copy of your backup kept in the sandbox (1,264 questions): opening it from the main menu took 11.7 s, opening a category 8.9 s, going back 8.9 s — the same on build 157 (15 s), so it grew with the library rather than coming from 158. A CPU profile showed the time in the duplicate finder (`libDupGroups` , app34): it compared every question with every other (≈800,000 pairs) and three parts of the database ask for it on every draw (the 🔁 Duplicates tab, More ▾, its 🔁 badge). Same rule, same groups, same order now — but only questions that share a word (or the same key) are compared, and the result is kept until a question, an answer or a "Not Duplicates" choice changes. On your data: the same 3 groups; open 11.7 s → 0.05 s, a category 8.9 s → 0.05 s. Nothing else was slow with your data (Players & History, Leaderboard, Season, saving: under 30 ms; a Setup redraw ≈0.3 s, the browser's own layout work, unchanged).
- **P2 · the main menu fits the laptop window.** With ⟳ Sync devices, its status line and "⬆ Up to Date", the menu was ≈25 px taller than a 1920×1080 screen's window (reproduced at 1903×1012 with your data and the title-case setting) and the bottom row sat on the dotted line. It now tightens padding, gaps and then the logo size only when it doesn't fit; 1366×768, 1536×864 and 1920×1080 without those extras are unchanged.
- **Q1 · bots you mark.** Start A Rehearsal? asks "Who marks the bots?" — **I mark them** (default, remembered on the device): the bot says its answer out loud and on the TV and waits for Correct / Wrong Answer (host bar: "👆 Mark it: Correct or Wrong Answer"); its Final answer waits for your verdict. **Bots mark themselves** keeps build 158's behaviour. Passing stays the bot's choice.
- **Q2 · bot personalities:** Sparky (Rookie), Byte (Pro), Quanta (Genius) — they announce the clue they pick, say the answer in character, pass in character, react when marked and announce a Daily Double wager (the device's built-in voice, at the game's volume, silent when the sound is off).
- **R1 · the host controls in the theme's colours.** With any theme other than Default: the Galaxy Tab remote (page and cards), the main menu, Prepare tonight's quiz, Setup's tabs, number buttons, text boxes and go-live bar, Question Database tiles, switches, pop-up notes and team avatars take the theme's colours (they had the default navy written in). Gold highlights stay; Default is unchanged (style.css only).

B · Found while building

- My first build-157 timing of the freeze said "instant" — the measurement ran before the library had loaded; the CPU profile showed 157 just as slow. Recorded so the cause isn't mistaken for a 158 change.
- The rehearsal dialog's "Marking" row first used the bot rows' class, which the build-158 bot test counts; it has its own class now.

C · Tests (page and APKs from commit 3a79ce7)

- **Full run: 206 suites** — browser 130 suites 3,872/3,873; laptop app (Electron 44) 76 suites 1,371/1,374.
- **Re-run alone on the same files:** `v169` (Present On This Screen from the menu, 0.2 s allowed) 12/12 twice — it also needed a retry on 157; `fix142d` 17/17 and `fix143d` 13/13 once the laptop app carried the build-159 tablet app (the EXE step). Recorded as timing under load / build order, not as clean passes.
- **`fix139d` 22/23 — not passed**, as in 158 and on the build-157 page: in this sandbox the laptop makes a new picture only every ≈0.4 s, so the slow TV keeps up and has nothing to skip; it still ends on the newest picture.
- **New suite `fix159` 25** (fails on 158): the duplicate finder gives the same groups as build 158's on 1,412 made-up questions and is ≥3× faster, the database opens from the menu in well under a second, "Not Duplicates" and edits update at once; the menu fits 1903×1012 with Sync and Up to Date and its bottom row ends above the dots; Christmas and Halloween leave no default navy on the main menu, Prepare, Setup, the board, a question, the Question Database, Players & History or the tablet remote, and Default is unchanged; "Who marks the bots?", the bots' names and lines, a bot waiting to be marked, the host marking it right and wrong, passing, the Daily Double line, the Final waiting for the verdict. `fix158b` / `fix158d` choose self-marking bots and pass (19, 7).
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6). **Picture audit** (`fix153c`) passed in the full run.
- **Reproducible:** a fresh clone at 3a79ce7 builds the tested page; the Galaxy Tab APK and the Mind It! TV APK, versionCode 159, are signed with the original key; the Windows EXE (Electron 44.9.3) carries the same page, `main.js` and both build-159 APKs; its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or Mi Box. In particular: the Question Database speed on your laptop (measured here with your backup's 1,264 questions); the theme colours in the Galaxy Tab's own web view (it needs a recent Android System WebView for the colour mixing — older ones keep the navy); the bots' voices and personalities through real speakers; the menu fit on your exact window.

Build 158 addendum (29 October 2026)

What you asked for (29 Oct, after build 157 went out): a screenshot of the Ready to Play card — "make the design and layout uniform please"; "on player menu theres no back button, i need to constantly open player menu so work on that, in fact go through the whole settings journey and optimize it, like even adding or swapping questions gets cumbersome when i have to constantly choose the round"; "go through all the menus and optimize it. make all the words in a tile in the middle too"; a screenshot of the Season tab — "this page needs to look better, also uppercase Dec"; a screenshot of the logo — "the background is blending into the logo and name"; "if i upload a template for a round and there are blank categories, then the remaining categories dont get replaced"; "for rehearsal mode, can we add bots ... multiple human players and maximum 3 bots of varying intelligence"; "the thing is player will be saying their answer out loud so how do we make the bots give me their answer"; then "release 158".

Same version: v1.69 · versionCode 158 (Galaxy Tab and Mind It! TV) · `APP_BUILD=158`. New module `app158.js` (`build.py` order `…,157,158,6`); the Ready to Play card (M5) is in `app157.js`. Also changed: `style.css`, `android/assets/start.html` (the tablet home's tiles and links). No server change (`desktop/app/main.js` and the Galaxy Tab server as in 157). Files from commit f5c5139. No Mac app (every 5th build: 160).

A · What changed

- **N1 · ← Back and Close.** Players & History, a player, a Historic Game, editing a player, the Question Database and Backups remember where you came from: "← Players & History" (and so on) goes back one step, the tab you were on included. Close clears the trail and returns to where you started — the main menu, Setup, or (opened from the tablet home) the tablet home, labelled "Close · Home".
- **N2 · words in the middle of every tile:** main menu, Prepare tonight's quiz, the tablet home, Question Database categories. The menu still fits 1366×768.
- **N3 · the Question Database remembers the round** you last added to (or the one Setup opened it for), on this device: "➕ Add to Round N · its name" in one tap; "Other round…" opens the list with that round ticked. Nothing is guessed before the first choice.
- **N6 · ⇄ Move to another round** lists the other rounds as buttons (name and empty slots; a full round greyed out "Full — no empty slot") instead of a box asking for the number.
- **N7 · Replace Round from Database with fewer categories.** `fillRoundFrom` only touched the columns the chosen template round had, so with 4 categories in the template, categories 5–6 kept tonight's old questions — though the question said "All 30 clues … will be replaced". The columns past the template are now emptied (name and questions; values and Daily Double spots kept), the card says "Fills 4 of 6 categories · the other 2 are emptied", Review & edit clues lists them to fill, and ↩ Undo still restores the whole old round.
- **N4 · the Season tab** redesigned: season pills (only with more than one season), a header with the season's dates in their own capitals ("Oct–Dec"), the finished season's Champion and Grand Rascala, "How points work", a podium for the top three and one list (rank, picture, name, nights, wins, points).
- **N5 · the logo:** a soft dark glow behind the wordmark and tagline; the tagline keeps its own capitals ("Enna Rascala….Mind It!" — the title-case setting had made it "…mind It!").
- **M5 · Ready to Play:** every tile one height, one radius, one line-icon style, centred words.
- **N8 · 🤖 Rehearsal bots.** Start A Rehearsal? offers 3 bot slots — Off / Rookie / Pro / Genius (fewer when the 10-team limit would be passed). Bots join after the safety copy is taken, so End Rehearsal brings back exactly the teams you had and nothing is recorded. A bot acts only through the host's own actions: on its turn on the board it picks a clue (Genius the biggest value, Pro top-down in a category, Rookie at random); on a question it thinks (Rookie 6–10 s, Pro 4–7 s, Genius 2–4 s, always before the clock runs out), **says its answer out loud with the device's built-in voice** (at the game's volume, silent when the sound is off; never a copy of a real person's voice), shows it on the TV and in the host bar, and is marked ✓ / × through the normal Correct / Wrong Answer (Undo works) — or passes when it doesn't know and another team can still answer. How often it knows: Rookie ≈ 35 %, Pro ≈ 60 %, Genius ≈ 85 %, easier on the top row and harder on the bottom row, and for questions marked Easy / Hard. A wrong answer is another answer from the same round. Bots wager on their Daily Doubles and in Mastermind It! (Genius bets to lock or cover second place), write a Final answer that appears where phones' typed answers do and is marked by the answer check. In Phone Buzzer Mode a bot buzzes through the laptop's buzzer server like a phone, so the first valid buzz still wins; a bot that doesn't know mostly stays quiet. Bots wait while the host has a dialog open. They don't use chips or hints; anything else a team would do, the host does for them as for a team without a phone.

B · Found while building

- The new logo glow first made the main menu 1 px wider than a 1366-wide screen (`fix155l`, `fix154p`); it is now sized to the logo.
- The laptop app's copy of the Galaxy Tab app (for updating the tablet over the Wi-Fi) was a manual step until now; `post158_build.sh` does it and the EXE check confirms it (`fix142d` / `fix143d` were red until it did).
- Three older suites expected the old flows and were updated: `qlib` and `fix138` (the Move box is now round buttons; the one-tap add button), `fix157` (the tablet home's Restore link now ends `&home=1`).
- Two test timings: `fix146`'s Save Champions Picture waited a fixed 2 s for a 1920×1080 PNG, which took longer on this machine today — on both the build-157 and build-158 page; it now waits for the save (up to 10 s), 67/67 on both. `fix139d` (slow TV) — see C.

C · Tests (page and APKs from commit f5c5139)

- **Full run: 205 suites** — browser 129 suites 3,858/3,860; laptop app (Electron 44) 76 suites 1,355/1,358 plus `fix155d` (the 💡 hint from a phone), which didn't finish while both halves ran side by side.
- **Re-run alone on the same files:** `fix146` 67/67 (test timing fixed), `fix145b` 35/35, `fix155d` 16/16, `fix142d` 17/17 and `fix143d` 13/13 (with the build-158 tablet app inside the laptop app). Recorded as timing under load, not as clean passes.
- **`fix139d` 22/23 — not passed.** The slow-TV check expects the slow TV (0.4 s a picture) to skip in-between pictures; in this sandbox today the laptop only made a new picture every ≈0.3 s, so the slow TV kept up and had nothing to skip (35–41 of 46). It fails the same way on the build-157 page, and the check that matters — the slow TV still ends on the newest picture — passes. The message now shows the picture rate.
- **New suites:** `fix158` 26 (Back from a player, a game and editing; Close to Setup, the main menu and the tablet home; centred tiles; Ready to Play tiles; the remembered round and one-tap add after a reload; tap-to-move; Replace Round with 4 of 6 categories and Undo), `fix158b` 19 (bots: the choice and the 10-team limit; pick, right, wrong with a believable answer, pass; the TV and host bar; waiting for a dialog; Daily Double wager; Final wagers, answers and verdicts; out loud and silent when muted; End Rehearsal; a real game never gets bots), `fix158d` 7 (laptop app, Phone Buzzer Mode: a bot buzzes and wins through the server, a phone that buzzes first beats it, a bot that doesn't know stays quiet).
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6). **Picture audit** (`fix153c`, every end-of-night screen on 5 devices with 2–10 teams) passed in the full run.
- **Reproducible:** a fresh clone at f5c5139 builds the tested page; the Galaxy Tab APK and the Mind It! TV APK, versionCode 158, are signed with the original key; the Windows EXE (Electron 44.9.3) carries the same page, `main.js` and both build-158 APKs; its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or Mi Box. In particular: the bots' spoken answers (the test checks the app asks the device's voice to speak, at the game's volume; the voice itself wasn't heard — this machine has no speakers, and the Galaxy Tab's voice depends on its text-to-speech setting); a bot in Phone Buzzer Mode with your real phones over your Wi-Fi; ← Back and "Close · Home" on the Galaxy Tab; the Electron 44 EXE on Windows.

Build 157 addendum (28 October 2026)

What you asked for (28 Oct, after build 156 went out): a screenshot of the main menu's Ready to Play card — "looks uneven, improve that"; "i dont see the sync button"; "put the option on tablet to restore from the backup file"; "make it .txt or json back up files"; "the syncs also arent happening" (both devices on 156; ⟳ Sync on the tablet and on the laptop changed nothing); then "eta for release".

Same version: v1.69 · versionCode 157 (Galaxy Tab and Mind It! TV) · `APP_BUILD=157`. New module `app157.js` (`build.py` order `…,156,157,6`). Also changed: `desktop/app/main.js` and `MindItHost.java` (the sync pieces), `android/assets/start.html` (the tablet home's Restore). Commit 06121b4. No Mac app (every 5th build: 160).

A · What changed

- **M4 · syncs with real data.** A sync starts with the tablet sending its whole copy (questions, players, Historic Games, pictures). Since build 139 a device on the Wi-Fi may send at most 2 MB in one message (laptop `main.js` and the Galaxy Tab server), and your copy is about 9 MB (58 pictures, 8.5 MB), so the laptop closed the connection, no answer came and the tablet stayed "Syncing…" for good — every later sync was ignored. Now the tablet sends a copy over 1 MB as pieces of at most 400,000 characters (`lib-sync-part`); both servers pass each piece to the quiz page under the same rules as a sync (only the device in control; only while the quiz page is open); the quiz page puts them back together (at most 200 pieces, 64 MB, 2 minutes) and merges as before. A sync with no answer in 90 s stops waiting and says so. The 2 MB limit for every other message stays.
- **M2 · ⟳ Sync devices on the main menu** (Ready to Play, laptop app only), with the sync status line.
- **M3 · Restore on the tablet.** Tablet home → Settings → Backup → ⬆ Restore From Backup File opens Backups & Restore (`mindit.html?open=restore`); the restore is the laptop's (asks first, safety copy, adds players and Historic Games, deletes nothing). A backup named `.txt` restores too, when what is inside is a Mind It! backup or profiles file (on the tablet and the laptop).
- **M1 · Ready to Play** lines up: chips in two even columns at one height (an odd last chip spans both), 📺 Connect a TV and ⟳ Sync devices as full buttons below.

B · Found while building

- **Why build 156's tests missed M4:** their "tablet" ran on the laptop itself and synced small data. The new `fix157d` connects over the laptop's network address — the door a real Galaxy Tab uses — with 9 MB of pictures; on build 156 it shows exactly your symptoms (the tablet's pictures never arrive, the tablet stuck on "Syncing…", ⟳ Sync devices on the laptop then does nothing).
- The tablet's first connection is opened while the page loads, before the new code; the pieces had to be added to that connection too (found by `fix157d` on the first try of the fix).

C · Tests (all on the final files, commit 06121b4)

- **Full run: 202 suites** — browser 127 suites 3,791/3,791; laptop app (Electron 44) 74 suites 1,351/1,351 plus `fix155d` (the 💡 hint from a phone), which didn't finish while both halves ran side by side; on its own on the same files it passed 16/16 — recorded as timing under load, not as a clean pass.
- **New suites:** `fix157` 28 (the Ready to Play card in three device cases at two screen sizes, the menu still fitting 1366×768; ⟳ Sync devices on the menu; the tablet home's Restore, Backups & Restore opening on the tablet, a .txt and a .json backup each restoring with the question first and nothing deleted), `fix157d` 7 (the real-Wi-Fi sync with 9 MB). Both were shown failing on build 156 first. `fix146t` 13/13 (the Galaxy Tab server compiled with the change).
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6). **Picture audit** 130 pictures, 0 issues (the main menu's new card included).
- **Reproducible:** a fresh clone at 06121b4 builds the tested page; the Galaxy Tab APK (with the new tablet home) and the Mind It! TV APK, versionCode 157, are signed with the original key; the Windows EXE (Electron 44.9.3) carries the same page, `main.js` and both build-157 APKs; its 4 parts join to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones or Mi Box. In particular: ⟳ Sync with your real data between your laptop and Galaxy Tab over your Wi-Fi (the test used the same size, 9 MB, over this machine's network address); Restore From Backup File with the Galaxy Tab's own file picker; the Electron 44 EXE on Windows (also new in 156).

Build 156 addendum (28 October 2026)

What you asked for (28 Oct): install gstack and run a full code review and QA ("make Mind It! demonstrably more stable, reliable and polished"); "in tablet, in the last screen, when you click on change theme, the options are cut off"; "Add a sync button, so that all data is synced between the devices in the same network, currently laptop is showing 3 games, while tablet is showing 2 games despite it being used as remote for all 3 games"; "also add a end game button in the apk same as the laptop" (during the game too); then (after the ideas list) "R1, R2, H" — "no add them to 156"; then "lets put the changes in build 156 along with the fixes i requested, and we do this today".

Same version: v1.69 · versionCode 156 (Galaxy Tab and Mind It! TV) · `APP_BUILD=156`. New module `app156.js` (`build.py` order `…,155,156,6`); `app155.js` is unchanged from the build 155 release. Also changed: `desktop/app/main.js` and `MindItHost.java` (one line each, F1). Commits 0ddfc1d, a925666, 2acad73, 5a51442, 077ce1f; `desktop/package.json` (Electron 44.9.3, electron-builder 26.18.0). No Mac app (every 5th build: 160); the guides are build 155's.

A · What changed

- **Q1 · menus fit the screen.** Dropdown menus open right-aligned under their button. On the tablet remote held upright the top bar wraps and 🎨 Theme sits at the left edge, so its list opened off the screen (all 14 options cut off, at every step, not only the last screen); held sideways the list was capped at 600 px and the last themes hid behind a scroll. A menu now moves back inside the screen when it opens (and through redraws and rotation), and the tablet remote's Theme list uses the screen's height.
- **Q2 · ⟳ Sync devices.** The laptop ⇄ tablet sync of questions, players and Historic Games (builds 113–126) ran when the tablet connected, and a reconnect within 10 minutes skipped it; a game is saved at the end of the night, after that sync, so the last game never reached the tablet. The laptop now adds a counter to the state it already sends the tablet; the tablet syncs whenever it changes. The laptop raises it 1.5 s after each game is saved (not in a rehearsal) and from ⟳ Sync devices (Player Profiles, the connected-devices list, ⋯ during a game). The tablet remote has an always-visible ⟳ Sync, and a request that arrives while it is already syncing runs once more afterwards. The merge is unchanged (newest copy of each record; deliberate deletions stay deleted; nothing else is removed). No change to either server.
- **Q3 · End game now.** ⋯ → End game now on the laptop and the Galaxy Tab host, any time during play (not on setup, standings, awards, winner, season or recap, nor once the game is over). After a confirm it stops the timer, closes any open question or wager and goes to the standings with the current scores; the normal end-of-game save follows. The tablet remote (controlling tablet only) shows 🏁 End Game Now when the laptop says it's allowed (`endNowOk` in the state it already sends) and sends an `endnow` command (ignored if repeated or for another game).
- **Q4 · the Volume slider has a name.** Build 155's accessibility check left one gap (the Volume slider in Undo & resets had no name for screen readers; promised for 156). Both Volume sliders (Undo & resets, Setup → Sound) now carry the text shown next to them as their label.

- **R1 · Electron 33.4.11 → 44.9.3** (electron-builder 25.1.8 → 26.18.0). 33 was out of support with ~45 advisories (review F2); 44.9.1 is the first 44 release with none (npm audit 0 on 44.9.3 and the whole build tree, which also clears F3). 44.10.0 (a day old) was passed over. Inside: Chrome 152, Node.js 24. `main.js` uses only long-stable calls (BrowserWindow, `setWindowOpenHandler`, permission handlers, single-instance lock) and needed no change. The main process's clipboard became asynchronous in Electron 44 (no `readImage`); the app doesn't use it — 📋 Copy picture for the chat copies from the page with `navigator.clipboard`, unchanged — but `fix137d` read the picture back that way and now uses `clipboard.read()` (the 1080×1350 picture is there).
- **R2 · sync status.** Laptop: when a tablet syncs it notes the time and both counts (the tablet reports its count after merging, command `syncst`), shown beside ⟳ Sync devices (connected devices, Player Profiles), with a note when the laptop has saved games since. Tablet remote: the laptop's counts ride in the state it already sends (`libN`); ⟳ Sync shows ⚠ when they differ; the full line sits in More controls. On the Galaxy Tab held upright the remote's button row was already full in build 155 (Theme, 🔒, Teams, Team Codes, One-handed, Add Phone Remote), and ⟳ Sync pushed Add Phone Remote to a third row (`fix119`); upright, ⟳ Sync now sits beside ● Connected as the icon only (named "Sync" for screen readers); sideways it keeps its label.
- **H · question stats.** Each saved game record keeps one line per question asked (`qres`: the question's key and whether a team scored on it; the Mastermind It! question counts as got if any team was right). Stats are counted from Historic Games, so they sync and follow deletions with the games; the Question Database merge is unchanged. Rows show the record; More ▾ → 📊 Question stats lists "nobody got these" and "got every time (2 games or more)", one line per question even with two copies in the database. Old games have no `qres`, so the stats start with build 156.
- **F1 · private team state needs the team's code.** `/api/play/state?team=<id>` added a team's private state (💡 hint mask, first letter, word lengths; Rewind It! wheel; chip state; whose turn to pick) even when the code was wrong or missing; team ids are public in the same reply. Now only after the code check passes, on the laptop and the tablet servers.

B · Found by the review and QA (gstack 1.91.290.0)

- `/review` (whole codebase by risk area — the skill reviews a branch's changes and there were none yet): F1 above; **Electron 33.4.11** in the laptop app is past support with ~45 published advisories fixed only from 42.11.16 (exposure is limited: own local page only, sandbox and context isolation on, external pages refused) — **upgraded in 156 (R1)** at your request; the Mac app stays on build 155 (Electron 33) until build 160; build-tool advisories (`tar`, `extract-zip` in electron-builder 25) were on the build machine only; cleared by electron-builder 26.
- Checked and sound: every host-only route on both servers refuses other devices; no file is served from a path a request chooses; phone wagers and answers are validated and one per team; saves on the Galaxy Tab are atomic with a backup copy; one countdown at a time; the reconnect logic never keeps two connections.
- `/qa-only` probes: 4 randomized games (~600 actions, 30 % double taps) — scores always equal the score journal, no double scoring, intervals/listeners/DOM back to baseline, 0 errors; 5 save → reopen cycles (board, mid-question, Blind It! wager, Mastermind It! wagers and marking) — everything restored; HTML typed into team, player, round, category, question, answer, quiz name and phone answers never ran (host, TV preview, phone page); timers.
- Not changed: the Mastermind It! timer defaults to **45 s** (build 139, your choice then); the QA brief said 60 s — it can be set in Setup.
- Found while building: the first full runs flagged `fix154p` (it had "Build 155" written in; the footer is right, now read from the page), `fix119` (the new ⟳ Sync pushed the upright tablet's Add Phone Remote to a third row — fixed, see R2) and `fix137d` (Electron 44's clipboard read-back — the test's, not the app's). The pictures for the owner then showed the ⟳ ⚠ icon wrapping when the counts differ; fixed before the final run.

C · Tests (all on the final files, commit e878f7e)

- **Full run: 200 suites, 5,141/5,147 checks** — browser 126 suites 3,787/3,787; laptop app (Electron 44.9.3) 74 suites 1,354/1,360. The six misses were in `fix132t` (handing the quiz to the Galaxy Tab, 5) and `fix139d` (the slow TV, 1), both timing checks, while both halves ran side by side; on their own on the same files they passed 57/57 and 23/23 (with `fix130m` 15/15) — recorded as timing under load, as in build 155, not as clean passes.
- **New and changed suites:** `fix156` 44 (themes on screen at every step both ways; End game now on the laptop and the Galaxy Tab host; the Volume slider's name; question stats), `fix156d` 21 (the laptop app over the real link: the game saved at the end reaches the connected tablet; ⟳ Sync both ways and mid-sync; sync status on both, with the difference shown; End Game Now from the tablet; the team code guards a phone's private state). The checks for the fixes (themes, the missed game, the team code, the Volume slider) and for sync status and the question record were shown failing on the earlier page first; End game now is new, so it was checked on the new page only. `fix119` now also checks the upright ⟳ Sync icon; `fix137d` reads the clipboard the Electron 44 way; `fix154p` reads the page's own build.
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6), the laptop app on Electron 44. No 60-minute soak (every 5th build: 160).
- **Picture audit** (`handover/scripts/b155/look155.py`): 130 pictures — laptop 1366×768 and 1920×1080, Galaxy Tab upright and sideways, a phone and the TV, 2–10 teams, board to the end of the night: 0 issues (sideways scroll, buttons off the screen or clipped, touch targets under 36 px, anything outside the TV picture).
- **Audience View audit:** 406/406, 405 screens, no issues.
- **Speed** (against build 155 the same way): about the same — memory after a game 14.2 MB (155: 14.1), the device snapshot under 0.12 ms, question stats over 200 games 2–3 ms.
- **Reproducible:** a fresh clone at e878f7e builds the tested page; the Galaxy Tab APK (versionCode 156) and the Mind It! TV APK (versionCode 156, the same file as earlier today) are signed with the original key; the Windows EXE (Electron 44.9.3, 106.6 MB) carries the same page, `main.js` and both build-156 APKs; it is now 4 parts for delivery (each under 30 MiB) and the joiner puts them back to the same file (Wine joiner test 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, phones, Mi Box S, TV or a Mac. In particular: the Electron 44 EXE on Windows (first launch, a saved game, 📋 Copy picture, the Wi-Fi server for phones, tablet and TV); the Galaxy Tab's 🎨 Theme list in your hands; ⟳ Sync and the status line over your Wi-Fi after a real night; End Game Now from the tablet in a real game; 📊 Question stats after real nights. The Mac app stays on build 155 (Electron 33, without the F1 fix) until build 160.

Build 155 addendum (27–28 October 2026)

What you asked for (27 Oct): "the last build of the week ... lets do all the checks and tests, i dont want any bugs or layout issues"; 💡 the hint ("works only on the offline mode ... it can even work on daily double but with the same rule that they get only half the points", "even include it in all modes except online", "no speed bonus", "hint should be available in full show too"); the Galaxy Tab host layouts; Mastermind It! wagers in two columns; sound clue loudness; 8 new achievements ("all approved and assign them to old games"); the phone place card; Connect a TV on the main menu; 🔁 Rematch; layout fixes L1–L5; Online Mode seasons and the Grand Rascala ("seasons and grand rascala is for 155"); the new end of the night ("make the reveal of the top 3 instead of top 2", "i want that workflow to be real easy and smooth"); the host flow A–F ("all 6 but for the start the show, there should be enough breathing room to show the categories", "how to play can be 5 seconds"); 💘 Valentine's Day and 🎲 The Risky Theme; phones as host ("optimize it for a standard android phone"); the guides ("make it like 10-15 pages and focusing only on online mode"); the career achievement renamed World Tour.

Same version: v1.69 · versionCode 155 (Galaxy Tab and Mind It! TV) · `APP_BUILD=155`. New module `app155.js` (`build.py` order `…,153,154,155,6`): 1 core (hint, achievements, Online Mode flag, seasons) · 2 host flow (Start the show, the panel during play, the end of the night, Rematch, the Galaxy Tab and phone host) · 3 layouts and looks. Also changed: `desktop/app/main.js` and `MindItHost.java` (the hint between phone and host), `MainActivity.java` (phones stay upright), the Galaxy Tab's `start.html` (14 themes, "this phone").

A · What changed

- 💡 **Hint** (every mode except Online Mode, Full Show included; needs team phones): the team answering first takes its one hint per game from its own phone, before any wrong answer or pass; only that phone sees it (first letter and word lengths), the TV shows 💡 on the team. Right = half the value (half the wager on Blind It!), wrong = the full value, no

Speed Bonus; a team the question passes to can't use it. Hints reset each game.

- **8 achievements**, given to old games when the app opens (once). The career one is 🗺️ **World Tour**.
- **Online Mode seasons and the Grand Rascala:** each Online Mode game is marked as such; a season runs October–December (and so on). Points per night = the number of players minus the players ahead (ties share the place's points); a score of 0 or less earns 1 point. The last player of the season is the Grand Rascala, unless everyone is level. Player Profiles → 🏆 Season shows any season.
- **The end of the night:** Mark (✓ / ✕ / 0 per team, ✎ to fix a typed answer, "Also accept" for alternates, the chat box in Online Mode) → ▶ Reveal all, lowest first (≈2.5 s each, ⏸ Pause; one team at a time is still there) → Final standings with the top three kept secret → one awards page (nothing that gives the top three away) → The Rascala! → the podium (3rd → 2nd → the winner) → Season standings (Online Mode) → 🏁 End Quiz & save, 🔁 Rematch. One step card on the laptop, the Galaxy Tab and phones, with a progress strip and Back on every step. Games with 3 teams or fewer go straight from the reveal to the awards (their standings are the podium) and are still saved.
- 🔁 **Rematch:** same teams, players and rules; new questions (never-played Question Database questions first); opens Prepare tonight's quiz on its Check page. Not offered in a rehearsal.
- **Host flow A–F:** ▶ Start the show (welcome, teams, categories with room to breathe, how to play in 5 s; ⏸ Hold / ▶ Resume), the answer in the now card, the team list folded to one line, Teams / Host Cam / Phones folded, Reset game moved into the ⋯ menu.
- **Galaxy Tab host:** portrait puts the extra cards in a swipe row and keeps the scoring buttons at the bottom; landscape keeps the board and the panel side by side; Mastermind It! wagers in two columns. **Phones as host:** stay upright, and say "this phone".
- **Looks:** 💘 Valentine's Day and 🎲 The Risky Theme (14 themes); sound clues loudness-matched; layout fixes L1–L5; the phone's place card; 📺 Connect a TV on the main menu.
- **The guides:** the Idiot's Guide is now 14 pages, Online Mode only.

B · Found while building

- The first full run (commit 36d1645) flagged 20 browser and 15 laptop-app suites. Real problems found and fixed: games with 3 teams or fewer weren't saved when the standings step was skipped; Recently Deleted was hidden by a layout fix; End Rehearsal, ✎ Edit Answer and "Also accept" were missing from the step card; the Online Mode chat boxes didn't take the cursor at the start of the Final; the Rascala! could be missed if skipped; the Risky Theme's control ring; small fold buttons; the tablet's Undo card was folded into the swipe row.
- The tablet remote's big button said "finish this step on the laptop" for ▶ Reveal all, although the tablet has its own ✓ / ✕ buttons; it now says what each new step does and why it waits.
- The picture audit (130 pictures, 5 devices, 2–10 teams) found the Galaxy Tab's 8+ team marks and the wager row's Save / Open code at 32–34 px; they're 44 px now.
- The other flagged suites expected the old end of the night (two awards pages, Back to setup, one reveal per tap, the old marking card); they were updated to the new flow, which they now check.
- The 60-minute soak drove the old flow; it now lets ▶ Reveal all run and checks the winner before moving on.

C · Tests (all on the final files, commit 1a7e26e)

- **Full run: 198 suites, 5,033 checks** — browser 125 suites 3,752/3,752; laptop app 73 suites 1,279/1,281. The two misses were both in handing the quiz from the laptop to the Galaxy Tab (`fix130m` the 2-minute expiry, `fix132t` the tablet taking over) while both halves ran side by side; run again on their own on the same files they passed 15/15 and 57/57 (`qc_summary_rerun155_handover.txt`). Nothing in that code changed in build 155. Recorded as timing under load, not as clean passes.
- **New suites:** fix155 38 (hint, achievements, seasons, the end of the night, Rematch), fix155e 51 (the step card, Reveal all, the podium, End Quiz), fix155a 28 (achievements and old games), fix155l 67 (layouts, wagers, themes, loudness), fix155h 28 (host flow A–F), fix155d 16 (the laptop app: hint path, the end of the night).
- **Picture audit** (`handover/scripts/b155/look155.py`): 130 pictures — laptop 1366×768 and 1920×1080, Galaxy Tab portrait and landscape, a phone (412×915) and the TV, 2–10 teams, board → question → wagers → mark → reveal → standings → awards → The Rascala! → podium → season. Checked for sideways scroll, buttons off the screen or with clipped text, touch targets under 36 px and anything outside the TV picture: no issues left after the fixes in B (the tablet's swipe row and the theme decoration run past the edge on purpose). The Rascala! screen itself was not pictured (the audit turns the automatic Rascala off; fix155e covers it).
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6). **60-minute soak** 14/14: 3 whole 8-team games driven from the tablet (one with a sudden-death tiebreaker), 67 questions, 292 tablet actions, 10 phones, 80 samples, the window hidden and shown 5 times — 0 false disconnects, no script errors.
- **Reproducible:** a fresh clone at the release commit builds the tested page; the Galaxy Tab and Mind It! TV APKs are signed with the original key (`5f4819e4…`), versionCode 155; the Windows EXE and the Mac app carry the same page, `main.js`, the build-155 tablet APK and the build-155 TV APK; the EXE's 3 parts join to the same file (Wine joiner 4/4); the Mac app's 5 parts join to the same zip.
- **Audience View audit:** 406/406, 405 screens, no issues.
- **Your backup** (a copy, in the sandbox; counts only): opens and plays on build 155 — 13/13, including 🔁 Rematch on your Question Database (120 new questions, none from the last game).
- **Speed:** no long tasks in 12 s of play, 6.8 MB of memory after a live game, the device snapshot 0.2 ms.
- **Accessibility** (laptop, Galaxy Tab, phone; every end-of-night step): every button and box can be reached by keyboard and has a name, except the Volume slider in the Undo & resets card (build 152), which has none for screen readers. The fix is one line, but it came after the full run and all four apps were built, so it waits for build 156 to keep this release exactly the tested build.
- **The Complete Idiot's Guide** (14 pages) was read against the final behaviour: Reveal all, the standings step skipped with 3 teams or fewer, the hint (not in Online Mode).

D · Not tried for real

Nothing was run on your laptop, Mac, Galaxy Tab, phones, Mi Box S or TV: the hint from a real phone, the end of the night on the Galaxy Tab in your hands, a phone as host, the Mac app on a real Mac (it is ad-hoc signed, so macOS asks once — see the Mac instructions), sound clue loudness through real speakers, and a season over several real Online Mode nights.

Build 154 addendum (27 October 2026)

What you asked for (26 Oct): a performance and code check; "a proper ui and user experience check … i feel we can improve on the host experience"; "see if the game set up page can be optimized, make the process of preparing the quiz more streamlined and easy, do this across all the devices"; a main menu "just to give a premium feel" ("only do the Laptop home screen as you have shown", with the centred wordmark and one shared header); Setup → Rounds & Questions in sub-tabs; sound questions ("A"), question stats ("H"); the category reveal at 3 seconds; "for the team that is last in the leaderboard put a small rascala emoji on them, live during the game"; The Rascala! voice (take 4, "B"); in Online Mode a random turn order ("on by default but can be switched off") and player nicknames; "add all of them [the achievements] and make it automatically reflect in the old games"; "avoid scrolling as much as possible". No Mac app, no guides and no soak in this build ("keep the mac and guide update to build 155", "soak test for 155").

Same version: v1.69 · versionCode 154 (Galaxy Tab and Mind It! TV) · `APP_BUILD=154`. New module `app154.js` (`build.py` order `…,152,153,154,6`): 1 core (category reveal, The Rascala! badge and voice, Online Mode turn order and nicknames) · 2 quiz preparation (laptop main menu, Prepare tonight's quiz, Rounds & Questions sub-tabs) · 3 clues and records (sound clues, question stats, 12 achievements, per-question log, old-games update) · 4 speed and host UX. The Galaxy Tab's `start.html` (Start New Game opens Prepare tonight's quiz) changed; the laptop app and Mind It! TV only carry the new page and build number.

A · What changed

- **Fixed first — a build-153 bug:** with the Audience View window open, build 153 rests the laptop's TV preview, which was the laptop host's only clickable board. **Pick a clue** now shows at the top of the host panel whenever the preview rests.
- **Laptop main menu** (`#b154m`): Prepare tonight's quiz (Enter), Continue last game (only with a saved game), Question Database, Players & History, Rehearsal, Settings; last night's podium and Rascala!; screens ready; backup status. Fits 1366×768 and up without scrolling.
- **Prepare tonight's quiz** (laptop, Galaxy Tab, phone, web; `handover/build154/prep/`): Same as last time / New quiz, step by step / Quick game; Questions → Teams → How you play → Check with Fix buttons; "How are you playing tonight?" sets rules, answering and screen together. All settings stay one tap away.
- **Rounds & Questions** split into Rounds, Questions and ⭐ Mastermind It! sub-tabs.
- **Sound clues** (`handover/build154/clues/`): MP3, M4A and WAV wherever clue pictures and videos work (upload, clue editor, Question Database, Excel templates by file name); the TV's gold sound wave; host ▶ Play / ↻ Replay / ■ Stop on the laptop and the Galaxy Tab; a Question Database media link now plays on the TV.
- **Question stats:** each Question Database question keeps right / wrong / no-answer counts.
- **Category reveal** 3 s per category (was 1.45 s).
- **The Rascala! badge** on the last-place team's TV score card(s) during play (none while all teams are level); **The Rascala! voice** (take 4 B, loudness-matched) at the reveal.
- **Online Mode:** 🎲 random turn order (on by default, a switch in the rules; drawn with the browser's crypto random; Tonight's teams shows the draw); nicknames from Player Profiles during the game, real names in Player Profiles, Historic Games, saves and the Excel exports.
- **12 achievements**; old games get Rascala Repeat and Seeing Clearly when the app opens (marked so it runs once); each new game saves a per-question log (`rec.qlog`).
- **Speed and code check** (`handover/build154/perf/report.md`): no hook errors, no duplicate labels, no dead functions; one forced double layout per redraw removed (about 18 % lighter on the laptop with the Audience View window open, 19 % on the Galaxy Tab hosting).
- **Host UX walk-through** (`handover/build154/ux/report.md`): the host controls spread over 2–3 columns with the Audience View window open; one Mastermind It! sealed-code help line and compact wager rows; corrected hints. Bigger items listed there for you to choose.

B · Found while building

- In the full run, `fix151u` M2 found that a direct Wi-Fi TV picture outside a capture threw once The Rascala! badge put a large picture on the score cards (the real capture path was fine). Fixed in the app (4d), then all files rebuilt and the full programme re-run.
- `fix150` and `fix149` expected the build-153 sound pack; The Rascala! voice is a 16th sound by design. Tests updated.
- A container restart interrupted the first full run; it was restarted from the beginning on the final files.
- **The Rascala! badge over the score (5–6 teams).** Checking the release pictures after the full run: with 5–6 teams (one tight row of score cards) the badge sat in the side-column spot, over the score. The publish run was cancelled during its emulator check (nothing published; the latest release stayed build 153), the badge moved to the card's top-left corner for that layout (one CSS rule), and `fix154` now checks 4–10 teams: inside the card, never over the name's or the score's text (fails on the old page at 6 teams, passes on the new). Every file was rebuilt; the browser suites and the Audience View audit were re-run (C).

C · Tests (all on the final files)

- **Full run: 192 suites, 4,847 checks, all passed** — browser 120 suites 3,526/3,526; laptop app 72 suites 1,321/1,321. Two container restarts interrupted runs: the first run (before fix 4d) was abandoned and the whole programme restarted on the rebuilt files; in the second, the laptop-app stage stopped after 42 suites and resumed with the other 30 on the same files (`resume154.sh` ; summaries `qc_summary_full154_desktop_part1/2.txt`).
- **After the badge fix (the final files):** the browser suites again, 120 suites 3,518/3,518 (`qc2` plays random games, so its count varies: 282 this time, 291 before; `fix154` 25 with the new 4–10-team check), and the Audience View audit again, 406/406. The laptop-app suites and the stress phases were not repeated: they passed on files that differ only by that one CSS rule.
- **New suites:** fix154 24 (category reveal, The Rascala! badge and voice, turn order, nicknames, host UX), fix154r 60 (sound clues, question stats, achievements, old-games update), fix154p 117 (main menu, Prepare tonight's quiz, sub-tabs).
- **Audience View audit:** 406/406, no issues.
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6). **No 60-minute soak** in this build (build 155).
- **Reproducible:** a fresh clone at the release commit builds the tested page; the Galaxy Tab APK and the Mind It! TV APK are signed with the original key (`5f4819e4…`), versionCode 154; the Windows EXE carries the same page, `main.js` , the build-154 tablet APK and the build-154 TV APK; its 3 parts join to the same file (Wine joiner 4/4).

D · Not tried for real

Nothing was run on your laptop, Galaxy Tab, Mi Box S, phones or TV: the main menu inside the laptop app window, sound clues through a TV's speakers, Prepare tonight's quiz on the tablet, the Online Mode draw on a real call. No 60-minute soak in this build (build 155).

Build 153 addendum (26 October 2026)

What you asked for (25–26 Oct): the Mi Box connecting without a camera ("how will the mibox app connect"), on any Android TV; "anything we can do to improve the game's functionality on teams" ("1 and 2"); "any improvements in layout for host controls or settings" (the Setup audit); "anywhere that we can improve layouts, or make it more premium, lets do it", the Audience View too; after an 8-team game: the laptop "a bit slow" while hosting the Audience View and a Teams call, up to 10 teams, the tablet remote's answer in portrait, final standings and achievements cut off on the TV; "for online mode auto set the perfect graphics mode"; podium finishes instead of win %; an End Quiz button on the laptop's final standings; item 10 (Mind It! TV updates from the laptop); the layout upgrades ("do them all and include it in 153 itself, all approved").

Same version: v1.69 · versionCode 153 (Galaxy Tab and Mind It! TV) · `APP_BUILD=153` . New module `app153.js` (`build.py` order `…,151,152,153,6`): the TV's "Add a TV" pairing on the laptop and the tablet · Video Call · 2b the real-game fixes · 3 the premium pass · 4 speed, Online Mode graphics, podium finishes, End Quiz · 5 the layout upgrades. The laptop app (`main.js` : the Add a TV hand-off and `/api/tv-app`), the Galaxy Tab app (`AddTv.java` , `MindItHost.java` , `start.html`) and Mind It! TV (`TvScan` , `TvPair` , LAN updates in `TvUpd` , the new start screen) changed.

A · What changed

- **Mind It! TV connects without mDNS** (`handover/tv153/`): (A) it searches its own /24 for the hosts' ports (32 at a time, at most every 30 s, only addresses on its own network); (B) it shows a 4-digit code that 📺 Connect Audience Screen → Add a TV on the laptop or the Galaxy Tab sends with the sender's own `http://…/tv` (right code only, 5 tries, then a new code and a pause).
- **Mind It! TV updates from the laptop:** the laptop app carries the same build's TV app (`app/tv/` + `tv-app.json`) and serves it read-only at `/api/tv-app` . The TV checks the laptop it last opened (home-network address, two fixed paths, no redirects) and GitHub, takes the newer build (the laptop on a tie or offline), and installs only on a press after the fingerprint, package, build and original-key checks.
- **Video Call (Teams, Zoom, Meet)** (`handover/vcall153/`): build 129's Teams Presentation Mode under one name and the same saved setting (`G.teamsMode`); 16:9 window; the laptop's own picture calm (no confetti, particles or rays; loops once; question and answer ~12 % bigger); Wi-Fi TVs and Mind It! TV get the normal picture.
- **Speed** (`handover/perf153/report.md`): with the Audience View window open the host's preview isn't drawn ("Showing in the Audience View window"); the timer ring is one Web Animation per run; clock-only changes reach Wi-Fi TVs at most every 500 ms. Measured at 4× CPU, 8 teams: timed question 486 → about 290 ms/s; Blind It! 16 → 26 fps.

- **Online Mode** sets Video Call and Motion Subtle on this laptop (device setting), says so with Keep Full motion / Keep Phone Buzzer, and puts the old Motion back on leaving.
- **Real-game fixes:** up to 10 teams (servers already take 12; saves unchanged; compact side scores for 9–10); Achievements, Game Highlights and Fun Stats fit every team; the Night Recap lists all teams; the portrait tablet remote shows the answer card.
- **Podium finishes** instead of win % (player page, Team Settings, both leaderboards, player editor; the Excel export keeps both). 🏁 **End Quiz** on the laptop's final standings, awards and winner screens (Cancel first, records once).
- **Setup audit S1–S10** (`handover/setup153/report.md` ; S8 by the Video Call merge), **the premium pass** (`handover/premium153/compare/`), **the Mind It! TV start screen** (mock-ups `handover/tv153/ui/`) and **the layout upgrades** (`handover/layout153/compare/` : Historic Games, past game page, profiles, leaderboard, Question Database editor, saved pictures at 9–10 teams, iPhone/iPad safe areas, touch Setup 48 px).

B · Found while building

- Video Call first duplicated the existing Teams Presentation Mode; merged under one name before anything shipped.
- `fix114` (tablet Setup buttons 44 < 48 px) failed after the premium pass: fixed in the app (48 px on touch), the test unchanged.
- `fix137` read the host's TV preview while Online Mode had the Audience View window open; the preview now rests then (on purpose), so the test closes that window first. Tests updated for intended changes: `icons` , `rail` , `fix151t` item 9, `v169` , `fix153p` , `fix153v` , `theme` , `fix138` , `fix139` .

C · Tests (all on the final files)

- **Full run: 189 suites, 4,625 checks, all passed** — browser 117 suites 3,304/3,304; laptop app 72 suites 1,321/1,321. In the full run, three older laptop-app suites (`g6` , `fix136od` , `fix137d`) needed test updates for intended changes (Quick Setup's fourth step; the host's preview resting while Online Mode's Audience View window is open). Updated and re-run on the same files, 31/31; the app was unchanged, so the rest wasn't re-run.
- **New suites:** fix153 20, fix153tv 33, fix153j 14, fix153d 10 (TV search and pairing), fix153v 21 (Video Call), fix153p 91 and fix153c 34 (premium pass), fix153z 63 (speed, Online Mode graphics, podiums, End Quiz), fix153l 90 (layout upgrades), fix153u 21 (TV updates from the laptop).
- **Audience View audit:** 405 TV screens (11 new scenes with 8–10 teams; normally and with Video Call forced on), no issues (406/406).
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid taps 4, health 6) and **the 60-minute soak 14/14**: 3 whole games with 8 teams, Final wagers from the phones, a sudden-death tiebreaker from the tablet, 67 questions in 54 minutes of play, 304 tablet actions, 82 samples of TV + tablet + 10 phones, 0 false disconnects.
- **Reproducible:** a fresh clone at the release commit builds the tested page; the Galaxy Tab APK and the Mind It! TV APK from it are signed with the original key (`5f4819e4…`), versionCode 153; the Windows EXE carries the same page, `main.js` , the build-153 tablet APK and the build-153 TV APK; its 3 parts join to the same file (Wine joiner 4/4).

D · Not tried for real

Nothing was run on your Mi Box S, Galaxy Tab, Windows PC, phones or TVs: the TV's network search and pairing code on your Wi-Fi, the TV updating from the laptop, the new TV start screen (code and mock-ups only; no Android emulator here), a real Teams / Zoom / Meet share with Online Mode, and the speed gains on your laptop during a call (measured in a slowed-down Chromium, not with Teams).

Build 152 addendum (24 October 2026)

What you asked for (24 Oct): template downloads in the Question Database's top row; items 1, 2, 5, 7, 8, 9 and 10 from the suggestions (Mind It! TV for the Mi Box, the laptop updating itself, the whole-database export and duplicates easier to reach, the tablet host page in portrait, a phone remote's category names, the guides); "have a check on the host controls" then "do them all" (H1–H10, End round early on Shift+E); the TV app as its own APK that updates itself ("the app in mibox is mainly to be the audience view, which will be controlled via tablet or laptop"; Mi Box S); the tablet's App Update stuck at "2.65mb/2.65mb"; "if you feel any layout can be improved for tablet or laptop then do it".

Same version: v1.69 · versionCode 152 · `APP_BUILD=152` . New module `app152.js` (`build.py` order `…,150,151,152,6`): 1 ⬇ Excel ▾ and the duplicates count · 2 the laptop app's update · 3 the portrait tablet's TV preview fold · 4 a phone remote's category names · 5 host controls H1–H10 · 6 the layout pass. The laptop app (`main.js` : App Update), the Galaxy Tab app (`MainActivity.java` , `TabUpd.java` , `start.html` : in-app update; `REQUEST_INSTALL_PACKAGES`) and a new app, **Mind It! TV** (`android-tv/` , package `com.mindit.tv` , versionCode 152, same original key).

A · What changed

- **Mind It! TV** (`handover/tv152/`): finds `_mindit._tcp` hosts (laptop or Galaxy Tab) on the Wi-Fi, opens `http://host/tv` full-screen in a WebView, remembers the session and reconnects, D-pad only, Leanback banner; updates itself (GitHub latest release → SHA256SUMS → package, newer versionCode, original certificate → PackageInstaller), only on its start screen. Needs WebView 105+ (container-query units on the audience page); a red notice below that.
- **Galaxy Tab update without the browser:** the browser hand-off stopped at 100 % on the owner's tablet. The app now downloads (GitHub, or the laptop on the LAN), checks the SHA-256, the package, the build and the original key, and installs through PackageInstaller; first-time "allow from this source" handled; browser kept as a fallback.
- **Laptop app update** (Windows portable EXE): download beside the running file, SHA-256 against SHA256SUMS.txt, restart into it on a second click (never mid-game), the new build tidies only an older Mind It! EXE in the same folder; a Mac opens the release page.
- **Host controls** (`handover/hostctl152/`): confirmations focus Cancel; End round early = Shift+E; T resumes after a pause; Pause / Resume / End; Mastermind It! marking first; the Timer card under the preview; the Danger zone; folded special-question panels; ⌨ Keyboard Shortcuts (⋯ or ?); click a score to adjust it; 14 px top bar.
- **Layout pass** (`handover/layout152/`): the Undo pop-up centred (its keyframes undid the centring); CORRECT / WRONG show the amount the same way; Undo & resets under the preview on wide wager screens; host-panel text ≥ 14 px; notices clear the portrait tablet's top bar; Resume keeps the timer ring's total.
- **Question Database:** ⬇ Excel ▾ (Quiz Template, Mastermind It! Template, the whole database); the 🔁 count on More ▾. **Portrait tablet host:** the TV preview folds behind "📺 Show TV preview". **Phone remote:** category names 10–12 px, no word cut.
- **Guides:** the Complete Idiot's Guide (189 pages) and Quick Start, cheat sheet and checklist for build 152, with 7 new or refreshed pictures; the players' guide unchanged.
- **Release script** (MindIt-Downloads `.github/scripts/mindit_release.py` , approved by the owner): accepts `Mind-It-Android-TV.apk` after checking package, versionName, versionCode = build, original certificate and the Leanback launcher; carries it over when a build doesn't restage it. The workflow YAML is unchanged.

B · Found while building

- The first full run (on the page before the last fix) had four older suites needing attention: `ext` (the Timer card moved under the preview, H4), `fin` (the Excel menu's hidden item has the same words as the template button it clicked), `fittv` (the portrait tablet's preview is folded by default), and `fix118` , a real slip: with the speed bonus on, L4 made Correct three lines tall. Fixed (the ⚡ hint shares the amount's line); the three tests now look in the new places with the same assertions; the binaries were rebuilt and the whole programme run again.
- `fix152u` needed to stop the runner's own laptop app first (it starts one with its test settings).

C · Tests (all on the final files)

- **Full run: 179 suites, 4,221 checks, all passed** — browser 109 suites 2,931/2,931; laptop app 70 suites 1,290/1,290.
- **New suites:** fix152 29, fix152h 39, fix152l 49, fix152tv 26, fix152tab 47, fix152u 21.

- **Audience View audit:** 306 TV screens, no issues (307/307).
- **Stress — all six phases** 24/24 and **the 60-minute soak 11/11**: 2 whole games with 8 teams (the first ran long with ⏱ One More Question, so a third didn't fit the hour), Final wagers from the phones, a sudden-death tiebreaker from the tablet, Night Recap on the TV, 59 questions, 270 tablet actions, 69 samples of TV + tablet + 10 phones, 0 false disconnects, no script errors.
- **Reproducible:** a fresh clone at the release commit builds the tested page; the Galaxy Tab APK and the Mind It! TV APK from it (`TZ=Asia/Dubai`) are signed with the original key (`5f4819e4…`), versionCode 152; the TV APK rebuilt from the same sources is byte-identical; the Windows EXE carries the same page, `main.js` and tablet APK; its 3 parts join to the same file (Wine joiner 4/4).

D · Not tried for real

Nothing was run on your Mi Box S, Galaxy Tab, Windows PC, phones or TVs: in particular the Android install screens (tablet and TV), the TV finding the quiz on your Wi-Fi, the WebView on the box, and Windows SmartScreen or antivirus on a downloaded EXE. No Android emulator here. The tablet and laptop first need build 152 installed by hand once; later updates are built in.

Build 151 addendum (24 October 2026)

What you asked for (23–24 Oct): items 2, 5, 6 and 9 from the suggestions (the trailer's exact takes, guide pictures and names, a UI/UX audit, tests that tidy up their virtual screens); Rewind It! started and stopped from the picking team's phone in Buzzer Mode, the Quizmaster everywhere else; Host Cam stays off ("actually make the camera off only"); "do all" for the 13 audit proposals; the Galaxy Tab in portrait and "make the whole layout on tablet look premium"; improvements found with the new review skills; the animation brief (vanilla JavaScript/CSS only, no React); the complete codebase audit. "1&2 go ahead, all approved" — publish when everything passes.

Same version: v1.69 · versionCode 151 · `APP_BUILD=151` . New module `app151.js` (`build.py` order `…,149,150,151,6`) in six sections: 1 Rewind It! on the phone · 2 the UI/UX audit (M1–M8, F9–F13) · 3 the skills review · 4 the Galaxy Tab audit · 5 animation polish · 6 the code audit. The laptop app (`main.js`) and the tablet server (`MindItHost.java`) changed: chip `rw` for the phone's wheel, lockouts that double, and the Wi-Fi TV heartbeat.

A · What changed

- **Rewind It! on the phone:** in Buzzer Mode with phones connected, the team that picked a Rewind It! question gets SPIN on its phone, the wheel turns freely on the TV, and the team's STOP lands it (the multiplier is drawn when STOP arrives, with at least one full turn to go). The laptop keeps "Spin For Them" / "Stop For Them". Both servers accept the phone's tap only from the team whose turn it is, once.
- **The UI/UX audit, all 13** (`handover/audit151/` , after-pictures in `implemented/`).
- **The Galaxy Tab audit** (`handover/audit151p/`): the remote's Rewind It! buttons, its REWIND IT! name and a winner card (the 3 bugs), portrait items 4–13 and the "Studio Glass" look; a phone used as the remote keeps its compact board.
- **Skills review:** team codes always on; the waiting wheel stands still with Reduce Motion / Motion Off; PIN and team-code lockouts double per address (5 → 60 min; 3 hours of nonstop guessing gets ~120 tries, was ~720).
- **Animations** (`handover/anim151/`): the winner / final-standings confetti falls again (a theme keyframe called `fall` had replaced it); BUZZ IN pulses; the wager and "your pick" glows pulse opacity instead of repainting; a dotted orb beside 10 long waits, never beside a failure. No dependencies, no WebGL.
- **Code audit** (`handover/audit151c/report.md`): Wi-Fi TVs on a slow link were closed while a big picture was still on its way, then sent everything again — the historic "Reconnecting…" loop; reproduced and fixed on both servers. Setup's off-screen theme samples pause (idle work 634 → 384 ms/s). The score toast's keyframe clash. 4 unused functions, 16 superseded bodies and 7 CSS rules removed (page −12.5 KB).

B · Found while building (all fixed before the final run)

- The audit's spaced, coloured payout signs made "Wrong −Đ15,000" spill past its row (the full Audience View audit: 62 spills) → smaller row text and gaps; `fix151p` checks every row up to Đ5,000 × 5.
- The tablet's portrait home layout also hit phones (425 px wide on a 390 px screen) → from 600 px only (`rh`).
- Studio Glass shrank the one-handed Back to Board to 74 px and wrapped the remote's header once the 📶 Teams chip shows → sizes restored (`hostreq` , `fix119`).
- `qc2` failed once in nine runs during development ("DD code accepted & question shown"); it passed in every full run since, including the final one.

C · Tests (all on the final files)

- **Full run: 173 suites, 4,019 checks, all passed on the first try** — browser 104 suites 2,750/2,750; laptop app 69 suites 1,269/1,269 (with the build-151 tablet app inside, so `fix142d` / `fix143d` pass).
- **New suites:** fix151 37, fix151j 18 (tablet server), fix151d 14 (laptop app, real phones), fix151u 33, fix151t 50, fix151p 12, fix151s 9, fix151a 32, fix151c 13, fix151cd 11, fix151cj 7.
- **Audience View audit:** 306 TV screens at 9 sizes, no issues (307/307).
- **Stress — all six phases** 24/24 (idle 2, network 6, load 4, lock 2, rapid 4, health 6) and **the 60-minute soak 14/14**: 3 whole games with 8 teams to the winner screen, Final wagers from the phones, a forced tie settled by a sudden-death tiebreaker played from the tablet, Night Recap on the TV, 66 questions, 298 tablet actions, 82 samples of TV + tablet + 10 phones, 0 false disconnects, no script errors.
- **Reproducible:** a fresh clone at `1190ece` builds the tested page (`6d044bfc…`); the APK from it (`TZ=Asia/Dubai`) is signed with the original key (`5f4819e4…`), versionCode 151, and carries that page; the Windows EXE (72,574,713 B, `769d1067…`) carries the same page, `main.js` and APK; its 3 parts join to the same file (Wine joiner 4/4).

D · Not tried for real

Nothing was run on your Windows PC, Galaxy Tab, phones or TVs. The tablet's new look (WebView, touch, the blur's cost), the phone's wheel, the animations and the slow-Wi-Fi TV fix need a look on the real devices. No Chromecast or Android TV receiver exists in the project yet. The sounds were checked for timing and level, not by ear.

Build 150 addendum (23 October 2026)

What you asked for (23 Oct): the 45-second trailer's audio layout (`MindIt_45s_Audio_Layout.md` : the cue sheet, the levels, the one-shots) — "this build will undergo full on stress test and everything". Every 5th build: the guides and the Mac app.

Same version: v1.69 · versionCode 150 · `APP_BUILD=150` . New module `app150.js` (`build.py` order `…,148,149,150,6`). Laptop app code unchanged; tablet app unchanged apart from the versionCode.

A · The sounds follow the trailer

The trailer gives each moment one sound and one level (each effect peak-normalised, then mixed at its cue level). The game now does the same, with impact −9 dB kept at build 149's 0.75 of the mixer:

Moment	Sound · level	Before (build 149)
Blind It!, Mastermind It!, sudden-death tiebreaker	impact –9	the same
Right answer	ding –11, then the score pop (tally –16) 0.15 s later	ding only, a little louder
Tiebreaker won	ding –11 (no pop: no score changes)	the same
A phone's BUZZ IN	buzz –12	two synthesised beeps
Locked-in wager	buzz –12	a little louder
New board, round summary, standings, recap, awards, match-up	whoosh –15	nothing
Standings	the leaderboard tally –16, 0.45 s in	nothing
Winner	impact –9, then the crowd (cheer –12)	cheer only, quieter
Theme change	sparkle –17	nothing
Question appears	sparkle –16	louder
Triple Gamble · Phoenix · Bottoms Up	dice –12 · phoenix –11 · –12	louder
Wrong, time's up, Rewind It!, the wheel's bell, The Rascala! (not in the trailer)	next to the ding: –12 (The Rascala! –11)	louder

A theme's own sound still wins, muted plays nothing, the synthesised sounds play until the pack is decoded; only the laptop (and its Audience window) makes the cue sounds, the Wi-Fi TVs get them through its mixer. **The exact takes:** the trailer's cue sheet names one file per sound, but the ElevenLabs flow keeps all four takes of each without marking which one the trailer used, and the one-shot files weren't attached; the game keeps build 149's picks from the same four takes. Send the `oneshots/` folder and the game can use the trailer's exact takes.

B · Guides and Mac

- **The five guides** (updated every 5th build; last at 147): the names as the app shows them (259 changed lines read by hand; the buttons you kept — "⭐ Go to the Final", "Go to the Final Round" — and the Final standings/score words stay), new sections for Rewind It! (12.1b, payout table) and The Rascala! (19.1b), the Mastermind It! brain and Blind It! sunglasses, the tagline, App Update on the tablet (4.2, 22.7), the sound pack and the trailer cues (21.2), Undo and statistics (15.7), the achievement list (60), the glossary. 3 pictures replaced and 10 added, all real captures of the final page with made-up team names. Complete Idiot's Guide 183 pages (was 176; the contents page finds all 40 page numbers), Quick Start 3, cheat sheet 2, checklist 2, players' guide 2. Not updated: the worked example's pictures (a real pre-148 game, captioned), the tablet and phone wager pictures (captioned), and the features one-pager and mobile players' guide (not in the release).
- **The Mac app** (every 5th build; last at 147): Apple Silicon, ad-hoc signed with `rcodesign`, 103,867,979 bytes in 5 parts that join back to the same zip; it carries the tested page, `main.js` and the build-150 tablet app.

C · Tests

- **New suite:** `fix150` 37/37 — every sound at the trailer's level (13 sounds), the soft lock-in ratio, right = ding + pop, tiebreaker ding only, a theme's own sound alone, BUZZ IN = the buzz and the beeps silenced only for that moment, other buzzer events silent, the whoosh on every listed screen and never into or back from a question or on a redraw, the standings tally, the winner impact then cheer (not in a Rehearsal), the theme sparkle (🎨 Theme and Setup; not when the page opens or the same theme is applied again), phones and the tablet remote silent, muted silent, no hook or page errors.
- **Full run on the final page: 162 suites, 3,792 checks.** Browser half: 95 suites, 2,548/2,548, all on the first try. Laptop-app half: 67 suites, 1,242/1,244 — two suites needed attention, both looked into before going on:
  - `fix132s` (1 check, every time): "the laptop goes quiet while a Wi-Fi TV plays the sound" read the laptop speaker's volume as 1.000. The laptop does go quiet: measured while a sound plays, its volume is 0.0002 and then 0. Chromium only updates a volume's reading while sound flows through it, and in build 150 nothing happened to be playing at the moment the test looked. The test now plays a short tick first; 24/24. (Build 149's page passed the old test because a sound happened to be playing at that moment.)
  - `fix130b` (2 checks): the slow Wi-Fi TV (emulated: 300 KB/s and a CPU 4× slower) was read 1.5 s after the next round and 2.5 s after the Mastermind It! reveal, and was still blank, waiting for the new picture. **Build 148's page behaves exactly the same here** (checked twice each with builds 148, 149 and 150), so it is not caused by this build: this workspace's Chromium slows that emulated TV's connection harder than the old one did. The test now waits for the picture (up to 15 s): Round 2's board arrived after 9.7 s, the reveal after 5.9 s; 20/20. Worth watching on your real TV: on a really slow TV a whole new screen can take a few seconds.
- **Audience View audit:** 306 TV screens at 9 sizes, no issues (307/307).
- **Speed on a slow TV** (`perf146.py`, a Wi-Fi TV with its CPU 4× slower, on the laptop app): **Default theme** — the TV kept 29–72 frames a second through every moment (board, question, timer, right and wrong answers, standings, winner, idle) and never stalled longer than 0.28 s; **Comic Book** (the heaviest scenery) — 23–42 frames a second, longest stall 0.33 s, and the TV lightened itself after 23 s as designed. Build 147's run (another machine) measured 2–3.5 frames a second and about 1 s stalls on Comic Book, so build 150 is at least as smooth; the new sounds don't touch the TV's picture.
- **Stress — all six phases** (each on a fresh laptop app; 24/24): **idle** — a question and the board left up 5 minutes each with the TV, the tablet and 10 phones: no false disconnects, nothing covered the screen 2/2 · **network** — the tablet offline for 20 s (a small "Connection Lost — Retrying…" line only, the TV unaffected), back by itself in 0.3 s with what it missed; the laptop page reloaded mid-game, all 10 phones still signed in 6/6 · **load** — 10 phones wagering at once (8 accepted, the 2 second phones refused), 400 back-to-back requests in 0.7 s (median 16 ms, slowest 48 ms), Final answers from 10 phones 4/4 · **lock** — a phone locked for 60 s: listed as asleep, same team after, no re-pairing 2/2 · **rapid taps** — the same command 3 times applied once, an out-of-date tap refused, 30 quick tablet actions scored once each, tablet and laptop agree 4/4 · **health check** — everything green in 3.6 s, offline/asleep phones and a missing tablet flagged, the game untouched 6/6.
- **The 60-minute soak: 14/14** — 56 minutes of real-time quiz on the laptop app with the tablet, a TV and 10 phones: 3 whole games to the winner screen (8 teams, ⭐ Go To The Final, Final wagers from the phones, a forced tie settled by a sudden-death tiebreaker played from the tablet and recorded in Historic Games), 65 questions (median gap 42 s), 309 tablet actions, 81 samples, the window hidden and shown 5 times: 0 false disconnects, no script errors on the tablet or any phone; laptop server memory 233 → 244 MB, the Quizmaster page 18 → 21 MB.
- **Reproducible:** a fresh clone at the release commit builds the tested page; the APK from it (built under `TZ=Asia/Dubai`) is signed with the original key (cert `5f4819e4…`), versionCode 150, and carries the tested page; the Windows EXE and the Mac app carry the same page, `main.js` and APK; the EXE's 3 parts and the Mac zip's 5 parts join back to the same files; the Windows joiner test 4/4.

D · Not tried for real

Nothing was run on your Windows PC, Mac, Galaxy Tab, phones or TVs. The sounds were checked for timing and level, not by ear. The Mac app was built and signed (ad hoc, Apple Silicon) here but not opened on a Mac.

Build 149 addendum (23 October 2026)

What you asked for (23 Oct): "D1-5 do them all" (The Rascala! character, the tie-break when every team is level, the tagline), a complete code check with optimisations, layout improvements, the tablet updating itself without the laptop ("put a update app function on the tablet"), the trailer's sounds as inspiration for the game sounds and new sounds where needed, then "replace the cat from the app and use this one for all Rascala references" (your cowboy-cat picture). "publish without my approval as long as the files are functional and tested"; the trailer's own audio files are for next time; release by 17:40.

Same version: v1.69 · versionCode 149 · `APP_BUILD=149`. New modules `app148.js` (the changes) and `app149.js` (data: the cat picture and the sound pack, generated by `handover/scripts/b149/assets149.py`); `build.py` order `…,147,148,149,6`; `style.css` build-149 block; `app23.js` (the Setup note quotes the tagline); the tablet's `start.html` and `MainActivity.java` (`openAppUpdate`); the laptop app's `main.js` (Drive restore time limit). First build made in Claude Code (the Cowork workspace was rebuilt from the repository: the build-148 page and APK rebuilt byte-identical to the released files).

A · What changed

Part	How
The Rascala!	your cowboy cat (WebP 86 KB, a PNG for TV browsers without WebP; the round badge shows its head) on the TV reveal, Historic Games, the game page and Career Achievements; no name plate; the line is the tagline
Tie-break	`b148Rascala` without the all-level shortcut: lowest score → most wrong → fewest right → shared, every time; games recorded by 148 keep their frozen result
Tagline	"Enna Rascala….Mind It!" (`.tc-skip`, never re-cased) under the big logo on the loading and intro screens, the reveal, the tablet home screen, the Setup note
Tablet update	`start.html`: the latest MindIt-Downloads release via the GitHub API (once per 20 h when online, or Settings → App Update); a carried-over APK counts as its own build; Update opens only that release's APK in the browser (`MainActivity.openAppUpdate` checks the address)
Sound pack	15 sounds (175 KB): 9 from your trailer library (ding, sparkle, buzz-in, whoosh, impact, tally, dice, phoenix, cheer) and 6 made for the game (wrong, time's up, Rewind It!, wheel bell, The Rascala!, Bottoms Up), chosen from 4 takes each by measurement and spectrogram, trimmed, normalised to −3 dBFS; played through the same mixer (volume, mute, Wi-Fi TV); a theme's own sound wins; the synthesised sound plays until decoded
Layout	the Blind It! / Mastermind It! wager screens keep clear of the scenery (per-theme space like the 148 wheel: Comic 30 %, Christmas 24 %, Movies 22 %, Summer/Diwali 16 %)

B · The code check (each confirmed in the code before fixing)

- Rewind It! 4×/5×: the laptop's WRONG ANSWER button showed −value×m; the score took −value×3. Fixed (the tablet reads the same button).
- Undo / Reset question / Reset round left the answer in the event log, so an undone wrong answer still counted (achievements, The Rascala!'s tie-break). Now marked undone and left out.
- After End Game the Special Question card stayed locked. Unlocked.
- Back from The Rascala! the winner fanfare and celebration played again. Not any more.
- The automatic reveal pushed the Champions Card away. It waits while the card is up.
- A Rewind It! wheel screen whose question was deleted in Setup got stuck. Back to the board.
- The tablet showed ✓ / ✕ during the spin (the laptop ignored them). Hidden.
- Laptop app: a Drive restore was cut off after 45 s whatever its size. Now 10 minutes.
- Faster: the Title Case pass builds its protected-name list once per moment and remembers results (3,000 lookups in 1 ms). The theme scenery's endless animations keep one clock across redraws instead of restarting.

C · Tests

- **New suite:** `fix149` 100/100 — the cat (WebP and PNG), no name plate, badge, achievement; the tie-break (7 scenarios incl. all level, frozen 148 games, records unchanged); the tagline on all 12 themes and 3 sizes, Title Case, Motion Off / Reduce Motion; every code-check fix; the wager-screen space on 6 themes; the sound pack (all 15 decode, every moment, theme override, muted, synth fallback, winner cheer, not in Rehearsal); the tablet update (newer / carried-over / same / offline / no network / not in a browser, the Java address check).
- **Full run** of the build-149 code before the sound pack: the browser half, 94 suites, 2,405/2,405; the laptop-app half stopped after 38 of 67 suites (all passed) so the build could be released sooner (your choice).
- **Re-run on the final page** (the sound pack came after the full run started): fix149 100/100, fix148 63/63, fix147 49/49, theme 28/28, qc2 297/297, fix133m 32/32, fix134 24/24, and the laptop-app sound suites fix132s 24/24, tg 26/26, ud 22/22, bu 21/21. The other 29 laptop-app suites ran again in build 150's full run (below).
- **Audience View audit**: 306 TV screens, no issues (307/307).
- **Not run this build:** stress phases and the 60-minute soak (they run in build 150, below), slow-TV speed. Build 149 was released without this report (your choice); it is published here with build 150.
- **Reproducible:** a fresh clone at the release commit builds the tested page; the APK from it is signed with the original key (cert `5f4819e4…`), versionCode 149, and carries the tested page.

D · Not tried for real

Nothing was run on your Windows PC, Galaxy Tab, phones or TVs. The sounds were checked by measurement and spectrogram, not by ear. The tablet's own update was tested with a stand-in for the tablet and the download page.

Build 148 addendum (23 October 2026)

What you asked for: at 09:10, "MIND IT! — SIGNATURE GAMEPLAY & ACHIEVEMENTS UPGRADE": Blind It! (the Daily Double, black sunglasses), Rewind It! (an optional cowboy-hat 1×–5× wheel instead of it, losses capped at 3×, no chips), Mastermind It! (the Final Round, a brain) and The Rascala! (last place: lowest score → most wrong → fewest right → shared), including every historic game; keep every internal name, rule and saved game. 09:15: "start working on publishing the app too". The character: a film character (Crime Master Gogo) wasn't used as a basis; you chose a tabby cat, **Naughty Pussy**, the tagline "Rascala! You are now a Naughty Pussy!", outfit 1 with a tiny top hat, no tooth on the right and no tear drop. 09:59: every team level → all joint Rascalas; you may pick on an old game's page; Rewind It! time-out like Blind It!; "True Daily Double" → Never Mind It!; "Go to final is fine as it is". 10:17: "do 1,2,3" (test halves side by side, no slow-TV measurement, stress phases without the 60-minute soak); guides every 5th build (next: 150).

Same version: v1.69 · versionCode 148 · `APP_BUILD=148`. New module `app147.js` (`build.py` order `…,146,147,6`), `B148N()` in `app0.js`, the names changed in string literals across the page, `style.css` build-148 block, `app3.js` (Rewind It! value), the tablet's `start.html`. Laptop app code unchanged.

A · How each part was done

Part	How
Blind It!	every player-facing "Daily Double" (string literals only — never comments, regexes, identifiers or saved data) → `B148N()`, which says Blind It! (or Rewind It! in a Rewind It! game); the sunglasses on the TV intro and the wager screen; "True Daily Double" → Never Mind It!
Mastermind It!	every player-facing "Final Round" → Mastermind It!; the brain intro when the Final starts and the brain on its wager and question screens; data kept: the quiz export's FINAL ROUND block, the Excel template's sheet, the "Final Round" category for imported questions; the host's "⭐ Go To The Final" kept (you)
Rewind It!	Setup → Special Question (`G.special`); a Daily Double square in a Rewind It! game opens the wheel (`rwspin`); the result is drawn with crypto random (1–5, rejection-sampled), saved on the square at once, then animated from that saved start on the laptop, the TV and Wi-Fi TVs; right +value×N, wrong −value×min(N,3); still a Daily Double inside (one team, no passing); chips off; the tablet gets the same buttons
The Rascala!	one function for every game; tonight's from the live counts, frozen as an achievement; older games worked out each time (proof from their saved awards, or "needs review" + your pick); the reveal after the winner; Historic Games, the game page, Career Achievements

B · Your backup (the 2 games)

your backup (2 Historic Games, checked here with the build-148 code; the file wasn't changed or uploaded anywhere; team, player and quiz names are left out of this report because it's published): **the 9 Oct game** — two teams finished level on 0; the awards saved with that game prove one had 8 wrong answers and the other 6, so the team with 8 is The Rascala! (decided automatically). **The 18 Oct game** — two teams finished level on 0; that game only proves neither had 5 or more wrong answers, so it needed review; you named your pick (10:31), and build 148 fills it in the first time it opens your history (Undo Pick on that game's page can change it).

C · Found while building

- **Saved data uses the old words** in a few places (the Excel template's FINAL ROUND block and sheet name, the "Final Round" category for imported Final questions, Historic Games' question round names): those stay, so templates, imports and old records keep working.
- **Wrong answers weren't saved per team** before build 148, so an old tie for last can't always be settled; from 148 they are.
- **A moment redrawn** (the TV redraws often in some themes) used to start again from nothing; the Rewind It! and Mastermind It! moments now keep their place.
- **Comic Book:** its city skyline sits in front of the lower part of the screen; the wheel is a little smaller there.
- **The Rewind It! wheel on 16:9 TVs** (found by the Audience View audit, fixed before release): with a long team name the cowboy hat above "Rewind It!" was cut off under the score cards — the screen's bottom space was counted twice, so its shrink-to-fit thought everything fitted. Now it shrinks properly, the hat is a little smaller and a long team name stops at two lines. The audit's two wheel scenes had been opening a square the test game marks as played (so they were checking the board); they now open the wheel and fail if it isn't up, and the audit now also flags a build-148 picture hidden under the score cards or the header.
- **The wheel and the themes' scenery** (seen on the 12-theme pictures after that fix, fixed before release): the space under the wheel screen is now set per theme, so the wheel, the result and the table sit above Comic Book's city, the Christmas houses, the Movies "Now Showing" sign, the Summer umbrella and the Diwali lamps. (Those themes' scenery also overlaps the bottom of the older Blind It! wager screen a little, as in build 147; that screen wasn't changed.)
- **The Rascala! lettering** (seen on the audit's pictures; you reported it at 13:21; fixed before release): the slanted closing quote of "Rascala! You are now a Naughty Pussy!" and the edge of the "!" in "The Rascala!" were cut off (slanted gold lettering loses whatever leans past its box). Both now have room for the slant and nothing else moves; checked at 6 sizes from a portrait screen to 4K with a very long team name.

D · Tests

- **New suite:** `fix148` 63/63 — Blind It! and Mastermind It! on the TV and the laptop with no old names left, Blind It!'s wager limits and scoring unchanged, the Setup card (default, choice, locked during a game), all five Rewind It! multipliers right and wrong (maximum loss 3×), equal chances (60,000 spins), one spin, one score, the result after closing and reopening, no passing, every chip off (and back with Blind It!), the TV picture for Wi-Fi TVs, Motion Off, the tablet's Spin / Show / Correct, The Rascala! scenarios 1–20 (one lowest, more wrong, fewer right, shared, three tied, negative and zero scores, Rewind It! / Blind It! / Mastermind It! losses, old games with and without proof, attribution, repeats, a backup restored, a reopened game, a shared title, counts, the Quizmaster left out, standings unchanged), the reveal (by itself after the winner, tagline, Continue, once), Historic Games, the game page pick, old names on old games, all 12 themes.
- **Older suites** updated where they checked the old names on screen.
- **Full run** of the release candidate (the two halves one after the other — side by side they overloaded this 2-processor machine, so that part of your "1,2,3" couldn't be used): **160 suites, 3,652 checks**. Browser half: 93 suites, 2,380/2,380, plus `roundnav`, which stopped on a mistake in its own test script (fixed: 28/28). Laptop-app half: 67 suites, 1,241/1,244 (`fix130a` passed on its automatic retry): `fix114` (2) still looked for the Final question's old lock message — test updated, 26/26; `fix119` (1) found the new Rascala line making a game page's first team row taller — fixed in the app, 12/12. The page then changed for the fixes in C (the Rascala line's place, the clipped Rewind It! hat, the wheel above the themes' scenery, the Rascala lettering); the 12 suites those changes touch ran again on the final page (md5 `d5564ae8…`): 298/298, all on the first try.
- **Audience View audit:** 306/306 screens clean on the final page, no page errors — all 34 Audience View screens (4 new: the Rewind It! wheel landed and spinning, The Rascala! alone and shared) at 9 sizes. It found the clipped Rewind It! hat (fixed, see C); its two wheel screens now really open the wheel, and it now also checks that a build-148 picture isn't hidden under the score cards or the header.
- **Speed, slow TV:** not measured in this build (your choice).
- **Stress phases** (no 60-minute soak and no 10-minute idle phase, your choice): 5 phases on the final laptop app, each on a fresh start: 22/22 — network (tablet offline 20 s, back by itself in 0.4 s; laptop page reloaded mid-game, 10 phones still signed in) 6/6 · load (10 phones wagering at once, 400 back-to-back requests: median 39 ms, slowest 119 ms) 4/4 · lock (a phone locked 60 s: same team, no re-pairing) 2/2 · rapid taps (the same command 3 times applied once, 30 quick tablet actions, tablet and laptop agree) 4/4 · health check (every connection green, offline and asleep phones and a missing tablet flagged, the game untouched) 6/6. Not run: the 10-minute idle phase and the 60-minute soak (your choice).
- **Reproducible build:** a fresh checkout of `main` (d09a80f) builds a page identical to the tested one (sha256 `45c4a2d9…`); the tablet APK from a fresh clone: built from a fresh clone of the repository at the final commit: page = tested page, versionCode 148, original key.
- **Android:** the APK carries the tested page (versionCode 148, original key). Before anything is published, the release workflow opens it on an Android 15 emulator, starts a game, closes the app and opens it again, and expects the saved game back; if that fails nothing is published. The result is in the Current Status and the release notes.
- **Code map:** modules 141 · changed functions 221 · hooks 278 · wrappers 328 · replacements 55 · issues 1 (new 0, known 1, fixed since baseline 0) · parse errors 0.

Build 147 addendum (22–23 October 2026)

What you asked for: at 16:49, premium cinematic transitions for the Halloween and Wizard themes on the existing animation engine, with before-and-after videos (made and sent: question reveal, correct, wrong, Daily Double, back to the board and winner for each). At 17:23 and 17:27 the brief that superseded it: upgrade all 12 themes to a premium television game-show standard — Default Mind It! first, the Comic Book redesigned in ink black, crimson, hero gold, paper cream and steel blue (no pink), each theme its own personality on one shared engine, 17 moments, frequent ones fast and the big ones cinematic, 60 fps transforms and opacity, offline, nothing blocking the game, Full / Subtle / Off and Reduce Motion, no new sounds, no new before-and-after galleries, released automatically. At 18:23: "all approvals are given, just ensure everything is working properly, with amazing animations and transitions".

Same version: v1.69 · versionCode 147 · `APP_BUILD=147`. One new page module `app146.js` (`build.py` order `…,145,146,6`) and `style.css` build-147 blocks. Laptop app and Galaxy Tab code unchanged.

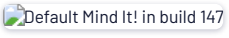
A · The 17 moments, and where they come from

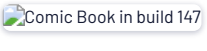
Moment	How it's done in build 147
1 Main menu · 17 back to the main menu	the theme's title moment (`t`) on the TV when the game returns to the title; Default's studio look and LED strips
2 Starting a new game	`t` on the intro (the wordmark and the show's subtitle)
3 Round introduction	`r` on each category card of the round intro
4 Team highlighting · 9 passing between teams	build 145's glide of the Now Playing ring (kept); the theme's colours on the ring
5 Selecting a tile · 6 question reveal	the square still grows into the card (build 145); the theme's reveal (`q`) as it lands
7 Correct · 8 incorrect	`c` on the team's score card; `w` on the card (the theme's own shake and colour)
10 Score up / down	build 145's count-up and delta (kept); `lead` when a team takes first place
11 Daily Double	the full-screen intro in the theme's colours plus `dd`
12 Back to the board	`b` from the square just played (Full)
13 Final intro · 14 final answer	`f` on the Final wager screen, `fa` on the answer
15 Standings · 16 winner	`s` after the podium rises, `v` on the winner

B · The themes

Theme	Colours (build 147)	Its moments
Default Mind It! (flagship)	midnight navy and royal blue, Mind It! gold; an LED wall behind the board, LED strips top and bottom that breathe; squares with a gold hairline; a layered question card with a gold inner frame; refined team panels	spotlight beams and a gold sweep for the title, round intros, the Final and the winner; a gold line traced round each new question; a gold light ring on a correct answer (with the gold sparks); gold light when a team takes the lead; beams on the Daily Double and the standings; confetti for the winner
Comic Book (redesigned)	ink black #111720, crimson #B92332, hero gold #E9B949, paper cream #F2E9D7, steel blue #315B83; crimson halftone squares with gold values, ink headers, a cream card with a crimson panel shadow; city, heroes and bursts redrawn in the palette; no pink	an ink panel wipes off the question; POW! on the scoring team; MISS! and a crimson-edged card on a wrong answer; page-turn category cards and board; a KA-POW Daily Double with speed lines; a hero-rays winner with the name stamped in
Christmas	forest green, deep red, warm gold	snowfall on the title, Daily Double and winner; frosted-glass question reveal and wrong answer; gold stars on a correct answer; a gold frame drawn for the Final
Halloween	as before	the preview's six (candle-light question with fog, jack-o'-lantern correct, the ghost on a wrong answer, lightning Daily Double, bats from the square, the moon and bat swarm for the winner) plus bats on the title, fog on the round intro, lightning and fog on the Final, embers for the standings
Ramadan / Eid	midnight blue, emerald squares, gold	eight-point stars drawn in gold (title, round intro, question corners, Daily Double, Final, winner); rising lantern light
Diwali	burgundy (was magenta), warm orange, gold	rangoli drawn in light; diya glow; golden trails; gentle fireworks for the Daily Double, the Final and the winner
Summer	deeper ocean blue, turquoise, sunshine	a wave wipe on the question; bubbles and ripples; a sunshine Daily Double; a sunset winner
Bollywood	burgundy, black, antique gold (the Star of the Show and the moment cards unchanged)	theatre spotlights; marquee lights round each category card; camera flashes on a correct answer; an awards-night winner
Sports	navy and championship gold (the pitch colours stay)	floodlights; broadcast band wipes; a scoreboard flip on a correct answer; gold confetti for the standings and winner
Wizard	as before	the preview's six (rune frame, wand-light, singe and smoke, the wax seal — now with a portal of runes behind the title — golden motes, the phoenix) plus spell-book page-turn category cards, a portal for the Final, motes for the standings
Gaming	black and midnight purple with cyan and violet (was mint green)	one clean scan line; a single small glitch on the title, category card and wrong answer; pixel bursts; a hologram grid for the Daily Double; hologram rings for the winner; no flashing
Movies	burgundy, black, warm gold	curtains opening on the title, the question and the Final; letterbox bars on the round intro; film strips across the board; spotlights; an awards-style winner

In the laptop app (the real app, Present mode, 1366×768; board, question, correct, back to the board, wrong, Daily Double, Final, winner):





C · Found while building

- **The speed check** found the first page much heavier on a slow Wi-Fi TV than build 146 (see Speed); fixed by the speed pass before the second full run.
- **The Wi-Fi TV stops and replays every effect when a new one starts;** the first wiring then left a moment's pieces standing still, or drew the whole moment again for every picture. The moments now run outside that list and their layers are put back after each picture; one that the new picture no longer has goes at once.

- **A TV redraw** (`renderStage` empties the TV) first meant drawing each running moment again; the layers are now kept with the TV's state and put back as they are, also when the TV's element is made new.
- **Comic Book:** the first redesign left a pinkish tint on a wrong answer and a large KA-POW over the Final's category; the logo colours and the clue value still used the old yellow and pink (now gold and crimson).
- **Movies:** the film wipe first showed one giant frame; now two strips of frames run across the board with the projector's light between them.
- **Subtle** moved the Comic Book category card; it now only glows, like every theme.

D · Tests

- **New suite:** `fix147` 49/49 — every theme × every moment on the Present-mode TV, each moment's layers drawn and cleared afterwards with nothing piling up across the 12 themes, a redraw keeping a moment as it is, nothing catching a tap, Subtle gentle light only, Off and Reduce Motion nothing, a lightened Wi-Fi TV nothing, the effects travelling to Wi-Fi TVs as plain data, the same game with Motion Full and Off, the Comic Book and Default palettes, switching through all 12 themes.
- **Frame-by-frame look:** every theme's 12 moments recorded on a frozen clock in the sandbox's browser (contact sheets in `work/b147/`).
- **Full run** on the final page (md5 `7129f22d…`), with the build-147 tablet app inside the laptop app: two full runs. **First** (on the page before the speed pass): webpage 92 suites 2,344/2,351 — `fix145` counted the Default theme's new LED-strip animations with its own fades (the test now ignores pseudo-element animations), `zones` found the new 1-pixel line under a score card's team name pushed it into the score with 8 teams (now an inset shadow, no layout change) and `theme` measured the question card's new top light too bright for its 4.5:1 text rule (softened); those suites and five related ones then passed (563/563); laptop app 67 suites 1,244/1,244. **Second, on the final page** (md5 `7129f22d…` , after the speed pass): webpage 92 suites 2,360/2,360 (`achphx` passed on its automatic retry); laptop app 67 suites 1,240/1,244 — `fix130b` passed on its automatic retry and `fix146d` missed four checks at one step, twice: it clicked the Go-to-setup confirm box a fixed 0.3 s after Setup and under the full run's load the box came later; on its own it passed 28/28, the test now waits for the box, and it passed 28/28 again. With that re-run: 159 suites, 3,604/3,604 checks on the final page.
- **Audience View audit:** 271/271 on the final page — every Audience View screen at 9 sizes.
- **Speed, slow TV 4×:** the laptop app hosting 8 teams and a Wi-Fi Audience Screen at 1920×1080, both slowed 4× (the TV drawn without a graphics card). The first build-147 page made that TV much choppier than build 146 (median of three runs each: the question reveal 53 → 13 frames a second, the standings 61 → 17, the winner 20 → 10). A speed pass followed: the spotlight beams became pictures that turn (no clip-path), the Default question's traced gold line a light running round the card's edges, the moments' layers leave by timer instead of an animation of their own, and on a Wi-Fi TV and the Galaxy Tab half the particles and every other beam; on a Wi-Fi TV a moment that is still the same keeps running between pictures instead of being drawn again (it had been drawn again with every picture during the standings: 1.6 s of work at 4×), and the Default studio drops the LED-wall dots and holds its LED strips still. Final page, Default (frames a second; build 146 median in brackets): timer 59 (59), idle board 55 (55), back to the board 62 (62), winner 22 (20), correct 52 (65), standings 50 (61), board appears 42 (64), wrong answer 29 (45), question reveal 19 (53). The question reveal and the wrong answer are still choppier than in 146 on that TV. Comic Book and Diwali were already 2–8 frames a second on it in build 146, even at rest (their drawn backdrops), and stay about the same (Comic 2–4, Diwali 4–7); a TV that slow lightens itself, as in 146. The Quizmaster page's longest task: 0.74 s (146: 0.82–1.03 s).
- **Stress phases and 60-minute soak:** 38/38 (with `STRESS_DIAG=1`), on the final build: idle 2/2 (no stall in the laptop app's main process or page), network 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6, and the 60-minute soak 14/14 — three whole games with 8 teams, 10 phones, the tablet remote and a TV in a 54-minute real-time quiz (64 questions, 305 tablet actions), Final wagers from the phones, a tiebreaker played from the tablet, the Night Recap and recap pictures; no false disconnects and no script errors; laptop server memory 236 → 240 MB, Quizmaster page 18 → 18 MB.
- **Reproducible build:** a fresh checkout of GitHub main (`53e3510`) builds the same page byte for byte (`e32f8f50…`); the tablet APK from a fresh clone: built from a fresh clone of the repository at the final commit: page = tested page, versionCode 147, original key.
- **Android:** the APK carries the tested page (versionCode 147, original key). Before anything is published, the release workflow opens it on an Android 15 emulator, starts a game, closes the app and opens it again, and expects the saved game back; if that fails nothing is published. The result is in the Current Status and the release notes.
- **Code map:** modules 140 · changed functions 215 · hooks 270 · wrappers 312 · replacements 55 · issues 1 (new 0, known 1, fixed since baseline 0) · parse errors 0.

Build 146 addendum (22 October 2026)

What you asked for: at 09:47, "1,2,3,4,5,7,13" from thirteen ideas, with item 13 "more on food, european football, indian cricket and geography, dont make the questions too easy and ensure the questions are jeopardy style"; at 10:00 the 🎲 names "inspired from the quizbiz names i have given" plus "The Avengers / The Quiztice League / Quiz Me From The Phone / Quiz On My Face / Question Tarantino / Quiztiano Ronaldo", matching the design language; at 10:18 a full design audit with your Mind It! UI/UX Audit skill instead of item 5 (each major screen beside two redesigns, playable previews and MP4s, no code change before approval); 10:50 "ok approved" (the 🎲); 11:07 "All Approved, and do all the premium options, make sure the design and style is maintained across the app"; 11:09 "everything is approved, the before and after pics are not needed"; 11:35 "ok run that test during your tests today" (switching the theme mid-quiz); 11:55 "can we make the theme switcher more easily accessible for the host, maybe in the host controls and he just has to choose from a dropdown list making it super easy?"; 14:13 the theme screen, Back to the Board and the Drive backup, and 14:19 "is it possible to add 147 to 146, just reduce the tests for it" (part 2); 15:10 "1,2,4,5,8 add those to 146, regular tests are fine" and 15:12 "keep soak" (part 3)

Same version: v1.69 · versionCode 146 · `APP_BUILD=146` . New page modules `app141.js` to `app145.js` (`build.py` order `…,140,141,142,143,144,145,6` ; `app144.js` is part 2, `app145.js` part 3). **The laptop app (`main.js`) and the Galaxy Tab (`MindItHost.java`) change too. A major change** (phones, servers and connections): the full run, the Audience View audit, the stress phases and the 60-minute soak were all run.

A · Your numbered items

- **1 · The 10-second blip** in build 145's idle stress phase: not reproduced in 8 idle runs (quiet, with a CPU load, with an ffmpeg load) — no stall over 250 ms in the laptop app's main process or its page. Most likely the build machine itself. The stress test now carries monitors (main-process lag, page lag, phone failures, tablet status) so any blip says where it came from; this build's run: not seen again in the stress run either: the idle phase with the new monitors (main-process stalls over 250 ms, Quizmaster page stalls) found none in 2 × 5 minutes (question and board), 2/2.
- **2 · A slow TV** (the laptop app hosting 8 teams and a Wi-Fi Audience Screen at 1920×1080, both slowed 4× — like a cheap smart-TV browser or an old laptop): measured twice (the TV's own lightening held off, and as shipped), nine moments each. The TV drew 36–65 frames a second while moments played (build 145 on the same check: 56–72), 57–60 on the idle board and with the timer running; the winner screen's falling dots 17–25 (build 145: 8). The Quizmaster page's longest single task was 0.13–1.0 s (build 145: 0.16–0.96 s). As shipped the TV did not need to lighten itself at 4× slower.
- **3 · Phones follow Motion** (Full / Subtle / Off): the laptop's and the tablet's phone servers pass the Quizmaster's setting on; a phone set to Reduce Motion still shows none.
- **4 · Wi-Fi Audience Screens get the motion:** each picture sent to a Wi-Fi TV carries the effects running on the TV page with their start times, and the TV plays them on its own copy — a slow Wi-Fi only delays an effect, it never restarts it. A TV set to Reduce Motion shows none.
- **5 → the design audit** (published to you as a private page; below).
- **7 · 🎲 team names:** 🎲 **Fun Names** in Quick Setup gives every team a different fun name; in Setup → 2 · Teams a 🎲 beside each name box gives that team a new one no other team has. 116 names, yours first (Question Tarantino, Quiztiano Ronaldo, The Quiztice League…).
- **13 · Ready-made quizzes** (sent in the chat as Starter Template files, not part of the app): "Spice, Sixes & Strikers" and "Bytes, Rhymes & Dynasties" (with your Halloween round). Every fact was checked by two independent checks and both files were loaded into the app to make sure every clue comes in.

B · The design audit, approved in full

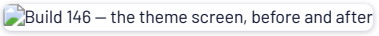
- **F1** a question's picture fits the space left in its card (measured: on a 16:9 TV the bottom 29% was cut off) · **F2** team names, the turn line, places and tags bigger on the TV (1080p names 23 → 30 px, 8 teams 20 → 25 px) · **F3** the revealed answer is the hero (82 → 150 px) · **F4** one team card on the phone · **F5** "Your Turn to Pick!" in words · **F6** calmer Daily Double plates · **F7** Correct and Wrong the same size · **F8** "Category" over the Final category · **F9** standings · **F10** Night Recap larger · **F11** Setup's right-hand column scrolls above the Go-Live bar · **F12** the tablet hides +30 Sec on the board · **F13** the phone's wager chips on one line, full-width keypad · **F14** "Next to Pick:" · **F15** score and place under the buzzer.
- **Motion: M1** the answer reveal (the answer rises and a line sweeps under it) · **M2** the turn banner slides in with one warm pulse · **M3** a picture rises into its frame.
- **The premium look** on every theme — glossy tiles, headers rimmed in the theme's accent, depth on the score cards, a glow on the question and the answer, picture frames, a spotlight instead of Daily Double rays — built from each theme's own colours, so Christmas stays Christmas and Bollywood stays Bollywood; **Comic Book** keeps its flat ink, **Gaming** its square neon edges, **Wizard** its parchment. The same depth on the laptop, the tablet remote and the phones; colour-blind mode untouched.

C · 🎨 Theme in the top bar (11:55)

- 🎨 **Theme ▾** next to Sound on the Quizmaster's bar, in the bar's own dropdown style: the 12 themes, each with a swatch of its colours, the current one ticked. One pick changes the look at once — the laptop, the Audience window and the Wi-Fi TVs — with a short note; the question, scores, turn and timer carry on. The Setup theme card is unchanged.

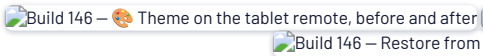
C2 · Part 2, added at 14:19 ("is it possible to add 147 to 146, just reduce the tests for it")

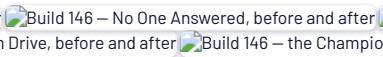
- **The theme screen** (14:13: "it looks tacky and not at all premium"): larger cards, each a clean preview of the TV in that theme — a tidy board, three score cards whose names fit, the theme's own backdrop at the sides — in the premium look from its own colours; under it the name, **In Use** on the current one and **Sounds**. Tapping a card opens the big preview as before. The Default card shows the default colours even while another theme is in use (it used to take the colours of the theme in use).
- **Back to the Board Automatically:** on by default and **3 seconds** (was 7); switched on once on every device with this build (it can still be turned off); the countdown says "3 s" (Title Case had made it "3 S"). No One Answered still waits 7 seconds (a separate countdown).
- **Google Drive backup, checked end to end:** (a) the waiting backup — about 7 MB for your data — was kept in the laptop's small settings storage, which it didn't fit, so the Drive status stopped being remembered; (b) it was built on the same beat that draws the round summary or winner screen, and standings → winner made two uploads; (c) the daily backup uploaded even when nothing had changed; (d) a fixed 45-second limit meant a big backup on a slow venue upload never finished and was re-sent every 3 minutes, also mid-question. Now the backup waits in memory, is built a few seconds later (the end-of-game screens share one upload), is skipped when nothing changed (Back Up Now always sends), gets 45 s + 15 s per MB (up to 10 minutes), waits 3 → 30 minutes between failures and never starts during a question; the status shows the size, the time and the next try. Same backup file, same Google script.
- **Tests for part 2:** a targeted run of 29 suites on the part-2 page (the ones touching the theme picker, Back to the Board, the Drive backup, touch sizes and the phone remote's board, plus general ones), 15:08–15:35: 913/915 — `flow` still expected 7 seconds (the test was then updated; 11/11 since), `fix146g` passed on its automatic retry (its "back on the board, it goes up" step; it now explains any miss, and has passed every run since); then the full run on the final page.



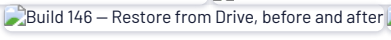
C3 · Part 3, added at 15:10 ("1,2,4,5,8 add those to 146, regular tests are fine"; 15:12 "keep soak")

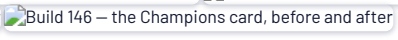
- **1 · 🎨 Theme on the tablet remote:** in the remote's top bar (after Connected, before Team Codes), the same 12 themes with their colours, the one on the TV ticked; only on the tablet in control. A pick sends a new link command, `theme`, which the laptop applies at once (acknowledged once; an unknown theme is refused) — the laptop, the Audience window and the Wi-Fi TVs change; the question, scores and turn don't.
- **2 · No One Answered closes after 3 seconds** (was 7), like Back to the Board; its question and the Board & Flow note say so.
- **4 · A slow Wi-Fi TV lightens itself:** a Wi-Fi Audience Screen that draws under 24 frames a second for two 4-second spells (after its first 10 seconds, only while visible and connected) switches to lighter motion — only the gentle effects, at the gentle level, and no heavy glows or shadows (the Now Playing ring stays). It tells the laptop or the tablet (new `lite` message in `main.js` and `MindItHost.java`); the laptop names that TV once and marks it 🪶 **Lighter motion** in 📺 Connect Audience Screen. Reloading the TV's page tries full motion again. The HDMI Audience window is never changed.
- **5 · Restore from Drive:** ☁️ **Restore from Drive** next to Back Up Now lists the newest backups in the Drive folder; ⭮ Restore downloads one and restores it the usual way (question first, Safety Backup). It needs **version 2 of the Google script** (`handover/extras/MindIt_GoogleDrive_Backup.gs`: answers `"v":2`; `action:"list"` returns the newest 7 files of the backup folder, `action:"get"` returns one of them and nothing outside that folder; any other action is refused). With the older script the app shows the 2-minute update steps and asks it for nothing else — the old script would have saved such a request as a file. `main.js` passes `list` / `get` through with a 60-second limit, the download a longer one by size.
- **8 · Champions card:** on the winner screen a **Champions Card** tile (beside Save Picture and Play Again) shows a card on the TV — the winning team, its players' pictures and stats (final score, correct answers, hot streak and biggest win when there are any; joint champions side by side) — with an entrance; it goes away when the winner screen does. 💾 **Save Champions Picture** saves a 1920×1080 PNG.
- **Tests for part 3:** the part-3 checks first ran on their own on the part-3 page (fix146 66/66, fix146t 13/13, flow 11/11, fix146d 26/28, fix146g 19/21). Every miss was in the new tests: the sandbox's two test TVs lightened themselves (1–2 frames a second) before the simulated one, and three checks compared text without the app's Title Case capitals. Fixed in the tests (the TVs' own switch held off where each step is checked, capitals ignored). In the full run `fix146` missed one check (the Champions Card tile's text includes its icon — a test fix that landed after the run started); run again on its own: 67/67.









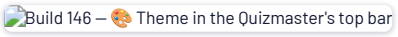


D · Found while building

- **The audit's phone pictures were taken without live game data**, so they missed the strip phones already show during a game (team · place · score · timer). Following them, the first draft added a second card that repeated it. Fixed before the full run: in a game that strip *is* the card; before the game the card shows the team.
- **Two of the twelve themes (Bollywood and Sports) were missing from the first premium-look check** (it listed ten). Both were checked on the board, question, answer and Daily Double, and keep their own style; the test now covers all twelve.
- **The Wizard and Comic Book themes** showed the reminder above the answer pale on the parchment/paper card; it now uses the card's ink.
- **8 teams:** the bigger TV names first lost to an older, stronger rule; fixed so 8 names fit at 25 px.
- **Part 3, found while building:** (a) the Champions card first became the laptop's big main button on the winner screen and took over Space — moved to a tile beside Save Picture and Play Again so Space and the layout stay as before; (b) in the first laptop-app run the sandbox's two test TVs (1080p drawn without a graphics card) measured 1–2 frames a second and lightened themselves — the switch working for real; the tests that check effects one by one now hold it off, and the slow-TV measurement is taken both ways; (c) the Drive test's big player photo dropped out of later backups, so its "slow Google with 6.5 MB" step really sent 78 KB — the photo is now a player's photo and the step requires over 6 MB; (d) "1 frames a second" → "1 frame a second"; (e) two older mentions of Back to the Board's 7 seconds (cheat sheet, Quick Start, the controls table) — now 3.
- **The full run found five more** (all from the new look, all fixed, then the whole browser half was run again): the premium score-card depth had covered your Now Playing white ring and the On Fire orange ring; 8-team side-column names touched the scores; a long answer no longer shrank to fit; the bigger TV labels spilled on a 1024×768 laptop and an upright tablet. The laptop-app half then found two: the 🎲 buttons were 36–40 px on a touch screen (now 48, like every Setup control) and a phone used as the remote got the tablet's bigger board (values cut off; phones keep their compact board).

E · Tests

- **New suites:** `fix146` 67/67 (the full run's one miss was a test changed after the run had started; run again on its own) (webpage), `fix146t` 13/13 (the Galaxy Tab's own server on the JVM) `fix146d` 28/28 (after one check learned the app's Title Case) (laptop app with a real phone, a Wi-Fi Audience Screen, the Audience window and, for part 3, a real tablet remote; includes your mid-quiz theme test — Setup → Theme → Halloween → Return to game, and the new 🎨 menu) and `fix146g` 21/21 (laptop app with a stand-in for your Google script: the backup and, for part 3, Restore from Drive). `bz` 23/23 now reads BUZZ IN from the button. `run_all.py` now has 158 suites.
- **Full run** on the final page (md5 `02f79838…`), with the build-146 tablet app inside the laptop app: the full run in two halves on the shipping page (md5 `02f79838…`): webpage 91 suites, 2,304/2,305 — `fix146` missed one check whose test had been changed after the run started (the Champions Card tile's text includes its icon); run again on its own 67/67 — and laptop app 67 suites, 1,240/1,244 — `fix138pd` still expected the old 7-second returns (No One Answered and Back to the Board now 3 s) and `fix146d` compared the slow-TV tag without the app's Title Case capitals; both tests were updated (no app change) and run again: 9/9 and 28/28. With those re-runs: 158 suites, 3,549/3,549 checks.
- **Audience View audit** (`avaudit` , every TV screen at 9 sizes): 271/271 — every Audience View screen at 9 sizes (TV, laptop, tablet, phone), on the shipping page.
- **Stress phases and 60-minute soak** (with the new monitors): 35/35 (with `STRESS_DIAG=1`): idle 2/2 (no stall in the laptop app's main process or page), network 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6, and the 60-minute soak 11/11 — two whole games with 8 teams, 10 phones, the tablet remote and a TV, 59 questions and 268 tablet actions, Final wagers from the phones, a sudden-death tiebreaker from the tablet, the recap pictures; no false disconnects and no script errors; laptop server memory 229 → 240 MB, Quizmaster page 16 → 18 MB.
- **Reproducible build:** a fresh checkout of GitHub main (`6cdf7fb`) builds the same page byte for byte (`cce0cdca…`); the tablet APK from a fresh clone: built again from a fresh clone of GitHub main: the same file (sha-256 `dd298a81…`), versionCode 146, original key.
- **Code map:** modules 139 · changed functions 214 · hooks 269 · wrappers 309 · replacements 55 · issues 1 (new 0, known 1, fixed since baseline 0) · parse errors 0.



Build 145 addendum (21 October 2026)

What you asked for: at 15:27, "is there any animations or transitions that we can improve in the app?" → five ideas → 15:30 "can you show me before and after for all of them?" → 15:36 "if possible show it to me in video to understand the changes better" → a 77-second before/after video (each item at real speed, then slow motion ×4) → **16:00:** "ok do them all, consider this a major upgrade so do all the tests and checks" → 16:21 "any more animation changes? give me as much as possible so that i dont have to keep asking" → eighteen more → **16:38:** "do them all and include it in build 145" → 16:39 "give me the before and after video, but go ahead without the approval" → 16:48 "motion wise let it be at top fidelity". Estimated time given: about 21:15, then about 23:30 Dubai (with the eighteen more).

Same version: v1.69 · versionCode 145 · `APP_BUILD=145` . New page modules `app139.js` and `app140.js` (`build.py` order `…,138,139,140,6`). **A major change by your choice:** the full run, the Audience View audit, the stress phases and the 60-minute soak were all run.

A · What moves now (`app139.js`)

- **1 · Tile → question:** the question card grows out of the square the team picked (0.43 s), its text fading in as it lands. Only from the board; a Daily Double keeps its own splash.
- **2 · Back to the board:** the board fades in (0.32 s) and the square just played shows its value once more and dims out where it was.
- **3 · Other screen changes** (round summary, standings, awards, winner, How to Play, Intermission, Final screens and the rest): a 0.34 s fade-in instead of a hard cut. Question, answer and Daily Double keep their own entrances.
- **4 · A team moving up a place:** its score card glows gold once and its place badge flips in — after the "+ amount" has floated away, so the two don't overlap.
- **5 · Pop-ups** on the laptop and tablet fade and scale in; one pop-up replacing another straight away doesn't animate again.
- **Never in the way:** every effect is kept with its start time and put back at the right moment when the TV redraws (it redraws when the timer starts, a phone checks in or a score counts), so none is cut short or restarted. The device's **Reduced Motion** setting turns all of it off. Timers, scores, turns, saves and phones are untouched; nothing new is saved.

A2 · Eighteen more (`app140.js` , same effect engine)

- **TV:** TIME'S UP stamps onto the card when the clock runs out · a wrong answer shakes the card once with a red edge · a correct answer bursts gold sparks from the team's card · the Now Playing light glides to the next team on a pass or a wrong answer · a picture question's picture pushes in slowly · No One Answered greys the card and slides the answer up · standings: 4th and below, 3rd, 2nd, a pause, then 1st — blocks rise and scores count · Night Recap rows in turn · a gold spotlight sweeps across the winner screen and the name rises · Intermission's countdown flips its digits.
- **Phones:** a new screen slides in · ✓ LOCKED IN on a sent wager and on a locked Final answer · a gold glow round the phone whose team picks next · a ripple where BUZZ IN is pressed and YOU'RE FIRST! on the winning phone.
- **Tablet remote, laptop, Setup:** the remote's screen fades as the laptop moves on and its buttons press down · Correct / Wrong flash the laptop's scoring card green / red; a card that wasn't there slides in · a Setup section fades in when you switch to it.
- **Motion: Full / Subtle / Off** (Setup → 5 · Audio & Presentation; Full by default — "top fidelity"): Subtle keeps only gentle fades; Off — or the device's Reduced Motion — stills everything, the older animations too. The laptop's TVs and the tablet remote follow it; phones follow their own Reduced Motion (telling them would need a change to the phone server).
- **Wi-Fi Audience Screens** get a copy of the TV page, not its animations, so they show the same screens without the motion (the short-lived effect pieces are left out of the copy).

A3 · Google Drive: the newest 7 (17:08 → 17:10 "1")

- The backup script template keeps 7 (`KEEP = 7` , was 40); the app's help text says so. Not tested against real Google (the Drive backup never has been; it runs in the owner's own Google account). The owner changes the one line in his live script.

B · Found while building

- **The draft measured the board before the TV had finished its layout**, so on the question screen the card started about 10% too high (visible in the before/after video, which was made from that draft). The effects now run after the whole TV layout (zones, fitting, scores) and measure against the whole screen; the new test checks the start lines up with the square on a 16:9 TV and on an upright TV with 8 teams.
- **Existing glitch found while filming:** on the zoned TV the timer ring's own "Time's up" was drawn at the countdown size and spilled out of the ring (since the zoned layout). Fixed and checked at 960×540, 1920×1080 and 800×1280: it sits inside the ring, centred.
- **Title Case** turned the phone's "LOCKED IN" into "Locked in": the stamps now carry `tc-skip` , like the app's other fixed texts.
- **The laptop's Wrong flash** was recorded after the answer had already been drawn, so it only showed on the next redraw; it is now recorded before.
- **A redraw cut effects short** in the first draft (the TV redraws several times in the first second). Fixed before the video: effects are re-applied at the right moment on every redraw (checked by the new tests).

C · Tests

- **New suites:** `fix145` 29/29 and `fix145b` 35/35 (webpage), `fix145d` 9/9 (laptop app with the real Audience window) and `fix145e` 14/14 (laptop app, two team phones and a tablet remote). `run_all.py` now has 154 suites.

- **Full run** on the final page (md5 `e5f854d4…`), with the build-145 tablet app inside the laptop app: full run on the shipping page (md5 `e5f854d4…`, with the build-145 tablet app inside the laptop app): 154 suites, 3,408/3,408 checks (109 min); fix127d passed on its automatic retry. Investigated: its 'old home-screen address' check opened a second remote page of the same iPhone while the first was still open; the laptop keeps one connection per device, so the second page connected and was dropped 4–5 s later — the same on build 144 (checked) — and the check at 4.5 s passed or failed by timing. The test now closes the first page first (as reopening from the home-screen icon does): 3/3 since. No app change.
- **Audience View audit** (`avaudit`, every TV screen at 9 sizes): 270 screens at 9 sizes, 0 issues (271/271)
- **Stress phases and 60-minute soak:** run (a major change by the owner's choice). Stress phases: net 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6; idle 1/2 on the first run — 135 s into the 5-minute idle board, the tablet and all 10 phones saw the laptop as unstable for about 10 s at the same moment and reconnected by themselves (no full-screen cover, nothing lost); idle re-run on a fresh laptop app: 2/2. 60-minute soak (laptop app, 8 teams, 10 phone pages, a tablet remote; whole games, a forced tie, the Night Recap, the Quizmaster window hidden and shown every 10 min): 14/14, 0 false disconnects. Next due: build 150, or sooner for a major change
- **Reproducible build:** identical (tools/verify_build.py: a fresh clone of GitHub main 2df3392 builds the tested page byte for byte, sha256 38adc55c…); the tablet APK from a fresh clone: a fresh clone of HEAD builds the tested page and an APK carrying it, signed with the original key (5f4819e4…), apk_check.py OK.
- **Code map:** modules 134 · changed functions 208 · hooks 252 · wrappers 301 · replacements 51 · issues 0 (new 0, known 0, fixed since baseline 0) · parse errors 0.



The before/after video (each item at real speed, then slowed down) was sent in the chat: `Mind_It_build_145_more_animations_before_after.mp4`.

Build 144 addendum (21 October 2026)

What you asked for (21 Oct, 13:14): "Do 5,6" — ↔ Move a whole category, and the docs brought up to date — from the list offered at 13:09 ("any upgrade suggestions?"); "show me 3" (category colours on the TV board), then **13:18:** "3 is rejected, it makes it look tacky". Before/after picture of 5 sent at about 13:19; **13:20:** "approved". Also settled: "For 7 it stays on the square", "wizarding name ok", "Gaming theme stays", "Loading is fine". Estimated time given: about 17:30 Dubai.

Same version: v1.69 · versionCode 144 · `APP_BUILD=144`. New page module `app138.js` (`build.py` order `…,136,137,138,6`). This is a minor change under your rule of 20 Oct (no stress phases or soak; next due: build 146).

A · ↔ Move a whole category (`app138.js`)

- **Where:** Setup → 4 · Rounds & Questions → Review & Edit Clues, on the first row of each category, after 🔄 Replace and ↕ Reorder.
- **What it does:** every round is a fixed grid, so the category swaps places with the one picked — in another round, or in the same round (its column changes). It asks first ("Swap "A" and "B"?"; Back changes nothing).
- **What moves:** the category name (and any other category-level field) and each question's contents — answer, alternates, host note, pictures and video, Question Database link. **What stays:** the values, the Daily Double squares and the played marks (your rule, 21 Oct: a Daily Double stays on its square).
- **Can't be picked:** the category itself, one with a different number of questions, one with a played question in a paused game (that category gets no ↔ Move either).
- ↩ **Undo move** in the toolbar of both rounds straight after; kept in memory only, never in the save. No change to the save format.



B · Docs brought up to date (item 6)

- Places where the docs and the code or repository disagreed: CLAUDE.md (git covered only part of the project; the Android build through the retired `build.sh`; the passwords "in build.sh"), CURRENT_STATUS §4/§8 (build 110, the MindIt-Downloads public/private question), MASTER_CONTEXT §2/§16.3/§17/§18 (build 110, the build-108 source zip), ARCHITECTURE header/§2/§9/§13 (the build-108 source zip, `build.sh`), and the Daily Double question that MASTER_CONTEXT called unanswered while GAME_RULES §25-I had it settled on 10 Oct. All rewritten to match the code; no code changed for this.

C · Tests

- **New suite:** `fix144` (webpage) 31/31. `run_all.py` now has 150 suites.
- **Full run** on the final page (md5 `1c8b2474…`), with the build-144 tablet app inside the laptop app: full run on the shipping page (md5 `1c8b2474…`, with the build-144 tablet app inside the laptop app): 150 suites, 3,321/3,321 checks (102 + 3 min). In the first run 147 suites passed first time and 3 couldn't run their comparisons with older builds because the new workspace was missing two things — the repository's git history (fix130c, fix133m: build-129/132 code) and the older published pages (fix140s: the build-139 page; run_all.py drops the web proxy, so it can't download them). Both were fetched, the three suites re-run on the same page and passed (fix140s 36/36, fix130c 7/7, fix133m 32/32), and `setup_workspace.sh` now fetches both. No retries.
- **Stress phases and 60-minute soak:** not run — a minor change under the owner's rule of 20 Oct (stress phases and the 60-minute soak only for a major change or every 5th build). Last run: build 141; next due: build 146
- **Reproducible build:** identical (tools/verify_build.py: a fresh clone of GitHub main ef86314 builds the tested page byte for byte, sha256 0fd85b27…); the tablet APK from a fresh clone: a fresh clone of HEAD builds the tested page and an APK carrying it, signed with the original key (5f4819e4…), apk_check.py OK.
- **Code map:** modules 132 · changed functions 206 · hooks 244 · wrappers 298 · replacements 51 · issues 0 (new 0, known 0, fixed since baseline 0) · parse errors 0.

D · Problems found while building

- **The Move dialog's cards** sat 4 + 2 per round at first (uneven); they now sit 3 + 3.

Build 143 addendum (21 October 2026)

What you asked for (21 Oct, 09:28): "7" — the tablet offers its own update — from the list offered at 09:26 ("any more upgrades or improvements?"). Before/after picture sent at about 09:37; **09:39:** "approved". Estimated time given: about 2:30 pm, then about 12:45 pm Dubai.

Same version: v1.69 · versionCode 143 · `APP_BUILD=143`. New page module `app137.js` (`build.py` order `…,135,136,137,6`). This is a minor change under your rule of 20 Oct (no stress phases or soak; next due: build 146).

A · The Galaxy Tab app (Java)

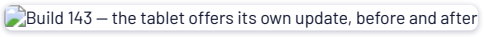
- `MainActivity.Bridge.appBuild()` → the app's versionCode.
- `MainActivity.Bridge.openTabletAppPage(host)` → `Intent.ACTION_VIEW` on `http://<host>/tablet-app`. The host must be a local IPv4 address (10.x, 172.16–31.x, 192.168.x, 169.254.x) with an optional port; the path is fixed. Anything else is refused (`false`).
- **Checked:** the APK compiles with both methods (present in `classes.dex`); the address rule run in Java on ten cases: `192.168.1.23:8765`, `10.0.0.5`, `172.20.1.1:80`, `192.168.137.1:8765` accepted; `8.8.8.8`, `example.com`, `192.168.1.23:8765/evil`, `javascript:alert(1)`, `172.32.0.1`, `192.168.1.23:8765?x` refused. Opening Chrome from the app has not been tried on a real tablet.

B · The tablet's Main Menu (`android/assets/start.html`)

- When a laptop is found (`probe` → `FOUND[host]`), `checkUpd(host)` asks it once for `/api/tablet-app` (3 s limit). If the laptop's tablet app is newer than `MindItAndroid.appBuild()` , the **Update Available** card appears under the found card: "Galaxy Tab app build N is on · This tablet has build M. Keep the laptop and the tablet on the same build." **Update** → `openTabletAppPage(host)` ; if that is refused, a message gives the address to open in Chrome. Apps before build 143 (no `appBuild`) and quizzes hosted on another tablet get nothing.

C · The tablet remote (`app137.js`)

- Only in the Galaxy Tab app (`IS_LINK_REMOTE` with `MindItAndroid.openTabletAppPage`). Once connected (`live`), it asks the laptop once for `/api/tablet-app` ; a newer build gives the `.lr-upd` banner under the top bar with **Update** and **Later**, in place of app125's `.lr-build` "Different builds" line (that line comes back after Later). The build compared is the app's own (`appBuild()` , else `APP_BUILD`). **Later** is remembered for the app session (`sessionStorage`). A refused open puts the address in the banner itself (a remote waiting for control shows no toasts — found by the new test).
- **Checked:** `fix143d` 13/13.



D · Tests

- **New suite:** `fix143d` (laptop app + a tablet browser with a stand-in bridge) 13/13. `run_all.py` now has 149 suites.
- **Full run** on the final page (md5 `d8ecccf0…`), with the build-143 tablet app inside the laptop app: full run on the shipping page (md5 `d8ecccf0…`): 149 suites, 3,288/3,288 checks, all on the first try (104 min). (qc2 counts 270 checks this time — its random whole games vary the number of checks from run to run; 0 failures.)
- **Layout audit** (`avaudit`): 270 screens at 9 sizes, 0 issues on the final page
- **Stress phases and 60-minute soak:** not run — a minor change under the owner's rule of 20 Oct (stress phases and the 60-minute soak only for a major change or every 5th build). Last run: build 141 (14/14); next due: build 146
- **Reproducible build:** identical (SHA-256 c541e502…abaa, commit 1c6cced); the tablet APK from a fresh clone: yes (versionCode 143, v2 + v3, certificate 5f4819e4…; `appBuild` and `openTabletAppPage` in classes.dex; the APK built with the private build.sh from the same files has identical contents).
- **Code map:** modules 131 · changed functions 206 · hooks 244 · wrappers 297 · replacements 51 · issues 0 (new 0, known 0, fixed since baseline 0) · parse errors 0.

E · Problems found while building

- **The first version compared with the page's build**, so a tablet page and a laptop on the same build never showed the offer even when the app itself was older (seen in the after picture). It now compares with the app's own versionCode.
- **The tablet showed two warnings** when the builds differed: the old Different builds line and the new banner. The banner now replaces the line while it is shown.
- **The "can't open the browser" message** used the remote's toast, which a tablet waiting for control doesn't show. It is now written in the banner.

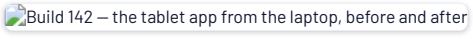
Build 142 addendum (20 October 2026)

What you asked for (20 Oct, 17:33): "1,4" from the eight ideas offered after build 141 — 1 · update the Galaxy Tab app from the laptop, 4 · an automatic Safety Backup at the end of every game. **17:35:** "all pics are approved for now". **17:41:** "i still have to enter my pin to connect laptop and tab, fix that too pls". Estimated time given: about 10:30 pm, then about 10:45 pm Dubai with the PIN fix.

Same version: v1.69 · versionCode 142 · `APP_BUILD=142` . New page module `app136.js` (`build.py` order `…,134,135,136,6`).

A · 1 · The Galaxy Tab app from the laptop

- **Where it comes from:** `desktop/build_desktop.sh win|mac [page] [tablet.apk]` copies the APK of the same build into `desktop/app/tablet/` with `tablet-app.json` (`build` = the versionCode in `android/AndroidManifest.xml` , `sha256`). The release EXE and Mac app were built after the APK, from it. Not in the repository (`.gitignore`).
- **The laptop app (`main.js`):** `GET /api/tablet-app` → `{ok, build, size, sha, path}` ; `GET /tablet-app` → a small page (title, "Build N · the tablet app that matches this laptop", one Download button, three steps); `GET|HEAD /tablet-app/Mind-It-Android-Galaxy-Tab.apk` → the file (`application/vnd.android.package-archive` , `Content-Disposition: attachment` , `Content-Length` , `nosniff`). Any device on the network may fetch them — it is the same public app as on the download page; nothing else is served under `/tablet-app/` . Without a bundled APK the details say `ok: false` , the page and the file answer 404 ("isn't included in this copy"), and the Quizmaster page offers nothing. Port 80 serves them too (the same handler).
- **The page (`app136.js`):** only in the laptop app (`LIVE.on` , not the Galaxy Tab's own page) it fetches `/api/tablet-app` once. 📲 **Connect Device** gets a line under the codes ("Galaxy Tab app build 142 is in this laptop app" · 📲 **Update the Tablet App**); the button opens 📲 **Update the Galaxy Tab App**: the QR code of `http://<Wi-Fi address>:<port>/tablet-app` , the address, three steps (scan; Download anyway; Update) and the Auto Blocker note. Connected Devices' **Different builds** warning (app124) gets the same button and says an iPad or iPhone should reload instead.
- **Checked:** `fix142d` 17/17 — the details match the bundled file and this laptop's build; the page and the APK from the Wi-Fi address (byte for byte), HEAD, three other addresses refused; the line, the panel (its code for the right address, three steps), the warning's button; a tablet browser opening the address and downloading the identical file. `fix142` — away from the laptop app (a browser, the Galaxy Tab's quiz page) nothing is offered and nothing is fetched.



B · The PIN between the laptop and the Galaxy Tab

- **The cause (the code is the source of truth; the guide said otherwise):** since build 128 a remote that pairs with the PIN gets its own key (`mindit.trust[hostId]`) and the laptop lets it in with that key alone (`trustCheck` , no PIN or game number). But the tablet's Main Menu (`android/assets/start.html` , `connectTo`) asked for the PIN whenever the laptop's game number (`SESSION` , new each time the laptop app starts) differed from the saved one — so every quiz night. The guide (5.1) said "after a laptop restart it reconnects by itself, without the PIN"; that was only true inside an open remote page.
- **The fix:** `connectTo` reads the laptop's `hostId` from `/api/link/info` ; if this tablet holds a key for it, it opens the remote page straight away (no PIN), and the remote page signs in with the key as before. If the laptop has forgotten the tablet (New PIN, Forget), the remote page says so ("isn't on the laptop's list of trusted devices any more"), removes the key, and the next Connect asks for the PIN. Laptops before build 130 (no `hostId`) still get the PIN.
- **Checked:** `fix142d` — pair with the PIN from the code; change the saved game number (as after a laptop restart); **Connect To Laptop → Connect** goes straight in (connected, listed on the laptop); without the key the PIN is asked.

C · 4 · A Safety Backup after every game

- **How:** a `finalizeGame` wrapper (outermost, after app41's Rehearsal rule): when this call records the game (`recordedThisGame()` false before, true after), 1.5 s later `safetySnapshot("After the game — <winners> won")` (app114; `exportFull()` , so it holds the game, its record in Historic Games, players and the Question Database).

Once per game id; never for a Rehearsal (never recorded). Same list and same limit (the last 10). The Safety backups card now says "after every game". On the laptop, the tablet's own quiz page and any browser.

- **Checked:** `fix142` 12/12 — one backup named after the winner, not again on the standings / winner screens, the record inside, a second game's own (joint winners both named), none for a Rehearsal, the card text; the Galaxy Tab's quiz page too. `fix142d` — in the laptop app.


Build 142 — Safety Backup after every game, before and after

D · Tests

- **New suites:** `fix142` (browser) 12/12; `fix142d` (laptop app + a tablet browser) 17/17. `run_all.py` now has 148 suites. `fix141` / `fix141d` accept build 141 or later.
- **Full run** on the final page (md5 `6d8b34cd…`), with the build-142 tablet app inside the laptop app: full run on the shipping page (md5 `6d8b34cd…`): 148 suites, 3,296/3,296 checks, achphx (Phoenix achievements) passed on its automatic retry — a one-off timing failure under load; run 4 more times on its own afterwards: 7/7 each (107 min).
- **Layout audit** (`avaudit`): 270 screens at 9 sizes, 0 issues on the final page
- **Stress phases:** not run (the same rule; last run: build 141, 24/24) · **60-minute soak:** not run — the owner's rule since 20 Oct, 17:56 ("lets limit the stress tests and 60 minute soaks to major changes or after every 5 builds"): build 142 is a minor change (new download addresses on the laptop, the tablet's Main Menu, one more kind of Safety Backup). Last run: build 141 (14/14); next due: build 146
- **Reproducible build:** the final page rebuilt from a fresh clone of GitHub `main` is identical: identical (SHA-256 04902d56…facd, commit 322e9d8); the tablet APK built from a fresh clone with `build_apk.sh`: yes (versionCode 142, v2 + v3, certificate 5f4819e4…; the APK built with the private build.sh from the same files has identical contents).
- **Code map:** modules 130 · changed functions 206 · hooks 243 · wrappers 297 · replacements 51 · issues 0 (new 0, known 0, fixed since baseline 0) · parse errors 0.

E · Problems found while building

- **The Different builds check in `fix142d`** first read the warning 1.2 s after drawing it, when the laptop's next device update had already redrawn the list without the made-up tablet. It now reads it at once (a test timing issue; the app was right).
- **The documents and the code disagreed** about the PIN (see B): the guide promised no PIN after a laptop restart. The code now does what the guide said; the guide adds when that changed.

Build 141 addendum (20 October 2026)

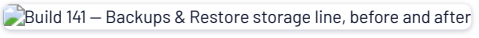
What you asked for (20 Oct, 09:36): "any upgrades or improvements?" Nine were suggested, with the places where the documents and the code disagreed. **09:39:** "do them all". Estimated time given: pictures about 1:30 pm, the release about 9 pm Dubai.

Approval: before/after pictures of the five you can see were sent at about 10:00 (Pre-show checklist, ⚑ Fix Later, the early-reveal question, the Night Recap, the Storage line). **10:04:** "ok all five". The other four change nothing on screen (a smaller complete copy, two release checks, two unused functions removed).

Same version: v1.69 · versionCode 141 · `APP_BUILD=141`. New page module `app135.js` (`build.py` order `…,133,134,135,6`).

A · 1 · Storage the browser keeps

- **Asking:** 2.5 s after the host page opens, `navigator.storage.persisted()`; if not yet, `persist()` once. Never on the TV, phones or remotes. The browser decides; nothing is shown when it says no.
- **Backups & Restore:** `afterHook("backupsDialog")` adds two rows to the last status box:
  - **Complete copy (pictures, videos, music):** "Saved HH:MM" (from `FS140.lastSave`), "Couldn't be saved — back up now" (`FS140.failing`) or "Not needed yet";
  - **Storage:** laptop app "Kept in the Mind It! folder on this computer"; Galaxy Tab "Kept in the tablet app's own files"; a browser "Protected — the browser keeps it" / "Not protected — the browser may clear it if space runs low, so back up regularly" / "This browser can't protect it — back up regularly".
- **Checked:** `fix141` (a browser that agrees, and one that refuses) and `fix141d` (laptop app: the folder line).


Build 141 — Backups & Restore storage line, before and after

B · 2 · The complete copy keeps Question Database pictures once

- **How:** `fsRefSlots(g)` finds game pictures identical to `PROF.images[imgIdOf(v)]` and writes them into the state record as `imgref:<id>` (the pointer the quick save has used since build 111), restoring the real picture in memory straight after packing. They are no longer copied into the media record.
- **Coming back:** on Resume, `normalizeState` → `imgRefsResolve` (app99, since build 111) puts the picture back from the profiles file; `fsMergeMedia` also carries the pointers when a resumed game is filled in from the complete copy. Build 140 resolves the same pointers the same way, so a copy made by 141 opens in 140.
- **Checked:** `fix141` — a quiz with a Question Database picture: the media record no longer holds it, the state record has the pointer, and after a restart Resume shows the picture.

C · 3 · The release check saves and reopens a game in the tablet app

- **The page reports** (Android only, once each, `console.info` → logcat): "MINDIT_FS page android build N", "saved android state+media N chars" / "failed android …", and at start-up "found android state N chars, media yes/no" / "found android nothing".
- **The check** (`.github/scripts/apk_save_test.sh`, run by `publish-release.yml` after the launch check): opens the app, taps **Start New Game** through the accessibility tree, confirms the home screen has gone, waits for "saved", force-stops the app, opens it again and expects "found android state". It answers the emulator's "isn't responding → Wait", "Viewing full screen → Got it" and "Allow notifications?" interruptions.
  - **ok** → a line in the release notes; **fail** → nothing is published; **skip** (the emulator couldn't drive the screen) → a warning, the release goes ahead.
- **Tried on build 140's published APK** (which doesn't report, so the expected result is a skip): five try-outs on build 140's APK (which doesn't report, so 'skip' is the expected answer): tries 1–3 stopped before the quiz page by the emulator's 'Process system isn't responding' dialog, Android's 'Viewing full screen' note and a quoting mistake in the script; tries 4 and 5 opened the quiz page (the home screen gone) and, as expected, found no report — try 4 also met the 'Allow notifications?' question, granted up front from try 5. In the browser, a fresh quiz page with a stand-in for the tablet's bridge saved its complete copy within 15 s without any tap and found it after a restart.
- **On the build-141 APK:** the release run on GitHub does it before publishing and adds its result to the release notes there; nothing is published if it fails.

D · 4 · The 60-minute soak plays a tiebreaker

- `stress110.py soak`: `SOAK_TIE_GAME` (default 2) levels the top two after the Final reveal; the tablet then plays ⚡ **Play Tiebreaker** to a winner, and the game record must show `suddenDeath`. In game 1 the tablet puts 📊 **Night Recap** on the TV (checked on the Audience window), then 🏆 **Winner Screen**.
- **Result:** 14/14 — 3 whole games to the winner screen in 56 minutes (game 1 with ⏱ One More Question and the 📊 Night Recap from the tablet on the real Audience window; games 2 and 3 with ⭐ Go to the Final; game 2's forced tie for first decided by ⚡ the sudden-death tiebreaker played from the tablet and recorded in Historic Games), Final wagers from the phones, 64 questions, 315 tablet actions, 0 false disconnects, no script errors. The first 60-minute run gave 13/14: the script pressed the tablet's next step (Show final

standings) before the tablet had seen the forced tie, so no tiebreaker was played (the app recorded joint winners, correct for a skipped tiebreaker). Script fixed (wait, then ⚡ Play Tiebreaker), checked in a 6-minute run (8/8, tiebreaker won and recorded), then the whole soak was repeated

E · 5 · Two unused functions removed

- `remoteBoot` in app110 (the phone remote's start-up from build 121, superseded by app116's) and the `wordmark` assignment in app34 (superseded by app36's). Both were replaced before they could ever run; the code map had them in its reviewed baseline.
- **Now:** `tools/codemap.py --check` : modules 129 · changed functions 206 · hooks 240 · wrappers 296 · replacements 51 · issues 0 (new 0, known 0, fixed since baseline 0) · parse errors 0. The baseline is empty.
- **Checked:** `fix141` (the logo still comes from app36 and the phone remote page from app116) and the full run.

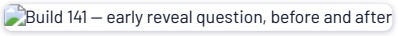
F · 6 · ⚑ Fix Later

- **The action:** `primaryActions` gains "⚑ Fix Later" (ghost, key **F**) on the question, answer, Final question and Final answer screens; "⚑ Flagged — Remove Flag" when the question is already flagged. Remotes (tablet and phone) get it through the same actions.
- **The record:** `PROF.fixLater` (newest first, at most 300): `{id,k,q,a,cat,where,quiz,gameId,libId,at}` , matched by `qKey(q)` , saved with `profSave()` (the players and Question Database file, so it is in backups too).
- **The list:** `afterHook("questionLibrary")` adds the "⚑ Fix Later (N)" tab while anything is flagged: question, answer, quiz · round · category · flagged date and time; **✎ Edit** (`editLibQuestion`) when the question is in the database, otherwise **➕ Add To Question Database** (new item, source "Fix later"), and **✓ Fixed**.
- **End of the game:** a dock card "⚑ N question(s) flagged to fix" with **⚑ Open Fix Later List** on the final standings, winner, awards and recap screens (flags from this game only).
- **Checked:** `fix141` (button, F key, toggle, the Final, saved and reloaded, the tab and its three buttons, the card) and `fix141d` (from the tablet and from the phone remote).


Build 141 — Fix Later, before and after

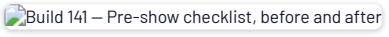
G · 7 · Reveal before anyone has answered asks first

- **When:** `b141Early()` — the question screen, not resolved, a Daily Double or rotate (taking turns) mode, nobody has tried or passed, no buzz winner. Classic turns never ask (revealing first is how Classic works).
- **Laptop:** the "Reveal answer" action's `fn` asks (`ask` : "Reveal the answer now?" · Keep It Hidden / Reveal Answer) when the press came from this computer — a click event, or a key within 150 ms (capture-phase keydown, so **R** asks too).
- **Remotes:** a remote command runs `fn()` without an event, so the laptop doesn't ask. The link snapshot marks the action `early` , and the tablet remote asks with its own confirm (app25 and app92). The phone remote reveals at once, as before.
- **Checked:** `fix141` (click, R key, Keep It Hidden, after a wrong answer, Classic turns, a Daily Double, a remote) and `fix141d` (the tablet: cancel keeps it hidden, OK reveals it at once; the phone remote reveals at once).


Build 141 — early reveal question, before and after

H · 8 · Pre-show checklist on Setup

- **Where:** `afterHook("arrangeSetup")` appends `.pre-check` to Setup's right-hand column; it is rebuilt every 3 s while Setup is on screen and no dialog is open.
- **Rows:** 📺 TV / Audience View (the Audience window, or TV screens on the Wi-Fi; **Open Audience View**) · 📱 Team phones and 🎛 Tablet remote (laptop app only; phones required only in phone-buzzer mode; 🔐 **Team Codes**) · 🔊 Sound (`audioStatus()` , the same check as Teams Presentation Mode; ▶ **Test Sound** plays the reveal sound) · ❓ Questions (empty questions in the rounds being played, and the Final; 📚 **Question Database**) · 💾 Backup file (made today; ⬇ **Back Up Now**) · 🛡 Storage (as in A).
- **The count:** "N of M ready" over the required rows; optional rows show "–".
- **Changed after the pictures:** the class names became `psc-*` — the first version's `.pc-row` collided with app18's Presentation Check. The action button moved under the value so the text isn't squeezed.
- **Checked:** `fix141` (the rows, an empty quiz, a browser that won't protect its data, questions filled in and Back Up Now, Test Sound, Open Audience View) and `fix141d` (the laptop app's phone and tablet rows; Open Audience View ticks the TV).


Build 141 — Pre-show checklist, before and after

I · 9 · 📊 Night Recap on the TV

- **The screen:** `G.screen="recap"` from "📊 Night Recap" (ghost) on the winner, final standings and awards screens. `renderStage` draws `renderRecap` from `recapData()` (app90): title, date, host; champion or joint champions with score and players; up to 8 standings rows (ties shown "="); questions played, Daily Doubles, biggest swing, game time; up to four highlights (the tiebreaker first, then chips, then achievements); the footer. If drawing fails it falls back to the winner screen and logs the error.
- **Laptop actions on it:** 🏆 Winner Screen · Back to standings · 💾 Save Standings Picture · 🎲 Play Again (New Questions). `gameOverNow()` is true on it, so phones and remotes see the game as over; `nowTitle` / `nowHint` describe it.
- **Layouts:** landscape and portrait (`@container (orientation: portrait)`), 4 to 8 teams (`rc-many` above 6). On an upright screen the champion and the whole table always show (`grid-template-rows:auto minmax(0,1fr)`), and the highlights get the rest: `b141FitOne` leaves out any highlight that doesn't fit whole, checked again by a `ResizeObserver` whenever the box changes size (TV window resized, Present mode) and once the fonts are in.
- **Checked:** `fix141` (1920×1080, 1080×1920, 1280×720 and 800×1280 with 8 teams, a two-line champion name and four highlights: every row shown and clear of the numbers, no highlight cut or under the footer, nothing outside the TV picture; a game saved on it reopens on it), `fix141d` (from the tablet and the phone remote, on the real Audience window) and `avaudit` (new scenarios `recap_4` and `recap_8`).


Build 141 — Night Recap, before and after

J · Tests

- **New suites:** `fix141` (browser) 49/49; `fix141d` (laptop app + tablet remote + phone remote) 12/12. `run_all.py` now has 146 suites.
- **Full run** on the final page (md5 `a2689b23…`): full run on the shipping page (md5 `a2689b23…`): 146 suites, 3,264/3,264 checks, all on the first try (101 min).
- **Layout audit** (`avaudit`): 270 screens at 9 sizes (including the new `recap_4` and `recap_8`), 0 issues on the final page. The first audit (on the page of the first full run) found 12 overlaps, all in `recap_8` at 800×1280 — fixed (see K)
- **Stress phases** (`stress110.py` , laptop app, 8 teams, 10 phone pages, a tablet remote): idle 2/2, net 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6 — 24/24 (laptop app, 8 teams, 10 phone pages, a tablet remote)
- **Guide:** 7 new pictures (the Pre-show checklist, the ⚑ Fix Later button and Question Database tab, the early-reveal question, the Night Recap on the TV, the winner screen's What's Next card with the Night Recap and the Fix Later card, the Backups & Restore storage lines); new text in Step 12, 6.4, 15.3, 15.10, 15.12, 19.1 and Chapter 22; a "New in build 141" list at the front; the cheat sheet gains ⚑ Fix Later (F) and 📊 Night Recap; the Complete Idiot's Guide is now 168 pages

- **Reproducible build:** the final page rebuilt from a fresh clone of GitHub `main` is identical: identical (SHA-256 0a424588…f795, commit d989f44); the tablet APK built from a fresh clone with `build_apk.sh` passes the certificate and manifest checks: yes (versionCode 141, v2 + v3, certificate 5f4819e4…; the APK built with the private build.sh from the same files has identical contents).

K · Problems found and fixed while building

- **The Night Recap on an upright screen with many teams (found by `avaudit` on the final page, after the first full run).** At 800×1280 with 8 teams and a two-line champion name, standings rows 7–8 sat under the four numbers and the footer on the last highlight (12 overlaps; every other size was clean). `fix141` 's size check had passed because it looked only for things outside the TV picture, with no highlights in its game. Fixed in `style.css` and `app135.js` as described in I; `fix141` now also checks 800×1280, overlaps and cut highlights, with a long name and four highlights (on the earlier page it fails at 1080×1920 and 800×1280). The recap screens were audited again at all 9 sizes (18 screens, 0 issues), and the full run was repeated on the fixed page.


Build 141 — Night Recap on an upright screen with 8 teams, before and after the fix

- **The soak didn't play its forced tiebreaker the first time.** After forcing the tie in game 2 the script pressed the tablet's next step — Show final standings, from a state that didn't show the tie yet — so the tiebreaker was skipped and the game recorded joint winners (right for a skipped tiebreaker). Fixed in the script (wait for the tablet, then ⚡ Play Tiebreaker); a 6-minute run played and recorded it (8/8), then the whole 60-minute soak was repeated (see D).
- **One old check expected build 140 exactly** (`fix140s` , G · a build-139 save opens in 140). It now accepts 140 or later, like `fix140` .
- **The checklist's first styling** reused `.pc-row` , already used by app18's Presentation Check, which boxed every row. Seen in the before/after pictures; renamed to `psc-*` .
- **The checklist's buttons squeezed the text** at the narrow Setup width. Moved under the value; the emoji were taken off the button labels.
- **The emulator save check** first gave "skip" three times: the emulator's "Process system isn't responding" dialog, Android's "Viewing full screen" note, and a shell quoting mistake in the script. All three are handled; it now also confirms the quiz page really opened, and keeps a screenshot every time.
- **The docs and the code disagreed** in four places (the code is the source of truth):
  - the master context's "Latest build" line still said versionCode 111 (and "it is 111 now") — the code is at 141;
  - the master context still listed the Undo pop-up, Go to the Final, One More Question and the sudden-death tiebreaker (all built in 140), and now the early-reveal question, ⚑ Fix later, the pre-show checklist and a recap TV slide (141), as "not approved" ideas — moved to "built";
  - "Mac (dropped)" in the master context and the changelog — the Apple Silicon app has been built for every release since build 123; corrected to "Intel dropped";
  - the status file's open question 3 still said the web link is republished every alternate build — since 18 Oct it is every build.

Build 140 addendum (19 October 2026)

What you asked for (19 Oct, 10:15): "3,4,5,6" from the list offered after build 139:

- 3: the full-game soak test;
- 4: refresh the Complete Idiot's Guide pictures;
- 5: the Final reveal on portrait screens with 8 teams;
- 6: the four ideas never approved before — the Undo pop-up, Go to the Final, "time for one more question" and a sudden-death tiebreaker.

Your answers for the tiebreaker: "Quizmaster's choice", "Out loud, Quizmaster marks", "Random from my Question Database". **For Go to the Final:** "Round summary, then Final".

Approval: before/after pictures of all five were sent at about 11:10, and you answered "OK all five". Estimated time given: about 7–8 pm Dubai.

A · ↩ Undo pop-up

- **How it works:** after every score (`applyScore`), the Quizmaster's page shows the journal entry's label at the bottom centre, with **↩ Undo**, for 8 seconds. Undo calls `doUndo` on that entry directly, with no `ask` dialog.
- **When it goes:** the next score replaces it, and it disappears once its entry is no longer in the journal (undone elsewhere, reset, new game). It is hidden in Present mode (`body.present`). It is never added on the TV, team phones, phone remote or tablet remote pages.
- **Changed after the pictures:** the pop-up no longer waits while the mouse is over it. A pop-up that appeared under a resting mouse would never have gone away.
- **Checked:** `fix140` , 78/78.


Build 140 — Undo pop-up, before and after

B · ⭐ Go to the Final

- **Where it is offered:** a ghost action on the board, and a button in Undo & resets next to End round early. Only when `plan().final` is set and rounds are still to come.
- **Asking first:**
  - A laptop click asks: the dialog names the skipped rounds and says scores are kept and that an open question or wager is closed.
  - A remote runs it straight away (no click event). The tablet asks with its own confirm: the `^⭐ Go to the Final` rule is added to both confirm rules.
- **What it does:** sets `G.toFinal={from,at}` and calls `goRoundSummary(true)` . On the round summary:
  - the next step is "Go to the Final Round" and `advanceRound` goes straight to `startFinal` ;
  - "Up next" names the Final, and the TV kicker says "Straight to the Final Round";
  - `roundLabel("final")` becomes "Round N · Final Round" with N = the last round played + 1.
- **Undoing it:** Resume cancels it, and so do a new game and `enterRound` .
- **Checked:** `fix140` .


Build 140 — Go to the Final, before and after

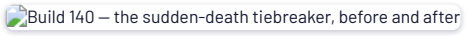
C · ⏱ One More Question

- **Arming it:** `G.lastQ={ri,ref}` . The next clue opened, or the one on screen ("Make This the Last Question" in Undo & resets), becomes the last one.
- **Ending the round:** once that clue is played and the board would come back (`backToBoard` , `closeClueNoScore` , or any render on the board), `goRoundSummary(true)` runs. The kicker reads "That was the last question".
- **On screen:** the TV shows "⏱ LAST QUESTION OF THE ROUND" in place of the Next strip. The laptop hint says so too. The tablet's big button says "🏁 Round Summary →".
- **Limits:** Back to board (keep open) leaves it armed. Cancel Last Question clears it. It is not offered with one question left.
- **Checked:** `fix140` .


Build 140 — One More Question, before and after

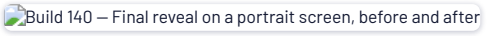
D · ⚡ Sudden-death tiebreaker

- **When it is offered:** only when at least two teams that aren't marked away share the top score, and either the Final reveal is complete or this is the last round summary with no Final. ⚡ Play Tiebreaker is the primary action, and "Show final standings" (skip) stays.
- **Choosing the question:** a random library item that is a text question (no picture, video, audio or choices), not in this game's rounds or Final (by `libId` and by question text), and not already asked in this tiebreaker. Questions never used in earlier games come first. If none is left, the Quizmaster asks one of their own.
- **Screens:**
  - **TV:** a new screen `tiebreak` with the ribbon, the tied teams' scoreboard only, and per phase the rule, the question, or the answer with a banner.
  - **Laptop:** a host-only question/answer card, plus Correct and Wrong for each team.
  - **Tablet remote:** a tiebreaker card with the question and the answer behind a tap, and a Correct / Wrong row for each team.
- **Ranking:** no points change. The winner ranks first among the teams level with it, through wrappers on `rankTeams`, `cmpTB`, `champions`, `tbText`, `tbLine` and `tbPlaces`. They apply only while the winner still holds the top score and shares it.
- **Records:** the game record gets `suddenDeath`, the winner's `tbWon` (so a history edit's `reRank` keeps it) and the qKeys of the tiebreaker questions shown.
- **Saved games:** a game saved during a tiebreaker reopens on it. A saved `tiebreak` screen with no tiebreaker state falls back to the Final reveal.
- **Checked:** `fix140`, including the 💾 standings picture line "Tiebreak: won the sudden-death question" and the tablet card.



E · Final reveal on a portrait screen, 6–8 teams

- **Before:** at 1080×1920 with 8 teams, the build-139 one-row rule (`fr-one`) squeezed the cards to 108 px wide. The standings were absolutely placed at the bottom, with about 800 px of empty screen between them.
- **Now:** a container query (`orientation: portrait`) for `.sx.fin-look.fr-two`:
  - the cards wrap in rows of four (three for six teams), centred, about 227 px wide at 1080×1920;
  - the standings sit in the grid flow straight under them, with larger rows.
- **Landscape:** unchanged.
- **Checked:** `fix140` at 1080×1920 and 800×1280 (two rows, standings within 80 px of the cards, nothing overlapping or off screen), and landscape 1920×1080 (still one row).



F · The soak plays whole games (test only)

- **What changed in `stress110.py soak`:**
  - Every game is now driven to the winner screen. Each team sends its Final wager from its own phone page (the wager pad: a quick pick, then Lock In). The tablet marks every Final answer and reveals them. If a tiebreaker comes up, the tablet plays it.
  - Then each game is checked: the winner on the TV; the record in Historic Games with the same positions as the final standings; the record in the laptop's `player-profiles.json`; and the recap card, the Historic Games recap and the standings picture.
  - Reading time is now 15–35 s per question (was 20–45 s).
  - So that whole games fit the hour, game 1 uses ⏱ One More Question late in Round 2, and game 2 uses ⭐ Go to the Final after six questions.
- **Result:** 12/12 — 3 whole games to the winner screen in 55 minutes (game 1 with ⏱ One More Question, games 2 and 3 with ⭐ Go to the Final), Final wagers from the phones, every game record and picture checked, 67 questions, 310 tablet actions, 0 false disconnects, no script errors

G · Guide pictures

- 22 pictures refreshed and 7 new (5 for the four new buttons, 2 for the complete save: the new line in Undo & resets and the red card); section 22.1 explains the complete save; the Complete Idiot's Guide is now 165 pages

H · Tests

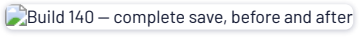
- **Full run** on the final page (md5 `0dc63aea…`), including the four new audit suites: full run on the shipping page (md5 `0dc63aea…`): 144 suites, 3,188/3,188 checks, all on the first try (98 min).
- **Layout audit** (`avaudit`): 252 screens, 0 issues (253/253)
- **Stress phases** (`stress110.py`, laptop app, 8 teams, 10 phone pages, a tablet remote): idle 2/2, net 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6 — 24/24 (laptop app, 8 teams, 10 phone pages, a tablet remote)

I · Not tried for real

- The Galaxy Tab itself, a Windows PC, a Mac and real smart TVs. The page is the same on all of them; the laptop app, the tablet remote page and the phones were tested on the build machine.
- The tablet's new save files (`gameWrite` / `gameRead`) were compiled and their page side tested against a stand-in bridge; the Java code itself has not run on a real tablet.

J · Code audit — your approval (19 Oct, 12:50) and how it was done

You approved all four improvements from the code audit (sent 12:34), to be built into **this build 140** — same build number, no separate build or release. Your instructions: don't stop the work in progress; ask before any change with substantial risk; run the complete regression afterwards, including laptop, Android, phone and TV connectivity; don't publish until everything is verified. No change needed a risk stop. One visual change, the red "isn't completely saved" card, was sent as before/after pictures at about 14:05 (you chose option 1 at 14:09: the card plus the one-time message, as built).



K · Audit 1 — the complete save (`app134.js`, `main.js`, `MainActivity.java`)

- **The gap:** `saveNow` (app1) keeps the game in localStorage (about 5–10 MB). A quiz with media that doesn't fit got the cut-down "lite" save: question pictures and videos, music, intro video, team photos and Final media set to null. After a crash, those were gone. Profiles, the Question Database and history were never affected (separate stores, app126).
- **Now:** after every quick save (debounced 1.5 s, at most 6 s), a complete copy is written as two records:
  - **media:** every embedded `data:` string of 8 KB or more, found anywhere in the game, keyed by its path;
  - **state:** the game with a marker at each path.
  - Media are written first, and only when they change (signature check), so a state never points at media not yet stored.

- **Where the copy goes:** every place the device has.
  - **Galaxy Tab:** `MindItAndroid.gameWrite/gameRead/gameClear` : files `game-complete-state.json` and `…-media.json` in the app's storage, same pattern as `profWrite` . Out of memory returns false instead of closing the app. Above 40 M characters, the browser database takes it.
  - **Laptop app:** `/api/host/fullsave` , host only, written atomically with fsync, previous copy kept as `.prev` . The files don't depend on the port.
  - **Every browser:** IndexedDB `mindit-game` with strict durability.
- **Start-up (`fsRecover`):** the newest copy that opens is used; damaged copies are passed over.
  - **Waiting to Resume:** a game offered for Resume is replaced by its complete copy when the copy is the same game (`saveId` , or `gameId` for saves from build 139) and at least as new. If the cut-down save is newer, the copy's media are put into it.
  - **Setup:** a Setup quiz loaded from a cut-down save is reloaded complete.
  - **Already resumed:** a game already resumed from a cut-down save gets its media back.
  - **No quick save at all:** if the quick save is gone but a complete copy exists, Resume is offered.
  - **Never overwritten:** the fresh Setup shown while a game waits for Resume is never copied over it (like app103's copy).
- **Warning:** only when no complete copy could be stored **and** the quick save is incomplete. It shows as a red card straight under the panel's first card, with 💾 Save Backup File and Try Saving Again, plus a message once per game (never in Present mode). Undo & resets says where the last complete copy went.
- **Start fresh:** the Safety Backup holds the complete game, and app114's cut-down one is skipped, so there is one backup, not two. New quiz while a game waits for Resume keeps that game's copy.
- **Checked:**
  - `fix140s` 36/36 (browser + tablet bridge): a SIGKILLed browser restarted; Setup reload; resumed without media; a state record cut short; media cut short; media newer than state; no media record; the browser database at its quota (100 KB) → card + message, then quota back + Try Saving Again; localStorage completely full; Start fresh; build 139 → 140 → 139 → 140 saves; the tablet's files, a restart, and a refused write.
  - `fix140sd` 19/19 (laptop app): SIGKILL and restart on another port (8765 taken), a state file cut short (.prev used), the data folder refusing writes, a phone address refused (403), bad requests (400), Start fresh.

L · Audit 2 — reproducible builds

- **`build.py` :**
  - no machine paths: inputs next to `build.py` , `--out` / `MINDIT_OUT` (default `dist/mindit.html`);
  - UTF-8 with `\n` line ends;
  - team badges in `assets/team-icons` ;
  - gzip without a timestamp;
  - the tightening step uses `node_modules` from `npm ci` and stops the build if it fails.
- **Locked tools:** `package.json` + `package-lock.json` (terser 5.51.2, csso 5.0.5, acorn 8.19.0).
- **`tools/verify_build.py` :** clones, runs `npm ci` and `build.py` , and compares fingerprints.
- **Laptop and tablet apps in the repository:**
  - `desktop/` : `main.js` , electron-builder settings with exact versions + lock file, `build_desktop.sh win|mac` ;
  - `android/` : Java, manifest, resources, home page, checks, `build_apk.sh` . The passwords come from the environment only; it never creates a keystore and stops unless the certificate is `5f4819e4…637b` ;
  - `.gitignore` keeps out keystores, `*.jks` , `release.pass` and `build.sh` .
- **Checked:**
  - two builds of the same commit are byte-identical;
  - the final page rebuilt from a fresh clone of GitHub `main` is identical: identical (SHA-256 979cf55b…3248, commit 09a4684);
  - the tablet APK built from a fresh clone with `build_apk.sh` passes the certificate and manifest checks: the release APK itself was built from a fresh clone (commit 09a4684): original certificate 5f4819e4…637b, v2+v3, apk_check OK, its page = the tested page, and its 22 entries are identical to the same build through the private build.sh;
  - the release laptop app and APK were built from the repository's `desktop/` and `android/` files: the EXE's and the Mac app's main.js equal desktop/app/main.js; the APK came from android/ in the clone.

M · Audit 3 — secure random codes

- **What uses it:** `secureRandom()` (app1) uses `crypto.getRandomValues` , with Math.random only if that is missing. It feeds:
  - `uid` and team codes;
  - the game key (`newGameKey` , `newState` , app12);
  - the remote PIN, the TV id, the tablet id and the Safety Backup id.
- **Formats unchanged:** 4 letters from `ACDEFHJKMNPRTUVWXY` , a 4-digit PIN, a 2-digit key.
- **Servers unchanged:** the laptop server already used `crypto.randomBytes` and the tablet server `SecureRandom` . Rate limiting and trusted devices are unchanged.
- **Checked:** `fix140r` 14/14 — calls counted; a zero-forced generator proves every value follows it; formats; the spread over 36,000 letters and 18,000 keys; the fallback when the generator is missing or throws; no Math.random in either server.

N · Audit 4 — code map and lint

- **The tools:** `tools/codemap.py` + `tools/codemap.js` parse the page's 128 source files in page order (acorn).
- **What it maps:** every hook, wrapper, replacement and temporary override (`CODE_MAP.md`): modules 128 · changed functions 205 · hooks 235 · wrappers 291 · replacements 53 · issues 2 (new 0, known 2, fixed since baseline 0) · parse errors 0.
- **What it flags:**
  - duplicate declarations (hoisting: the last one wins everywhere);
  - replacements that drop earlier wrappers or hooks, at load time or while the page runs;
  - double replacements;
  - wrappers that run before their function exists;
  - hooks on missing functions;
  - names defined nowhere, unless guarded by `typeof` (inside `try` they are marked as failing silently).
- **`--check` :** fails on anything not in the reviewed baseline.

- **Found on today's code:** no duplicate declarations, dropped wrappers or undefined names. Two double replacements:
  - `remoteBoot` : app110 is superseded by app116;
  - `wordmark` : app34 is superseded by app36.
  - The earlier versions are dead code. They were left in place (no refactor) and recorded in the baseline with the reason.
- **Checked:** `lint140` 12/12 — the page's sources, plus made-up modules containing each kind of problem, all found, with no false alarms.

0 · Problems found and fixed while integrating

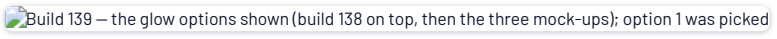
- **The complete copy could be overwritten while a game waited for Resume.** At start-up, the laptop shows an empty Setup while it offers the saved game, and that Setup is saved. Found while testing a killed browser. Fixed: no complete copy is written while Resume is offered, the same rule app103 already used for its copy of the save.
- **The quiz was not always recognised after a crash.** The quick save on disk could be older than the one in memory and missing the new quiz id. Fixed: every new quiz gets its id at once (`newState`), and saves from build 139 or earlier are matched by game id.
- **Start fresh backed up the wrong thing.** Build 125's Safety Backup copies the quick save, which by then can be the empty Setup on screen (a gap since build 125) or the cut-down save. Fixed for 140: Start fresh backs up the complete unfinished game, once.
- **New quiz could delete a waiting game's copy.** New quiz while a saved game still waits for Resume keeps that offer (app110). Fixed: it now keeps the game's complete copy too.
- **The warning card first landed in the wrong place.** It went under "More controls", where it couldn't be seen, or took the round's status line. Seen in the before/after pictures. Fixed: it is added after the panel's layout, straight under the first card, in red.
- **Stray requests from other local web servers (found by the full run, `qc2`).** A page opened from some other local web server sent a request meant only for the laptop app; that server answered 501, which showed in the console. Fixed: the page talks to the laptop's save files only after the laptop app's own server has answered. The full run was restarted on the fixed page.
- **Three old checks expected build 139 exactly** (`fix139` , `fix139t` , `fix139d`). They now accept 139 or later.
- **Old statements in the docs.** The build instructions and architecture notes said the page is about 2.7 MB (it is about 3.6 MB). They also said the build uses this machine's paths, the order from `docs/BUILD_ORDER.txt` , and the badges from the laptop app's folder, and that a failed tightening step carries on. All of that is corrected: the code (`build.py`) is the source of truth.

Build 139 addendum (19 October 2026)

Your requests (18 Oct, 23:20): "do them all" (the seven upgrades offered after build 138: back-pressure for TVs on weak Wi-Fi, a media memory limit on the Galaxy Tab, a stuck-device safeguard, a team-code guessing limit, forgetting screens that have left, a smaller request limit on the laptop, and the 60-minute soak test), "make the default timer for final question at 45 seconds", "check the layouts especially when there is more than 7 players", and "the gold pulsing light is very similar to on fire player". At 23:21 you asked to see the glow options and picked **1 · White spotlight**. Estimated time given: about 8 hours, done around 7:30 am.

A · The "now playing" light: a white spotlight (your pick)

- **Before:** the team in play had a gold ring and a gold glow that pulsed, and 🔥 Streak Flames have an orange ring, an orange glow and flames, so the two looked alike. When the team in play was also On Fire, the flames' `!important` shadow replaced the Now Playing glow completely; only the gold tag and the lift showed it.
- **Options shown with before/after pictures** (`Build_139_glow_options_before_after.png`): 1 white spotlight, 2 electric blue, 3 gold stays and On Fire turns red. You picked 1.
- **Now:** a white ring and a soft white glow for Now Playing on every layout (`.stage/.sx/.sx.zoned/.sx.q-look/.ts-scores .lectern.control`); the two halo layers from build 138 carry it (calm and peak are both white), so the light still breathes slowly without repainting. The gold "Now Playing" tag is unchanged.
  - Now Playing **and** On Fire: the white ring round the orange ring and glow (5+ in a row: round the yellow "blaze" ring).
  - A Returning Champion in play: the white ring round its gold one.
  - Comic Book draws its own black outline on every lectern; the white ring still shows there, and On Fire keeps the comic outline, as before.
- **Checked (`fix139` , 20/20):** the computed border and shadows on the TV and the Quizmaster's preview; On Fire without white; the double ring; the other teams without a ring; all 12 themes.


Build 139 — the glow options shown (build 138 on top, then the three mock-ups); option 1 was picked

B · The Final Round timer defaults to 45 seconds

- `newState()` sets 45 (was 60), and every place that fell back to 60 now falls back to 45 (the laptop's timer, the TV's number, Setup, the "Timer Ready" check, the tablet's preview).
- **Saved games:** `normalizeState` now moves a saved game whose Final timer is still the old default 60 to 45, once (defaults version 8), like the 30 → 45 s question timer in an earlier build. Any other value is kept, a Final already running keeps its time, and 60 chosen after the change stays.
- **Checked (`fix139`):** a new game, the Setup box, the running Final on the laptop and the TV, the five save cases, and Online Mode. The older checks `ext` and `rules35` expected 60 and were updated to 45.

C · The layout check with more than 7 players

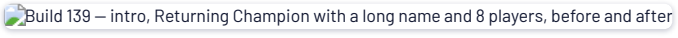
- **How:**
  - a copy of `avaudit.py` with every Audience View screen at 8 teams with long names, at 9 sizes: with no players, with 8 players a team, in Online Mode, and with 4 teams × 12 players;
  - the laptop at 1900×975 and 1366×768;
  - the tablet remote at 1280×800, 800×1280 and 768×1024;
  - team phones at 390 and 360 wide;
  - the 💾 standings and recap pictures.

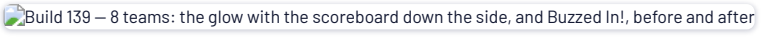
  8 teams is the most a game allows. Done by a second agent in parallel, on build 138's page; nothing was changed while it looked.
- **Found and fixed** (before/after pictures: `Build_139_final_reveal_…` , `…_champion_intro_…` , `…_side_glow_buzz_…` , `…_pictures_…` , `…_tablet_laptop_before_after.png`):
  1. **Final reveal, 6–8 teams:** the live standings (absolutely placed at the bottom) covered the answer cards by up to about 200 px at 1920×1080. With 6+ rows the standings now use two columns; with 6+ cards the cards stay on one row, a little smaller (`app132#frfit` , decided by the counts, because the cards zoom in and the screen is arranged after the hook runs). 2–5 teams: unchanged.
  2. **Intro, Returning Champion:** with several players the card grew upwards over the logo (up to 432 px). Now the players are on their own smaller line (at most 2 lines), the team name is at most 2 lines and smaller when long, and the card sits a little lower.

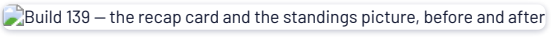
3. **Side scoreboard (7–8 teams):** the Now Playing glow sat 47 px to the right (31 px at 720p): `.sx.sc-side > *{grid-column:1}` also placed the halo layers in grid column 1. Now `grid-column:auto` for them, and the glow sits exactly on the lectern.
4. 💾 **Recap card, 7–8 teams:** the chip and achievement boxes pushed the stats onto the footer and past the bottom. The boxes now stop where they fit.
5. 💾 **Standings picture:** more than 6 champion players were drawn off-centre (centred on all, 6 drawn). Now centred on what is drawn, with a "+N" circle.
6. **"Team Buzzed In!" with 7–8 teams:** cut off at both ends with a long name in the side layout. Smaller there, with an ellipsis if it still doesn't fit.
7. **Tablet remote, Final marking:** rows shrank and centred, so Correct/Wrong sat at a different place on every row, 40 px tall. Rows now span the card; buttons are 48 px.
8. **Laptop at 1366×768:** the 8 status chips took 7 lines and pushed the action buttons below the fold. Two columns with 7–8 teams.

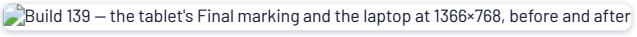
- **Looked fine:** every other TV screen at 8 teams on all 9 sizes (board, questions, answers, Daily Double, Triple Gamble, Phoenix, Bottoms Up, round summary, standings, Final wagers/question/standings, winner, awards, How To Play, category reveal, buzzers, matchup), Online Mode, the laptop's Setup with 8 × 8, Final wagers card, Adjust Score, the tablet's question screen with 8 statuses, and team phones (they never list the teams; a 44-letter name wraps).
- **Checked (`fix139l` , 16/16):** no overlap in the Final reveal with 6 and 8 teams at 1920×1080 and 1280×720 (and 4 teams unchanged), the glow rectangle equal to the lectern's in the side layout, the buzz banner on screen, the champion card clear of the logo with a 44-letter name and 8 players, two-column chips. The layout audit on the final page: 234 screens, 0 issues (235/235); and with 8 teams × 8 players at all 9 sizes, 234 screens, 0 issues (235/235) — on build 138's page that audit flagged the intro, the Final reveal and Buzzed In!.


Build 139 — Final reveal with 8 teams, before and after


Build 139 — intro, Returning Champion with a long name and 8 players, before and after


Build 139 — 8 teams: the glow with the scoreboard down the side, and Buzzed In!, before and after


Build 139 — the recap card and the standings picture, before and after


Build 139 — the tablet's Final marking and the laptop at 1366×768, before and after

D · Reliability on the Wi-Fi

- **Found during this work, fixed:** in build 138 the laptop app had no `error` listener on its WebSocket connections. A message over the size limit (a TV: 32 KB) or a broken frame from **any** device on the Wi-Fi threw in the main process — Electron's error box — and the app stopped answering everything. Reproduced on build 138 (a test TV sent 40 KB: no answer afterwards). Now that connection closes and the app carries on (`fix139d`).
- **TV back-pressure** (laptop `main.js` and tablet `MindItHost.java`): a TV gets the next picture only when the previous one has gone out (laptop: under 256 KB still buffered; tablet: its send queue empty) and it has confirmed all but the last two; otherwise it is marked owed and gets the newest picture when it catches up (on its confirmation, when its queue empties, or at most 4–5 s later). Each picture is the whole screen, so skipping the ones in between loses nothing. Sound packets are skipped for a TV with over 1 MB waiting.
  - Tablet server (`fix139j`): a slow TV (it takes 0.35 s a picture) got 7 of the 40 pictures sent in 2.1 s, ending with the newest (#41) — the ones in between were skipped.
  - Laptop app (`fix139d`): a TV that keeps up got 80 pictures; a slow TV (0.4 s a picture) got 23 and still ended on the newest (#83 = #83).
- **Stuck-device safeguard:**
  - **Tablet:** every connection has its own send queue and writer thread; sends used to write to the socket directly, usually while holding the server's global lock, so one device that stopped taking data could freeze the server. A closing message ("blocked", "moved", an error) still goes out before the socket closes. The heartbeat closes a device whose writing has made no progress for 20 s; more than 96 MB waiting closes it at once; never the quiz page on the tablet.
  - **Laptop:** a phone remote, tablet or TV with more than 1 MB waiting that doesn't go down for 30 s is closed (it reconnects by itself); over 96 MB closes it at once; never the quiz page.
  - Checked: while a TV takes no data, the tablet server answered within 0.05 s and the other TV kept getting pictures; the stuck TV was disconnected. Same in the laptop app.
- **Media memory on the tablet:** pictures and videos for Wi-Fi TVs are kept within 120 MB of data in all, 40 MB each (about a 30 MB file); the ones not on screen go first, least recently shown first. A file over the limit, or with no room next to what is on screen, isn't kept: the TVs show the screen without it, the quiz page is told once (`aud-skip` , a note on the tablet), and it is never asked for again — so it can't loop (the page sends each file once). Checked with a 41 MB file, thirteen 10 MB files one after another, and all thirteen on one screen (`fix139j`), and the note on the tablet's quiz page (`fix139t` , 9/9).
- **Team-code guessing limit** (laptop and tablet): 20 **different** wrong codes from one network address in 10 minutes → 5 minutes' wait, during which even the right code is refused with `wait` and the phone says "Too many wrong team codes from this phone. Wait 5 minutes, then try again." A phone repeating the same mistyped code every 1.5 s counts once (the phone page keeps checking with the code it was given); other addresses are unaffected.
- **Request and message limits:**
  - **Laptop:** other devices may send 2 MB (was 30 MB), checked against the announced size before anything is read (the Galaxy Tab's hand-over routes keep 30 MB; the quiz page on the laptop keeps 150 MB). Link messages from other devices: 2 MB (was 64 MB), through a second WebSocket server for non-local connections.
  - **Tablet:** link messages from a remote after its hello: 2 MB (was 64 MB).
- **Screens that have left** (laptop and tablet): an offline screen that isn't named, blocked or the sound screen is forgotten after 7 days away, and at most 12 such screens are kept (pruned at start-up and when a screen connects).
- **Checked:** `fix139j` (tablet server on the JVM, 26/26), `fix139d` (laptop app, 23/23), `fix139t` (tablet quiz page + server, 9/9).

E · Tests

- **Full run** on the final page (md5 `b1098ad2…`): full run on the shipping page (md5 `b1098ad2…` , 94 min): 139 suites. 136 passed first time, none needed a retry. Three failed because the tests were out of date, not the app, and passed once the test was corrected: `fix130c` 6/7 → 7/7 (its CSS check counted build 139's approved restyles as build-130 rules), `desk` 17/18 → 18/18 (it expected the 60-second Final) and `fix139j` stopped at its last step → 26/26 (the test program didn't expect the tablet to close a connection while it was still sending 3 MB). In all 3,053 checks, all passing.
- **Stress phases** (`stress110.py` , laptop app, 8 teams, 10 phone pages, a tablet remote): idle (1 min each, question and board) 2/2, net 6/6, load 4/4, lock 2/2, rapid 4/4, health 6/6: 24/24 on the final build
- **60-minute soak:** 2/2 — a 60-minute real-time quiz with a tablet remote, the TV window and 10 phones: 60 questions, 377 tablet actions, 93 samples, the Quizmaster window hidden and shown 5 times, 0 false disconnects, no script errors; laptop server memory 223 → 233 MB, the Quizmaster page's JS heap 15 → 16 MB (no game was finished inside the hour)

F · Not tried for real

- The Galaxy Tab itself (its server ran on a computer's Java; the release check opens the APK on an Android 15 emulator), a Windows PC, a Mac, real smart TVs and real weak Wi-Fi (slow and stuck TVs were test programs that read slowly or not at all).

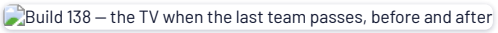
Build 138 addendum (18 October 2026)

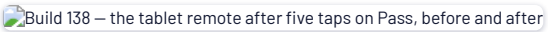
Your requests: "do a code check across all and optimize it if needed" (16:59), then the passing and automatic-return specs (19:07–19:22, section A), all in this build. Estimated time given: published by about midnight.

A · One go per team, the automatic No One Answered and the 7-second return (your three specs this evening, merged into this build)

- **Your requests:**
  - "Fix Unlimited Passing & Add 'No One Answered' Option";
  - its correction "Automatic No One Answered Flow";
  - "combine it all and put it in build 138";
  - "Automatic Return to Board After Answer Reveal" (merged into this build).
- **What the code did before** (the code is the source of truth, and it differed from the spec's starting point):
  - the laptop's own buttons already stopped once every team had had a go: "No one got it — next question", and "NO ONE GOT IT" on the TV;
  - the **Galaxy Tab / iPad / iPhone remote** kept showing ✓ Correct, ✕ Wrong and → Pass after that, and the laptop accepted them (`linkHandle` → `passTurn` / `answerWrong` had no such check). This made the "unlimited passing" real;
  - `nextTeam()` didn't skip teams that had already had a go, which mattered after the team was changed by hand;
  - the host override ("A different team answered?") could mark a team wrong twice, taking the penalty twice;
  - Classic turn order allowed any number of goes per team.
- **Now** (`app131.js`, plus small patches to `app3`, `app4`, `app25` and `app116`), on normal board questions:
  - **One go per team:**
    - `cu.tried` (passes and wrong answers, existing) decides who is eligible: available and not tried (`qEligible`);
    - `answerCorrect`, `answerWrong` and `passTurn` refuse a team that has had its go;
    - `setActive` refuses it, and Make Active is greyed;
    - the host override lists only eligible teams;
    - remotes get `no_go` / `tried` refusals, with a message on the tablet;
    - after a pass or a wrong answer the question goes to `nextEligible()`, the next eligible team in the usual order;
    - Daily Double, Triple Gamble and the Final keep their own rules.
  - **Automatic close:**
    - when no eligible team is left (`qExhausted`, checked after every redraw), `noOneAnswered()` stops the timer and shows the answer screen;
    - the TV's banner under the answer now reads **NO ONE ANSWERED** (it read "NO ONE GOT IT", and only in turn order);
    - after **7 s** (`NOONE_MS`), `closeClueNoScore()` marks the question played, gives no points and keeps the deductions. The next team after the owner is highlighted in turn order; in Classic the picking team stays, as before;
    - the countdown is stored in the game (`cu.noOne.until`), so a reload or a hand-over finishes it on time.
  - **No One Answered button:**
    - in the action bar and the Scoring card, replacing "No one got it — close clue";
    - it asks first on the laptop; the tablet and the 📱 phone remote ask on their own screen.
  - **The 7-second return.** The code already returned to the board by itself after a correct answer: `app80`, 3½ s, on by default, with a "Stay here" button and a setting in 5 · Audio & Presentation. Your spec described the old flow as manual, so this was reported to you at 19:33. Rather than adding a second timer:
    - that feature now waits **7 s** (`AUTO_BOARD_MS`);
    - it also starts when a Daily Double or Triple Gamble ends with a wrong answer (`app131#rv`, once per question);
    - its setting is now called "Back to the board automatically after each answer".
    - No One Answered always closes after 7 s (`NOONE_MS`), whatever the setting.
    - Next Question, Back To Board Now, "Stay here" or Undo before then cancel it. Each timer checks that the same question is still showing, so the board comes back once.
    - Only the quiz device runs these timers (not remotes, phones or TVs).
    - The team in control was already set when the question was scored (unchanged).
  - **Statuses:** "Answering Now · Passed · Wrong Answer · Not Attempted · Away" and "N of M teams can still answer" on the Scoring card. The tablet remote shows the same, and when none are left: "All teams have attempted or passed on this question." In that case its big buttons go.
  - **Phone Buzzer Mode:** Reopen Buzzers keeps the teams that have had their go out.
- **Choices made where the specs left room** (stated to you at 19:18–19:25):
  - **7 s for No One Answered too.** The first correction said about 5 s; the newest message says 7 s for every revealed answer.
  - **The existing back-to-board setting and "Stay here" stay.** If the setting is off, correct answers wait for Next Question as before; No One Answered still closes by itself.
  - **The tablet shows the No One Answered close as its big "↩ Back to Board"** (`app92`): "No one answered — no points".
  - **A manual "Reveal Answer" doesn't start the 7 s** while teams still have their go, because the question isn't decided yet.
  - **Reopen Buzzers** reopens only for teams that haven't had a go.
- **Checked (`fix138p`, 30/30):**
  - two teams both wrong; four teams all pass; mixed wrong and passes;
  - a correct answer after two passes, with the real 7 s measured (7.1 s after the correct answer);
  - one team left; No One Answered early (Cancel and OK);
  - tablet commands after every go (refused);
  - Daily Double and Triple Gamble;
  - a reload in the middle; the next question; a manual adjustment;
  - Next Question before 7 s (no second jump); Undo during the countdown;
  - Classic.
  - **In the laptop app with a real tablet remote:** `fix138pd` 9/9 — an iPad remote on the laptop's link: three Passes moved the question 0 → 1 → 2 → 3, the statuses showed on the tablet, a Pass after every team's go was refused with "Every team has had its go on this question.", and laptop and tablet were back on the board 7.3 s after the last wrong answer; a Correct from the tablet returned once.







0 · How the check was done

- **Read:**
  - every module the page is built from (`app0` … `app129` , `reuse.js` , `starter_book.js` , `style.css`);
  - the laptop app's main process (`desktop/app/main.js`);
  - the Galaxy Tab's built-in server (`MindItHost.java` , `Json.java`).
- **Ran:**
  - a whole simulated game with every error hook watched: no hook, app or page errors; about 12.6 ms per redraw; the page's memory went from 10.1 to 11.2 MB; 27 timers, all cheap;
  - CPU profiles of the Quizmaster page with the Audience View open, in Chromium and in the laptop app (all of its processes);
  - the tablet server on a computer's Java (as the other tablet suites do), fed malformed and oversized requests.
- **What came out:** the fixes and speed-ups below. The code check itself changes nothing about how the game plays; section A, your specs, does.

1 · Bugs fixed

1 · Six buttons did nothing in the laptop app (Windows and Mac).

- The laptop app (Electron 33) has no `prompt()` . It throws "prompt() is not supported." These buttons used it:
  - ▪ New Category, Move To Category, Rename and Delete Category (when the category has questions) in the Question Database;
  - ⇄, which moves a question to another round;
  - ✎ Edit Answer on Mark Each Team.
- Build 137's addendum noted that `prompt()` does nothing in the laptop app, but only replaced it for Online Mode's chat answers.
- **Fix:** `askText()` in the new module `app130.js` is the app's own text box. It opens with the current value selected; Enter or OK saves; Escape, Cancel or a click outside closes it without saving; Escape doesn't also close the Question Database underneath. It offers the existing categories as suggestions. `app27` and `app32` use it.
- **Checked:**
  - in the laptop app, `prompt('x')` throws "prompt() is not supported.", and + New Category works through the box;
  - on the published build-137 page, with `prompt()` throwing as it does in the laptop app, + New Category does nothing (25 → 25 categories);
  - on build 138, all six buttons work.

2 · ✎ Edit Answer was undone within a second when team phones were connected.

- The laptop app's phone link replaces `G.finalAnswers` twice a second with what the phones sent, so a correction written there was overwritten.
- **Fix:** a correction is kept in `G.finalAnswersEdited` (the phone's answer stays as the original in `G.finalAnswersOriginal`). `finalAnswerFor()` reads the correction first, so the suggestion, the TV reveal and the game record use it. The link snapshot, which phones and Wi-Fi TVs read, carries it too. A new Final Round starts with no corrections.
- **Small change:** saving an empty box now leaves the answer as it was. Build 137 blanked it.
- **Checked in the laptop app:** three phones answered the Final; "burj kalifa" was corrected to "Burj Khalifa"; 3.5 s later, with the phone link on, the correction was still there, the suggestion said correct, and the TV reveal showed “Burj Khalifa”.
- Online Mode is unchanged: there, answers are corrected in the Mark Each Team boxes, and it has no ✎ button.

3 · Laptop app: a large save could damage accented letters and emoji.

- The laptop app decoded each 64 KB piece of an incoming request on its own. A character whose bytes fell across two pieces became "��".
- It was reproduced with build 137's `main.js` : a 940 KB save came back with 14 damaged characters.
- Your 10 October backup has no damaged characters (no U+FFFD).
- **Fix:** `readBody()` decodes the whole request as one UTF-8 stream (`req.setEncoding("utf8")`).
- **Checked:** a 1,826 KB save full of é ’ — 😀 東京 comes back identical, with 0 damaged characters. The laptop's own profiles were then put back.

4 · 📶 Devices always said "Tablet not connected".

- It read `LINK.devices` and `linkDeviceCount()` , which haven't existed since the connection rework.
- **Fix:** it reads the trusted-device list that the laptop sends to the page (`window.DVSTATE`). Phone remotes don't count as the tablet.
- **Checked in the laptop app:** with the tablet connected it says "● Connected · ✓ Tablet connected". With only a phone remote online it says "Tablet not connected".

5 · Galaxy Tab hosting: Wi-Fi TVs were disconnected every 15 seconds.

- The tablet's heartbeat pinged every socket every 15 s and closed any that hadn't answered the previous ping. TV sockets have their own 5-second beat (`tvBeat()`), so the heartbeat closed them every time, and they reconnected.
- **Fix:** `heartbeat()` skips TV sockets.
- **Checked on the tablet server:** a TV stays connected for 35 s and answers 7 pings. Build 137's server closed it after 12.8 s.

6 · Unexpected or malformed data.

- **Galaxy Tab** (an uncaught error there closes the app):
  - deeply nested JSON overflowed the stack. Nesting now stops at 256 levels: 120,000 `[` get an HTTP answer and the server keeps answering;
  - a WebSocket frame claiming a negative length reached a 2 GB allocation. It is refused now. Before its hello, a device may only send messages up to 64 KB (TVs: 32 KB, as before);
  - requests from other devices are limited to 2 MB, refused before anything is read. The hand-over routes allow 30 MB and the tablet itself 150 MB, as before;
  - a request with more than 100 header lines is closed;
  - a request's body is read as it arrives. Before, a buffer of the announced size (up to 30 MB from any device, or a negative size that failed) was set aside first.
- **Laptop app:**
  - a "null" message on the link socket threw in the main process, which shows Electron's error box. Such messages are now ignored, as the TV socket already did;

  - a malformed address threw outside the error handling. It now gets "400";
  - the "try the next port" listener stayed attached after the server started, so a later network error would have started another server. It is removed once the server is listening.
- **Both: remote PIN guessing.** 20 wrong PINs from one network address within 10 minutes make that address wait 5 minutes; even the right PIN waits. Before, any device on the Wi-Fi could try all 9,000 PINs. A remote that used to be trusted, or the wrong session, doesn't loop: the remote stops and asks for the PIN.
- **Checked:**
  - tablet server: `fix138j`, 13 checks; build 137's server fails 7 of them;
  - laptop app: `fix138d`. Five malformed messages caused no error in the main process, and the Quizmaster page's own link stayed connected.

7 · Laptop app: the "isn't responding" box.

- Each hang of the page opened another box.
- Now there is one at a time, and it closes by itself when the page answers again (`AbortController`).
- Not tested (making a real hang takes 30 s or more). Checked by reading the code.

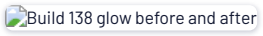
8 · Laptop app: handing the quiz to the tablet.

- **Cancel during the last step:** pressing Cancel while the final snapshot was downloading no longer carries on with the hand-over.
- **A late confirmation:** after the 2-minute wait ran out, a confirmation that the tablet sent late now restarts the hand-over. Before, it was left waiting.
- The hand-over suites in the full run cover the normal path (see Tests). These two edge cases were checked by reading the code.

2 · Optimisations

9 · The "now playing" glow.

- The pulse animated the lectern's `box-shadow`, which the browser repaints every frame, all the time a team is in control.
- It is now two layers on the lectern whose opacity the compositor cross-fades:
  - the calm layer carries the lectern's own outer glow (read from the theme, and re-read after a resize, because the glow is sized in `cqw`);
  - the peak layer is the old brightest keyframe (`0 0 3.4cqw rgba(255,182,39,.85)`).
- The lectern keeps its inset highlight. Streak Flames keep their own glow. The layers follow the lectern exactly (within 0.02 px), with and without the TV safe area, after a resize, in the Quizmaster's preview and in Wi-Fi TV frames.
- **Measured** with the Audience View open, CPU of the Quizmaster page and the Audience View, in milliseconds per second:
  - Chromium: 59.9 → 3.5;
  - laptop app, all its processes: 819 → 224 ms per second for all its processes (the page's renderer 734 → 86; the graphics process 59 → 111, because the build machine has no GPU and draws the moving layers in software — on a laptop with a graphics chip that part is close to free).
- **The look** (picture): the calm and brightest moments are the same as before. Halfway through, the amber ring fades out, where before it melted into the glow.



10 · The page is sent compressed.

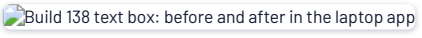
- Phones, tablet remotes, Wi-Fi TVs and the tablet's own page used to download the 3.5 MB page in full every time. The laptop app re-encoded it on every request.
- **Now:**
  - the laptop app keeps the page gzip-compressed: 3,575,122 → 1,811,855 bytes (gzip level 9, made once at start);
  - so does the tablet server: 3,575,122 → 1,821,031 bytes (made once, on the first request);
  - both send an ETag, so a reload gets "304 Not Modified";
  - a browser that doesn't ask for gzip gets the page as before.

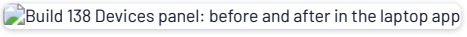
11 · Fewer writes to `trusted-devices.json`.

- A paired phone remote polls twice a second, and each poll rewrote the file.
- It is now written at most once a minute per device, or at once when the device's name changes. The time it was last seen is still kept in memory and saved with the next write.
- **Checked:** 12 polls in 6 s gave 1 write.

12 · Dead code removed.

- `app4`: two lines after a `return`.
- `app6`: an `if(false){…}` block.
- `app26`: an always-true `||true`.
- `app129`: two unused variables.





3 · Looked at and left as it is (suggestions for later)

- **TVs on weak Wi-Fi:** a slow TV can fall behind; there is no back-pressure.
- **Picture and video rounds hosted on the tablet:** no memory limit for very large media.
- **Tablet server:** socket writes happen under the server's global lock, so one stuck device can slow the others (needs a watchdog).
- **Team codes:** there is no guessing limit (PINs now have one).
- **Audience displays:** the list of displays that have left isn't pruned.
- **Laptop app:** the limit for requests from other devices is still 30 MB (the tablet's is now 2 MB).

4 · Tests

- `fix138` (webpage, Chromium, with `prompt()` throwing as in the laptop app): 32 checks.
  - The text box; the six buttons; build 137 doing nothing.
  - Edit Answer surviving the phone refresh, in the snapshot, reveal and record.
  - Both glow layers: position, shadows and animations.
  - The rule presets are build 137's, and there are no hook or page errors.

- `fix138d` (laptop app): 21 checks. The large save; malformed messages; the compressed page and ETag; + New Category; Edit Answer with three phones; Devices; trusted-devices writes; the PIN limit; no main-process errors.
- `fix138j` (tablet server): 13 checks.
- `qlib` : now types into the new box.
- **Full run:** full run on the shipping page (md5 `a5b16b1c…`): 134 suites, 2,953/2,953 checks, `fix127d` passed on its automatic retry (a 4.5-second reconnection check; 22/22 again when run on its own) (90 min). New: `fix138` 32/32 and `fix138p` 30/30 (browser), `fix138j` 13/13 (tablet server), `fix138d` 22/22 (with the Wi-Fi TV check) and `fix138pd` 9/9 (laptop app + tablet remote); `qlib` 26/26.

Build 137 addendum (18 October 2026)

Your requests:

- "any upgrades? anything you would suggest for the online teams mode": six proposals with before/after pictures.
- "do them all, and from now on weblink gets republished everytime".
- Also in this build: item 1 from the build-136 list, the Android emulator launch test in the release workflow ("i will do 1 in a bit"). It was offered to you as optional.

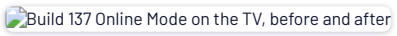
0 · Two build-136 statements were wrong (found while making the proposals)

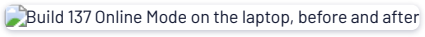
- **How to Play.** The build-136 addendum ("How to Play says the same") and the guide ("How to Play on the TV explains it the same way") were wrong.
  - Build 136 changed `howToLines()` , but the TV hides that list and shows the rule cards of `afterHook("renderStage","app72#1")` instead.
  - Its Final Round card still said "answers on their phone".
- **The cursor.** The build-136 release notes said "the wager card puts the cursor in the first box".
  - In the webpage version, the first box was a sealed-code box (`.wg-input.code-row`).
  - The laptop app has no code boxes, so it was right there.
- **Why the tests passed.** `fix136o` checked `howToLines()` , and its focus check accepted any `.wg-input` .
  - Both checks now look at what is shown: the TV's `.ht-card` texts, and `.wg-input:not(.code-row)` .
  - On the published build-136 page the corrected suite fails exactly these two checks (25/27); on build 137 it passes (27/27).

1 · Online Mode for Teams calls, part 2 (`app129.js` ; only while Online Mode is on)

- **Teams Presentation Mode on pick.**
  - The `applyRulePreset` wrapper calls `setTeamsMode(true)` , which opens the Audience View and, 150 ms later, `presentationCheck()` .
  - It does this right away at Setup. Mid-game it waits: `ask` gets a temporary wrapper around its callback, so nothing happens until **Apply**.
  - Classic and Full Show never switch it on.
  - Not in the Galaxy Tab app (`window.MindItAndroid`): it has no second window (`window.open` returns nothing there), so the Presentation Check would only show a blocked Audience View.
- **Speed Bonus window.**
  - Picking Online Mode sets `G.speedBonusSecs=15` when it is 10, and marks it with `G.olSpeed15` . Another preset puts back 10 only if it is still 15.
  - A new quiz with Online Mode remembered starts at 15 (`newState` wrapper).
  - Checked with `speedBonusFor` : an answer 12 s in earns the bonus at 15 s and not at 10 s.
- **How to Play.** `afterHook("renderStage","app129#howto")` replaces the texts of two cards:
  - **Pick & Answer:** "Teams take turns choosing a category and value. Unmute and answer out loud on the call before the timer runs out."
  - **Final Round:** "Everyone sends a wager, then their answer, to the Quizmaster in a private chat message — the scores can change completely."
- **Final answers by private chat.**
  - **TV:** `.ol-fnote` on the Final question ("📩 Send your answer to the Quizmaster in a private chat message"), re-fitted by `fitClueText` .
  - **Where answers are kept:** `G.finalAnswersChat` (team id → text), apart from `G.finalAnswers` . In the laptop app the phone link replaces `G.finalAnswers` twice a second; checked, a typed answer stays.
  - **Reading them:** `finalAnswerFor(i)` returns the phone answer, else the chat answer. So the existing suggestion (app32), the reveal line on the TV (app3) and the game record's `finalAudit` use it unchanged.
  - **Mark Each Team** gets an input per team (not revealed, no phone answer):
    - Enter or Save calls `olSetChatAnswer` , then the cursor moves to the next team without an answer;
    - the first box gets the cursor once per Final;
    - drafts survive a redraw (`OL_DRAFT`);
    - a changed answer clears that team's mark and audit;
    - the `prompt()` -based "✎ Edit Answer" is removed for these answers, because the box replaces it (`prompt()` does nothing in the laptop app).
  - **Reset:** `startFinal` and `resetProgress` clear the chat answers.
  - **Quizmaster remote:** `linkSnapshot` gives the remote the merged answers.
- **Wagers.** In `wagerCard` / `ddWagerCard` (Online Mode only):
  - removed: `.keybox` , `.code-row` , the booth button and "· N phones connected";
  - the auto banner says the wagers come by private chat;
  - **All in · Half · 0** (`Min` when the minimum isn't 0) call `hostSetWager` like Save, so the same checks and auto-reveal apply;
  - the Daily Double card gets **Half** (at least the minimum) between Minimum and Maximum;
  - the first-box focus in `app128` now skips `.code-row` .
- **Copy picture for the chat.**
  - A tile (or button in narrow layouts) after **Save picture** on the standings and winner screens, in Online Mode.
  - `navigator.clipboard.write([new ClipboardItem({"image/png": recapPNG() promise})])` : the promise keeps the click's permission while the picture is drawn. If that fails, it retries with the finished picture, then falls back to `offerDownload` with "Picture Saved Instead".
  - **Laptop app:** `main.js` now grants `clipboard-sanitized-write` in its permission handlers. It was refused before, which was measured: `NotAllowedError` . It is write only; reading stays refused. `BUILD` (the hand-over protocol, 132) is unchanged.

- **Styles:** `.ol-fnote`, `.ol-fans`, `.ol-quick`, `.end-tiles.ol-four` (2 × 2 tiles when there are four).





2 · The release workflow opens the tablet app on an Android emulator

- **Scripts in MindIt-Downloads:**
  - `apk_launch_test.sh`: install, find the launcher activity, tap the icon (`monkey`), wait 20 s; the process must be alive, the resumed activity must be in `com.mindit.app`, and there must be no `FATAL EXCEPTION`;
  - `apk_launch_result.sh`: ok → notice, plus a line in the release notes; fail → error; skip / no result → warning.
- **In `publish-release.yml`:** after validate, when the build has a new APK, `reactivecircus/android-emulator-runner@v2` runs (API 35, google_apis, x86_64).
  - An app that doesn't install or doesn't stay open stops the release.
  - An emulator that doesn't start only warns.
  - Both workflows pass `actionlint`.
- **Tried on the published apps** (runs 39117358542 and 39117817701, try-out branch `apk-launch-test`):
  - build 134 fails: "Unable to instantiate activity ... com.mindit.app.activity: ClassNotFoundException", and the launcher stays on screen;
  - build 136 opens: `com.mindit.app/.MainActivity` on screen, process alive.
- **By hand:** Actions → "Test the tablet app on an Android emulator" → tags.
- Build 137's own APK goes through the same check when the release workflow publishes it; the result is shown on the release (the last line of its notes) and in the workflow run.
- **Branches left behind:** deleting branches is refused from the build machine's session, so you can delete these on GitHub; neither holds anything that isn't in `main`.
  - In MindIt-Downloads, the try-out branch `apk-launch-test` (merged into `main`).
  - In the source repository, a branch `master` that I pushed by mistake (the same commit as `main`).

3 · The web link is republished every build

- **New rule** (owner: "from now on weblink gets republished everytime"), now in CLAUDE.md.
- Build 136's page was republished straight away (version 173). republished with the build-137 page (version 174; the saved copy holds exactly this page)

Tests: full run on this page: 129 suites, 2,837/2,837 checks, all first try. `fix137` 31/31 · `fix137d` 10/10 · `fix136o` 27/27 (25/27 on the published build-136 page) · `fix136od` 6/6.

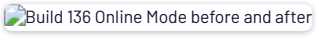
Build 136 addendum (18 October 2026)

Your requests:

- "Do 2,3,4,5" (from the build-136 upgrade list), and "make mindit-downloads public".
- **Online Mode:** "Mind It! — Replace Family Night with Online Mode", then "i will use it on Teams calls" and "do those changes, combine everything and put it in build 136".

0 · Online Mode replaces Family Night (`app128.js`)

- **Presets:** Classic · Full Show · **Online Mode**. Family Night is gone, not kept alongside.
- **Settings:** Online Mode keeps every Family Night setting except your seven:
  - Phoenix Chip **off**;
  - Speed Bonus, Streak Flames, Returning Champion, Board Fill Animation, Pass-On Bonus (25%) and the Final Round **on**.
- **Sticky:** while it is selected (`G.ruleMode="online"`, remembered for the next quiz), it stays selected when other switches change. The Phoenix switch is hidden, and the Phoenix Chip stays off even if an old save says otherwise (`ruleGet` and `udEnabled`).
- **When picked in Setup:** 2 question rounds + the Final Round (automatic values: Round 1 200–1,000, Round 2 400–2,000) and Standard answering.
- **For Teams calls** (players can't reach the laptop's Wi-Fi):
  - The Final Round wager screen shows "📩 Send your wager to the Quizmaster · private chat message" instead of the QR code.
  - The Daily Double screen asks the team to tell the Quizmaster.
  - No "Scan to join" code.
  - How to Play says the same.
  - The laptop's wager card says so and puts the cursor in the first empty box.
- **Unchanged:** Classic and Full Show (compared with the build-135 page). The same Final Round screen in Full Show still shows the QR code in the laptop app.
- **Checked:**
  - the TV shows no QR code in Online Mode, in the browser and in the laptop app;
  - no Phoenix Chip at the start of Round 2 (Full Show gives one);
  - an Online Mode game starts with the Speed Bonus.



1 · Lighter backups (item 2)

- **What was in a backup:** `exportFull()` wrote a copy of the game with every question picture in full, next to the player pictures, where the same pictures already are. Measured on your 10 October backup: 7 question pictures (13 places in the quiz), 3.4 of the 10.5 million characters a build-135 backup of that data takes.
- **Now (`app127.js`):**
  - The copy of the game in the file points at the identical picture in the same file, with the `imgref:` pointer the autosave has used since build 111.
  - The undo copy of a replaced round is left out once the game has started. A restore drops it then anyway; since build 111 it is usually gone already.
  - The game on screen is never changed.
- **Applies to:** Back Up Now, Hand Over, Save Backup File, Safety Backups and Google Drive backups. All of them use `exportFull()`.
- **Restoring:** a restore merges the pictures first, then puts them back into the quiz (`importAny` → `mergeProfiles` → `normalizeState` → `imgRefsResolve`).
  - Tested on the build-119 page from the rollback zip, the published pages of builds 132, 133, 134 and 135, and 136: every picture is back in the quiz, byte for byte.

- **Build 111 had decided the opposite** ("backup files stay complete … so they restore on any build"). Build 136 backups still hold every picture, but a build older than 111 can't put a shared one back. None of those builds is in use, and the rollback build 119 restores them (tested).
- A build-135 backup restores in 136.

- **Your data:** the 10 October backup, restored into build 136, writes 7.1 million characters instead of 10.5 million.

2 · 🗜 Make photos smaller (item 3)

- **What it shrinks:** only the *original* kept for re-cropping a player photo (`photoOrigId`), when it is bigger than 800 px. That is the size new photos have been kept at since build 111. Encoded like a new photo (`picEncode` : WebP where the browser can make it, else JPEG).
- **Same picture IDs:** pictures move to the tablet as "add if missing", so a tablet keeps its own copy. Restoring an older backup brings the larger copies back. Both are harmless.
- **Never touched:** the photos shown on screens (512 px), question pictures (they fill the TV), GIFs, see-through pictures, and anything that changed while it was being checked.
- **The steps:** it checks first and asks with the numbers. Then it makes a safety backup ("Before Make Photos Smaller"), replaces the pictures and saves. If the safety backup can't be made, nothing is changed.
- **Your data:** on the 10 October backup, 13 of the 14 originals shrink, 2.5 MB → 1.2 MB. A backup of that data then goes from 10.5 to 5.7 million characters.

3 · Where the data is (item 4)

- **Backups & Restore → Right now** has two new rows:
  - **Player Data:** this laptop's file / the tablet's file / this browser's database / this browser, then size and last save. "Not saved yet" or "Loading…" when that is the case.
  - **Pictures:** how many there are and their size.

4 · Warnings on builds 129–134 (item 5)

- **How it works:** `releases/notices.json` and a new `notices` step in the release workflow (`mindit_release.py notices`) put "Don't install this release's Galaxy Tab app" at the top of those releases' notes. A release that already has the warning is skipped.
- **Unchanged:** files, tags and dates.
- **Why the workflow does it:** releases can't be edited from the build machine's session.
- Tried first against stand-in releases (all six get the warning once; a second run changes nothing); applied by the build-136 publish and checked afterwards.

5 · MindIt-Downloads public

- Repository settings can't be changed from the build machine's session, so you switch it yourself: Settings → General → Danger Zone → Change visibility.
- What becomes public with it was listed for you before you switched.
- Done by the owner on 18 October; a download link opened without signing in (checked from the build machine).

Tests: Final full run of **127 suites on the page that ships (Online Mode included): all 2,813 checks passed, every suite on its first try** (83 minutes) — the same page that is inside the Windows, Mac and Android apps. Before Online Mode was added, a full run of the 125 suites then in place passed 2,767 of 2,768: the one miss was the build-111 check that backups hold no pointers, which build 136 changes on purpose (that check now asks that every picture is inside the file, a shared one stored once: 20/20), and `fix130a` passed on its automatic retry (it has missed "all three show the same picture" in earlier builds too). The storage check `storq` now opens the page from a site address: from a file:// address the test browser sometimes gave a new page an empty localStorage while the other pages still saw it (seen with a probe), which made the migration step read nothing; over http it passed 5 runs out of 5. New suites: **`fix136o` (26/26)** — the presets, your seven settings, the rest as Family Night had them, 2 rounds + Final with the usual values, the Phoenix switch hidden and off, it stays Online Mode when you adjust, Quick Setup's button, Classic and Full Show unchanged against the build-135 page, remembered, a game starts with the Speed Bonus, Standard answering, no phone QR on the TV, How to Play, the wager card's cursor; **`fix136od` (6/6)** — in the laptop app the Final Round shows the QR code in Full Show and the "Send your wager" box in Online Mode, no join code, and no Phoenix Chip at Round 2 in Online Mode (Full Show gives one); **`fix136` (27/27)** — lighter backups, restored in build 119 (the rollback build) and 132–136 with every picture back byte for byte, Make Photos Smaller, the new rows; **`fix136d` (7/7)** — the same in the laptop app with its player file. **Apps:** Windows EXE 69,529,814 B (page and `main.js` inside identical; joiner tested under Wine 4/4); APK versionCode 136 signed with your original key (v2 + v3), `apk_check.py` and Google's `aapt2` read the name "Mind It!" and `.MainActivity` ; Mac zip 100,806,283 B (arm64, ad-hoc signed, page and `main.js` inside identical).



Build 135 addendum (18 October 2026)

Your requests:

- "Do them all, comics background added" — the eight backdrops, the remote build warning, the guide screenshot swap, steadier tests, tidied docs.
- **Android:** "Critical Android APK Launch Failure & Packaging Fix", with screenshots showing the name "application" and "application closed because this app has a bug".
- **iPhone:** "Fix iPhone Web App Storage Full Error & Player Profile Saving".
- "do all the fixes in this build".

The Comic Book picture you sent was not used. It is a fan artist's signed artwork of Marvel and DC characters, and the app is published on a public GitHub Pages site. The Comic Book backdrop is an original drawing instead.

1 · The Android app (builds 129–134)

- **Root cause, confirmed in the file:**
  - A compiled Android manifest stores each text attribute twice: a *raw* string index and a *typed value* (TYPE_STRING, data = string index). Android reads the typed value.
  - Since build 129, `axml_fgs.py` adds `android:foregroundServiceType` to the compiled manifest. It inserts one string into the string pool and moves every later *raw* index up by one, but it left the *typed* values where they were.
  - On the device every string therefore pointed one entry too early:
    - the app's label read "application", exactly as on your screen;
    - the activity class read "activity", which doesn't exist, so the app closed at once;
    - most permissions shifted to the neighbouring name.
  - `aapt dump badging` , androguard, the build checks and the release workflow all print the raw strings, so all of them showed "Mind It!".
  - **Confirmed independently with Google's `aapt2 dump xmltree` **, which prints the typed values. The build-134 APK reads label "application", activity "activity", service "service". The build-135 APK reads "Mind It!", ".MainActivity", ".HostService".
- **Fix:**
  - `axml_fgs.py` now moves the typed values too, including CDATA.

- New `apk_check.py` (standard library only) reads the manifest like a device. It checks:
  - every TYPE_STRING value equals its raw string;
  - the name is "Mind It!" and the icon is a resource;
  - the launcher activity is `.MainActivity` (MAIN + LAUNCHER, exported) and is a class in `classes.dex`;
  - the service is `.HostService` with connectedDevice;
  - all permissions, package, versionName, versionCode, min/target SDK;
  - `resources.arsc` is stored and aligned, and the assets are present.
- It runs at the end of `android/build.sh`, so the build stops on a fault. The GitHub release workflow (`mindit_release.py` validate) also runs it, so a faulty APK can't be published.
- On the build-134 APK it reports the device-side label "application" and fails. On build 135 it passes.

- **Unchanged:** package `com.mindit.app`, the original signing key `5f4819e4…`, minSdk 24, targetSdk 35. It updates over the broken install and the tablet keeps its data.
- **Not verified on a device:** no Android device, emulator or system image is available on the build machine.

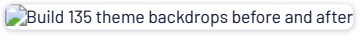
2 · iPhone "Storage Full" and player profile saving

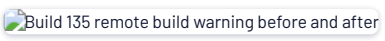
- **Root cause, reproduced in Chromium with Safari's quota simulated:**
  - `profSave()` wrote the whole player database (players, every photo as base64, the Question Database, history) to localStorage as one JSON string. It called **any** exception "storage is full".
  - The iPhone/iPad remote page (`/tablet?remote=`, Connect Device → iPad / iPhone) ran `libSyncNow` on every welcome. That pulled the laptop's complete database: your 10 October backup is about **7.0 million characters**, about 14 MB as Safari counts it, against Safari's 5 MB per site. It then saved it 2–3 times. Nothing on that page ever reads the copy.
  - The laptop page also showed the false message once its database outgrew the browser's quick copy, though its file was written.
  - A stale quick copy on the laptop could, in theory, be saved over the newer file in the first second after opening.
- **Fix (`app126.js`, with small hooks in `app19.js` and `app25.js`):**
  - **1 · Remote page:** no database sync on the phone/iPad browser remote. The Galaxy Tab app (it has a file) still syncs, and re-syncs automatically at most every 10 minutes.
  - **2 · Storage chain:**
    - **Android:** tablet file first.
    - **Laptop:** the laptop file. It is read before anything is written, every write is checked (`{ok:true}`), and an out-of-date quick copy is removed once the file confirms.
    - **Browser:** a large database goes to IndexedDB `mindit-profiles` (`meta` = everything but pictures, `img` = one entry per existing picture id). The write is read back before the old localStorage copy is removed, and a small pointer `mindit.profiles.store` records where the data is. Small databases stay in localStorage exactly as before.
  - **3 · Error kinds:** browser quota (only a real `QuotaExceededError`), storage blocked (Private Browsing), database couldn't open, data couldn't be packed, laptop/tablet file not written, laptop file couldn't be read. Each has its own message, shown once per session. The same classification covers Star Of The Show clips, the rehearsal safety copy and the autosave note.
  - **4 · Nothing is lost:** the data stays in memory, retries run at 5 s, 15 s, 30 s and 60 s, and a bar offers **Try Again** and **⬇ Save Backup File**. It saves when the page is hidden or closed. If a stored copy can't be read, nothing is saved over it.
  - **5 · Fewer writes:** a save with unchanged content writes nothing, and on a large database saves within 150 ms are written once.
  - **6 · New photos:** JPEG unless they really have transparency (Safari can't make WebP), within about 360 KB. Existing pictures and IDs are untouched.
  - **7 · Webpage offline cache (`sw.js`):** the page is kept once under `./index.html` instead of one 3.4 MB copy per address.
- **Not verified in real Safari:** no WebKit is available on the build machine. The iPhone remote and Safari's limit were emulated in Chromium.

3 · Backdrops, remote warning and the rest

- `app125.js`: eight original inline-SVG scenes through `sceneTheme()`: a scene band along the bottom, sky items and movers, keyframes with theme prefixes. Each stays within about 14 KB of new drawing.
  - Each was checked behind the quiz at 1920×1080: still with reduced motion and with Subtle intensity, and every layer in the picture.
  - The host's link state now carries `build` and `hostWord`, so the remote shows **⚠ Different Builds** at iPhone width too.
- **Guide:** the build-120 screenshot with real first names was replaced by one with the demo cast. The guide also gained text for the tablet fix, the storage messages and the backdrops.
- **Tests:**
  - `imp` / `libx` now wait for each import step.
  - `fix134` / `fix134d` follow the page's own build number.
  - The Intermission loudness comment is corrected to about 7 dB (measured).
- **Docs:** the stale "next task" section and the build-132 publishing note in CURRENT_STATUS are fixed.

Tests: Full run of **123 suites: all 2,732 checks passed** (82 minutes), on the final build-135 page — the same page that is inside the Windows, Mac and Android apps. One older suite passed on its automatic retry: `fix128d` (build 128) reads the 📱 Add Phone Remote QR code from a screenshot, and once it couldn't read it. the check now also tries the picture with a white margin and with OpenCV's second QR detector; it then passed 35/35 five times in a row, first try each time. An earlier full run, on the first build-135 page, stopped in two suites: `heic` (the new photo saving had pushed the iPhone HEIC conversion aside — fixed, 5/5) and `fix130c` (its reference page had moved out of the downloads repo — fixed, 7/7); the 12 photo, storage and sync suites were then re-run on the new page, 134/134. New suites: **`fix135` (27/27)** — the eight backdrops (each within its size budget, still with reduced motion and with Subtle intensity, every layer in the picture), the remote's build warning (also at iPhone width) and the tablet server passing on its build; **`storq` (17/17)** — the webpage with Safari's 5 MB limit simulated: a 6.8-million-character database moves to IndexedDB and comes back identical (same player and picture IDs), an older build's 2.2-million-character copy moves safely, the true messages (browser limit, Private Browsing, database won't open, unexpected error — never "full" unless it is), Try Again saves what was waiting, ten saves become one write, the phone remote page never asks for the database, JPEG within budget; **`fix135d` (7/7)** — the laptop app's server and the warning; **`fix135s` (9/9)** — the real laptop app with a 6.8-million-character database (file written and confirmed, an old quick copy never overwrites it, a failed write keeps the change and Try Again saves it) and an iPhone remote with Safari's limit (no database copy, no warning, also after a reconnect). **Apps:** Windows EXE 69,525,950 B (page and `main.js` inside identical; joiner tested under Wine 4/4); APK versionCode 135 signed with your original key (v2 + v3), `apk_check.py` and Google's `aapt2` read the name "Mind It!" and `.MainActivity`; Mac zip 100,802,327 B (arm64, ad-hoc signed, page and `main.js` inside identical, symlinks kept).





Build 134 addendum (17 October 2026)

Your request: "sting 1 is fine · use loop 1 · and do all the below upgrades", then "go ahead with all, dont wait for my approval". The upgrades:

1. the build number in About & Credits;
2. a warning in Connected Devices about mixed builds;
3. a magical castle backdrop for the wizarding theme, with owls and magical creatures;
4. your image as the comic book backdrop — **not done: the image never arrived**;
5. remove the old build-131 app files from the downloads repo;
6. make the downloads repo public;
7. steadier test runs.

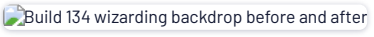
What changed — exactly these files

- `app124.js` (new):
  - **Wizard / Magical School backdrop.** An original inline-SVG night scene, replacing the candles and crest: a castle on a cliff with lit, flickering windows, a moon and a lake. Flying across it are three owls (wings flap), a distant dragon, a phoenix that rises off the top before reaching the logo, and a small group of bats. Everything sits behind the quiz.
    - The castle is drawn generically: no film's castle, crest or character.
    - With reduced motion or Subtle intensity, nothing moves and every creature stays in the picture.
  - ☕ **Intermission music.** `LOOP_MP3` is the theme's 19.2 s loop (Quiz Room Riff, cut on exact samples 195422–1042125, level evened out). It is 308,079 B, SHA-256 `d9177370…`.
    - It plays while Intermission is on, where the quiz's sounds play. Remotes never play it.
    - Level: volume × 0.337 (about −25 LUFS at the default volume). It fades in over 2 s and fades out over 1.5 s.
    - It follows the volume slider, Sound Off and your own theme music.
  - **The build number.** `APP_BUILD=134`, shown in About & Credits ("1.69 · Build 134") and Setup → About.
  - **Mixed builds.** Connected Devices shows **⚠ Different Builds** when a connected remote reports another build. A remote that reports none (133 or older) is shown as "an older build".
- `app25.js`: a remote now says its build when it connects.
- `app121.js`: the Rehearsal Report uses `APP_BUILD`.
- `style.css`: the warning box.
- **Laptop app `main.js` and tablet `MindItHost.java`:** they pass each remote's build on to the quiz page. Their transfer `BUILD` stays 132, so builds 132, 133 and 134 can still move a quiz between each other.
- `tests/run_all.py`: the laptop-app suites now wait until the app answers on its debugging port with its page open, instead of a fixed 13 s.

Before/after pictures: `BUILD_134_BEFORE_AFTER_WIZARD.png`, `BUILD_134_BEFORE_AFTER_BUILD.png`, `BUILD_134_BEFORE_AFTER_DEVICES.png`.

Tests: Full run of **119 suites: all 2,678 checks passed** (81 minutes). Two older suites passed on their automatic retry: `imp` (a dialog was re-drawn while its button was being clicked) and `libx` (the import check read the dialog before it had finished). Both were run twice more afterwards and passed 8/8 each time; nothing in build 134 touches them. The new suite **`fix134` (24/24)** checks the loop file byte for byte, starts and stops the Intermission music (fade, volume, Sound Off, own theme music, remotes stay silent), the build number in About & Credits, Setup → About and the Rehearsal Report, the wizarding backdrop (every creature in the picture and behind the quiz, still with reduced motion) and the tablet server passing on each remote's build. **`fix134d` (8/8)** does the same for the laptop app's server and the warning ("this laptop runs build 134"). **Apps:** Windows EXE 69,499,660 B (page and `main.js` inside identical; joiner tested under Wine 4/4); APK versionCode 134 signed with your original key (v2 + v3); Mac zip 100,768,202 B (arm64, ad-hoc signed, page and `main.js` inside identical, symlinks kept).

Not tested: the real laptop, Galaxy Tab, phones and TVs. The Intermission music hasn't been heard on real speakers, the castle backdrop hasn't been seen on a real TV browser, and the warning hasn't been seen with a real Galaxy Tab on another build.







Build 133 addendum (17 October 2026)

Your request: "use 5a and release it" — the intro theme preview you approved (*Option A – mix 5a*) becomes the app's intro, and the release goes out. Nothing else in the app changes.

What 5a is:

- the ElevenLabs music *Quiz Show Riff* (Option A) for the riff and the quieter groove;
- your ElevenLabs *Dark Villain, Cinematic* take of **"MIND IT!"**, with the voice's speed, pitch and wording unchanged; the music dips under it;
- then the music stops for the Mind It! explosion, made louder and more booming;
- one bar later, the last riff.

What changed — exactly these files

- `app12.js`:
  - **The three sounds.** 5a is split into the app's three parts, at the levels the app plays them at the default volume (music ×1.0, voice ×0.704, explosion ×1.0). Playing them at their places rebuilds 5a. The parts are listed in the table below; nothing of the old three sounds is left in the page (1,233 distinctive pieces checked).
  - **Timings:**
    - the voice enters at 8.55 s of the music (`ROCK_VOICE`; `VOICE_AT` 5.35 for the sting);
    - `VOICE_END` 1.06, so the explosion lands at **9.63 s**, right after "IT!";
    - the loading loop is the riff's two-bar hook, 1.619–4.819 s, cut on exact samples where the join is seamless;
    - `INTRO_END` 13.04: the intro page and the 🎵 sting now last until the last riff has finished (the intro page about 8.7 s instead of 6.8 s; the sting about 9.8 s).
  - **Build 132's dip under the first line is off** (`MUSIC_DIP_DB` 0): the new music dips under "MIND IT!" by itself, as in 5a.
  - **If the music can't be loaded**, the voice comes in at once instead of after 5 s of silence.

- `app123.js` (new): ★ About & Credits → About card → *Intro Theme — Music and announcer voice created in collaboration with ElevenLabs* (the credit ElevenLabs asks for). Before/after: `BUILD_133_BEFORE_AFTER_CREDITS.png`.
- `app121.js`: the 📋 Rehearsal Report says build 133.
- `main.js` and the tablet server are unchanged, so their `BUILD` stays 132 and a 132 device and a 133 device can still move a quiz between them.

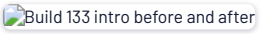
Part (in [illegible])	Size	Length	SHA-256
`ROCK_MP3` (music)	314,140 B	13.04 s	`c8bfc9a0…`
`VOICE_MP3` ("MIND IT!" + its room tail)	31,808 B	1.91 s	`3913b3cb…`
`BOOM_MP3` (explosion)	33,062 B	2.01 s	`f9951ce4…`

How it was checked (build machine):

- **The intro rendered offline through the app's own code and sound chain** (Chromium, the real `continueIntro` / `playSting`, default volume), then compared with the 5a file you approved:
  - every part (riff, groove, "MIND IT!", explosion, last riff) lands at the same moment, to the sample, with correlation 0.994–0.998;
  - once the levels are matched, what's left is MP3 coding (−20 to −24 dB);
  - the app's sound chain plays everything 1.1–1.4 dB louder, and the explosion about 0.6 dB less than that (it was the same for the old intro).
- **Balance in the app:** the explosion is 9.6 LU over the riff (5a: 10.3) and "MIND IT!" 4.2 LU over the riff (5a: 4.3).
- **Loudness in the app at the default volume:** −15.3 LUFS (build 132: −14.2). The music is about 1.7 dB quieter, and the explosion louder, than in build 132.
- **Before/after:** `BUILD_133_BEFORE_AFTER_INTRO.png`.

Tests: Full run of **117 suites: 2,635 of 2,637 checks passed** (83 minutes). The 2 misses were older checks that assumed build 132: `fix132s` waited a fixed 9 s for the 🎵 sting to end on a Wi-Fi TV, and the new sting lasts 9.84 s; `fix132r` expected the Rehearsal Report to say build 132. Both were updated and re-run: 24/24 and 19/19. `desk` passed on its retry (the laptop app was still starting the first time). The new suite **`fix133m` (32/32)** replaces `fix131v`: it checks the files byte for byte, renders the intro and the sting through the app, and covers the loading loop (the repeat is identical and the join has no click), the no-music fallback, loading → intro → game in real time (the intro page stays while the last riff plays), Skip Intro, the winner screen, the Rehearsal Report and the credit at laptop and phone widths. **Apps:** Windows EXE 69,187,013 B (page inside identical; joiner tested under Wine 4/4); APK versionCode 133 signed with your original key (v2 + v3); Mac zip 100,452,011 B (arm64, ad-hoc signed, page inside identical, symlinks kept).

Not tested: the real laptop's speakers, the Galaxy Tab, phones, a TV (HDMI or 🔊 Sound On This TV), and the iPhone webpage — the sound was measured only, not heard on a real device. The ElevenLabs licence for an app and for public or paid events depends on your ElevenLabs plan (Scale or Business; offline commercial use is excluded on every plan).





Build 132 addendum (17 October 2026)

Your request: "do them all and for 2 dip the music slightly" — the six upgrades offered after build 131:

1. 🔊 sound on a Wi-Fi TV;
2. a clearer first line in the intro — **your choice: dip the music slightly** under it;
3. 📋 a Rehearsal Report from the real devices;
4. a shorter TV address;
5. the remote lock as a small icon;
6. handing the quiz back from the laptop to the Galaxy Tab.

What changed

- **Page:**
  - `app120.js` (new): TV sound;
  - `app121.js` (new): the report;
  - `app122.js` (new): the hand-back;
  - `app12.js`: the dip, and the custom sting routed to the TV;
  - `app117.js`: the audience panel — the sound button and the short address;
  - `app119.js`: the lock icon and the short-address check;
  - `app118.js`: the transfer texts for both directions;
  - `app116.js`: the remote's offer message and its name as "Laptop";
  - `app40.js`: the Connect ▾ items;
  - `app63.js`;
  - `style.css`.
- **Laptop app** `main.js`:
  - port 80;
  - the TV sound relay;
  - `/api/host/diag`;
  - the hand-back source with its fencing;
  - `BUILD` 132.
- **Android:**
  - `MindItHost.java`: the TV sound relay, the hand-back target, and two fixes;
  - `MainActivity.java`: `openHostPage`, and more in `hostStatus`;
  - `HostService.java`: the Wi-Fi announcement status;
  - versionCode 132.

The dip (item 2): the music goes down 5 dB from just before the first line to the end of the pause, then comes back for "MIND IT!" and the explosion. Measured on the offline render:

- the first line now stands about **+6.6 dB** over the music (build 131: about +2 dB);
- "MIND IT!" stands about **+7.3 dB** over the music;
- everything outside the dip is identical to build 131.

The two preview MP3s (before / after) are in the chat.

Bugs found while building, and fixed

1. **A Galaxy Tab whose quiz had moved to the laptop could never host again** (builds 130–131). The tablet's server kept its "moved" record forever:
   - it kept sending every phone, remote and TV to the laptop — also for a new quiz days later;
   - the tablet's own quiz page turned itself into the laptop's remote as soon as it opened.

   Now:
   - the record is closed (kept as history) when the quiz comes back, when a quiz is started there, or when a set-up quiz is open there more than 6 hours after the move;
   - only the quiz that actually moved turns a page into a remote.

   The same applies to the laptop after it hands a quiz to the tablet. `fix132t` checks both.
2. **The tablet's server crashed while listing devices if a trusted remote wasn't connected** (builds 130–131). The cause was a Java conditional that unboxed a missing number. The crash cut off:
   - the remote that had just connected;
   - the tablet page's own remote link — again and again.

   Seen in testing: an iPad dropped 13 times in 8 seconds. Fixed; `fix132n` reproduces it — it fails on the build 131 code and passes now. The laptop was never affected.
3. The optional **theme music** (Setup → Theme music) played from the laptop even when a TV had the sound; it now goes to the TV too.
4. The audience panel's buttons stretched when the sound button was added; fixed before the screenshots (see the before/after sheet).

Transfer compatibility: `BUILD` is now 132 on both sides. A 131 device and a 132 device politely refuse to move a quiz between them, so update the laptop and the tablet together. Everything else works across 131 and 132.



📸 Before and after (all layout changes, plus the new screens):

Check	Result
`fix132s` (new, laptop app) — **TV sound + short address.** The laptop answers on port 80: `<ip>/tv` opens the TV page, the bare address lands there too, and Quizmaster-only addresses are refused on both doors. The panel shows `<ip>/tv` with the long form underneath. **Sound On This TV:** the TV asks for OK first, while the laptop keeps the sound. OK pressed: the TV plays, and the laptop's speaker gain goes to 0. The sting reaches the TV (66 pieces of sound, up to −11 dB at the TV's speakers); the TV goes quiet after it (silence isn't sent); a single effect arrives within a second; the theme music plays there too. The TV drops: the sound is back on the laptop within seconds, with a note. It reconnects and asks for OK again. Another TV takes the sound (one at a time); Sound Back Here; a TV can't send sound or commands into the quiz	**24/24**
`fix131v` (updated) — the intro: the page carries your clip; music and explosion are identical to build 131 outside the dip; about −5.0 dB inside it; the dip is in the intro and the sting	**22/22**
`fix132r` (new, laptop app) — **Rehearsal Report:** ready in seconds; the file name; this device; the app file's SHA-256; system, port and short address; the Test Hosting rows; the TV with its own browser, size and error list; remotes, team phones, sound, tablet discovery and errors. **No** team codes, PIN, device keys or folder names; a plain page with no scripts; Save Report File; also in Connect ▾	**19/19**
`fix132t` (new, laptop app + simulated Galaxy Tab) — **Hand Quiz to the Galaxy Tab**, after a build-130 take-over: the menu item; the list of Galaxy Tabs; the offer; Not Now; Cancel; accept. The laptop pauses scoring; the tablet takes exactly the same game with the countdown running; both phones move and keep their team, and stay there; the tablet hosts again (its earlier "moved" record closed); the laptop becomes the remote in control, named "Laptop"; the iPhone remote follows and waits; names and the TV follow. The laptop retires and refuses or redirects late phones; host generation 2; the laptop as a remote scores on the tablet. The laptop's start screen says where the quiz went, it doesn't turn itself into a remote, and a new quiz on it hosts normally. **Failure before the hand-over** (the quiz can't be opened on the tablet): the laptop keeps hosting with the same scores and countdown, the phone stays, and the tablet goes back to being the remote with one tap. **Tablet app stopped just after the hand-over:** the laptop accepts no scores meanwhile; the tablet restarts and finishes by itself, with the same scores, and the phone follows	**57/57**
`fix132n` (new, tablet server + Chromium) — a trusted iPhone goes away; an iPad then connects and stays connected (no drops); the tablet's own link stays up; Connected Devices lists both. The same check on the build 131 tablet code: 4/7 (the iPad dropped 13 times in 8 s)	**7/7**
Full regression	**117 suites, 2,618 / 2,621** (83 min). The 3 misses were older checks that expected the build 131 look, and were updated for the changes made on purpose: `fix130t` expected the tablet's TV address with `http://` in front (now shown without it); `fix130c` flagged the one existing style changed on purpose (room for the quiz name beside the 🔒 icon on narrow remotes — shown in the before/after); `fix119` counted the 🔒 icon as part of the tablet's button row (it now sits beside Connected). Re-run: 13/13, 7/7, 12/12. `fix130a` passed on its second try: its Wi-Fi-drop step closed the TV's connection while already offline, which sometimes left it closing for seconds; the step now closes first, then goes offline, and passed first time in 3 runs straight after other suites (41/41). **Layout audits:** host screens 90 views, 0 issues; Audience View 234 screens, 0 issues

Check	Result
Windows EXE	69,077,219 bytes, SHA-256 `96a5f52a…` ; the tested page (md5 `1433e6fd…`) and `main.js` (md5 `e185b083…`) inside; parts 23,025,740 + 23,025,740 + 23,025,739 bytes + `Join Mind It - build 132.bat` ; the joiner's fingerprint test under Wine 4/4
Android APK	versionCode **132**, versionName 1.69, 1,746,714 bytes, SHA-256 `ce65e019…` ; v2+v3, original key `5f4819e4…` ; foreground-service type kept; the tested page and the new tablet code (`openHostPage` , hand-back files) inside
Mac (Apple Silicon)	arm64, ad-hoc signed, the tested page and `main.js` inside; zip 100,344,693 bytes, SHA-256 `712700c7…` ; 5 parts re-join identically; **not run on a real Mac**

Tested here vs. needs real devices

- ✅ **Tested on the build machine:**
  - the laptop app's engine (Electron) with the page and `main.js` ;
  - the tablet's server code on the JVM with the tablet page in Chromium (bridge simulated);
  - TVs, phones and remotes as Chromium pages at real sizes.
- 🟡 **Needs your devices:**
  - **TV sound on a real TV browser** — whether it unlocks with the OK button, its delay, and whether it plays smoothly over your Wi-Fi;
  - Windows letting Mind It! use port 80 (if anything else uses it, the long address still works);
  - the Galaxy Tab app opening its own quiz page on Host It on This Tablet (`openHostPage` in Android's WebView);
  - the hand-back with a real Galaxy Tab;
  - the report's tablet part (`hostStatus` on Android);
  - **listening** to the intro dip on real speakers.
- **Rehearse at home:** Rehearsal Checklist rows 35 (short address), 37 (TV sound), 38 (hand back) and 39 (report).

Live acceptance: **PENDING USER LIVE TEST**.

Build 131 addendum (17 October 2026)

Your spec: "Replace Intro Vocals Only (Global Update)". Your new voice clip replaces the old intro voice everywhere; everything else stays exactly the same. **Your choices:**

- **Use the fitted version** (after listening to the before/after previews).
- **Republish the web link with 131** (outside the alternate-build turn, so every platform gets the new voice; the next turn is build 133).

The limitation (reported before building, as your spec asks): the old voice ran 3.1 s from its first word to the explosion; your clip runs 5.4 s, with a 1.2 s silent pause between its two lines. The music, the loading loop and the explosion had to stay unchanged, so the clip fits only if:

- the pause is shortened to 0.3 s;
- the voice starts at the music's build, about 1 s earlier than the old voice (it can't start any earlier, because the loading music is still looping before that point).

As a result, the first line plays over the build, where it stands about 2 dB above the music; the old first line, in a quiet gap, stood about 14 dB above it. "MIND IT!" stands about 7 dB above the music (old: about 9 dB) and ends right on the explosion.

What changed — exactly these files and references

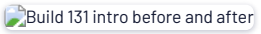
- `app12.js` , `VOICE_MP3` : the embedded intro voice, the only copy anywhere, replaced by your clip. The clip was edited in two ways only:
  - the silent pause was cut from 1.23 s to 0.30 s (the cut is inside digital silence, so no spoken sound was touched);
  - the level was raised +11 dB with a gentle peak limiter (−1.5 dBFS) to match the old voice.

  The file is 72,351 bytes, SHA-256 `e97315fd…` ; your original is kept in the private source zip.
- `app12.js` , timing:
  - `ROCK_VOICE` 5.75 → 4.8 and `VOICE_AT` 2.55 → 1.6 (where the voice enters);
  - `VOICE_END` 3.09 → 4.04, so the explosion stays at **8.86 s of the music**, as before;
  - `VOICE_PAUSE` 1.46 → 2.57 and `VOICE_MINDIT` 1.89 → 2.87 (used only by the no-music fallback);
  - `continueIntro` now starts the voice at "the build or now, whichever is later", joining the clip at the same point, so it can never drift from the music.
- **Unchanged:**
  - the music `ROCK_MP3` and the explosion `BOOM_MP3` (same bytes);
  - the loop points, the music's start point, the music / explosion / voice volume settings, the loading time and the loading loop's jump to the build;
  - the intro's length; Skip Intro.
- **Every place the voice plays** goes through this one asset:
  - the loading screen → intro;
  - 🎵 Play The Mind It! Sting (Audience View menu, Setup → 5 · Audio & Presentation, the host panel's 🎵 button);
  - the theme sound preview;
  - the winner screen.

  The TVs never play sound (it comes from the laptop or tablet). If someone has uploaded a *Custom "Mind It!" sting* in Setup, the 🎵 buttons and the winner screen play that file instead, as before. Remove it there to hear the new voice on them.
- **No stale copy:**
  - the project, Android, desktop and documentation files were searched for the old recording: only the built page held it, and that page has been rebuilt;
  - the voice is inside the page, not a separate file or address, and is decoded into memory each time the app opens;

- the Windows, Mac and Android apps carry the page inside them;
- the hosted webpage's offline copy is replaced the next time it's opened with internet.

- `AndroidManifest.xml` versionCode 131. `main.js` and the tablet server are unchanged, so builds 130 and 131 can still hand over a tablet quiz to each other.



📸 *Before and after: the intro rendered with the app's own sound settings (music + explosion in grey, identical in both; the voice in colour). The two preview MP3s are in the chat.*

Check	Result
`fix131v` (new). **The page:** carries your clip byte for byte; none of the old voice's 252 distinctive pieces are left; music and explosion are the same files as build 130; loop points, levels and loading time unchanged; the explosion stays at 8.86 s of the music. **Offline render** (Chromium Web Audio, the app's gains): music + explosion identical to build 130 sample for sample (max difference 0); the voice enters at the build; "IT!" ends 0.01 s before the explosion; the decoded voice is 4.45 s with first line, 0.3 s pause, then MIND IT!. **In the app:** loading → intro → game, with the loading music started once and the voice once; the explosion 4.06 s after the build, as in build 130; 🎵 sting with the explosion 5.71 s after the start, as in build 130; played again reuses the same voice; Skip Intro stops music, voice and explosion at once; the winner screen plays it	**20/20**
Full regression	**113 suites, 2,526 / 2,527** (77 min). The one failure was in a test, not the app: `fix130a` 's Wi-Fi-drop step relied on the browser's offline mode to stop the TV reconnecting, and twice in this run (right after `fix129d`) the TV reconnected straight away. The step now blocks new connections itself. Re-run after `fix129d` three times: **41/41** each time (it had also passed 4 out of 4 times on its own). No app file changed. The layout audits were not re-run, because no layout changed
Laptop app (the Windows app's engine, on the build machine)	the real laptop app (Electron 33, the engine inside `Mind It!.exe` and the Mac app): sound running, loading → intro, music started once, voice once, decoded voice 4.45 s, voice at the build, explosion 4.06 s after the build — **7/7**
Windows EXE	69,063,574 bytes, SHA-256 `b4719d0b…` ; tested page inside (md5 `2885a82e…` , no old voice); `main.js` unchanged from build 130; the three parts re-join to the same file; joiner fingerprint test under Wine 4/4
Android APK	versionCode 131, versionName 1.69, 1,722,138 bytes, SHA-256 `99e32a90…` ; v2+v3, original key `5f4819e4…` ; tested page inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page + `main.js` inside; zip 100,326,041 bytes, SHA-256 `e66db8ce…` ; 5 parts re-join identically; **not run on a real Mac**

Tested here vs. needs real devices

- ✅ **Tested on the build machine:**
  - the webpage in Chromium;
  - the Windows app's own engine (Electron) running the page;
  - the tablet's page in Chromium;
  - the offline renders.
- 🟡 **Needs your devices:**
  - **listening** to it on the real laptop, Galaxy Tab, Mac and TV speakers (nothing here was listened to by a person — the checks are measurements);
  - whether the first line is clear enough over the music's build in your room;
  - Android's WebView playing it (the APK carries the same page, checked by its fingerprint).
- **Rehearse at home:** Rehearsal Checklist row 29 now starts with "the intro (the new voice, then the explosion)".

Live acceptance: **PENDING USER LIVE TEST**.

Build 130 addendum (16 October 2026)

Your two specs: "Universal TV Audience Display" and "Live Quiz Host Transfer: Galaxy Tab → Windows Laptop". **Your choices:** both in one build; any TV on the same Wi-Fi may show the audience screen (no pairing code); and all five smaller upgrades.

What changed

- 📡 **Audience Screens (no cable).** Any TV, projector or computer with a web browser on the same Wi-Fi opens `http://<laptop or tablet address>/tv` and shows the audience screen. It's in 📺 Audience View ▾ → Connect Audience Screen, on the setup screen and in 📲 Connect ▾. Several screens can be connected at once, at 720p, 1080p or 4K, with an optional safe margin for TVs that cut the edges. The panel shows each screen's name, status, connection quality and last update, with Test, Refresh, Rename, Margin, Disconnect / Reconnect and Forget. It works whether the laptop or the Galaxy Tab hosts. HDMI and the Audience View window are unchanged.
- 💻 **Take Over Tablet Quiz.** A quiz running on the Galaxy Tab moves to the laptop while it runs, after the tablet's Quizmaster taps Approve Transfer. Scores, the question on screen, Daily Doubles and wagers, chips, the countdown and the Final all move. Team phones keep their team without typing a code again, remotes and TVs follow, and the tablet becomes the Quizmaster remote. It moves from tablet to laptop only, as your spec asked, with no automatic take-back.
- 🧪 **Test Hosting** (Connect Device → Run Test) checks that phones, remotes and audience screens can reach the laptop or tablet. A 🔋 **battery chip** appears below 20% (red at 10%), with one reminder; it never shows on the TV. 🔒 **Lock** on the remote ignores taps until you press and hold. Connected Devices shows a ▂▄▆ **signal** and speed for each remote. The Windows joiner now checks the joined file's **SHA-256 fingerprint**.

How it works (short)

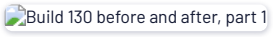
- *Audience screens:* the host draws the audience picture exactly as for the HDMI window (or on a hidden page when no window is open). It sends that finished picture over its own connection, `/audience`, which can only receive; it never sends the game data. The server accepts only hello, screen size and test replies from a TV, limits their size and rate, and refuses any picture that contains script. A TV asking for the Quizmaster's data gets "refused" (403). So hiding is not done with styling: answers, wagers, PINs, team codes and players are simply never sent.
- *Google Cast* was looked at and not used. A Cast receiver needs a paid Google developer registration and a secure (https) web address, which a local quiz on venue Wi-Fi doesn't have. A TV's own web browser does the same job with no extra hardware, and HDMI stays as before.
- *Take over:* nine stages:

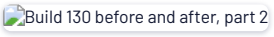
  1. ask;
  2. approval on the tablet;
  3. the tablet prepares a snapshot;
  4. the laptop checks it (version, fingerprint, contents);
  5. the tablet pauses scoring (at most 30 seconds);
  6. hand-over (the tablet saves "committed" before it answers);
  7. the laptop takes the quiz;
  8. phones, remotes and TVs move;
  9. done.

  Only one device hosts at any moment. After the commit, the tablet refuses game changes and points every device to the laptop. The laptop's quiz gets a new host generation number, and a command that was already applied is ignored if it arrives again. **Before the commit**, any failure leaves the quiz on the tablet, and it resumes by itself. **After the commit**, the tablet never takes the quiz back by itself. If the laptop can't be sure the hand-over finished, it asks: *Host It Here* or *Discard*. Both devices keep the snapshot, and the laptop saves its own previous game as a Safety Backup first.

Found and fixed during testing (none of these reached a download)

- **How to Play changed look:** the new Test Hosting styles reused three style names the audience *How to Play* screen already uses. The before/after pictures showed it. The new styles were renamed, and How to Play is now **pixel-identical to build 129**. A new check, `fix130c`, compares How to Play with the published build 129 page. It also requires every new build 130 style to use names that only build 130 code uses.
- **Test Hosting's battery line** used the reading from up to a minute earlier; it now reads the battery when the test runs.
- **Moving a quiz:** on the build machine, these went wrong and were fixed:
  - an iPhone remote could grab control before the tablet;
  - remotes lost their names when they moved;
  - a laptop restart could re-apply an old snapshot over newer scores;
  - a restarted laptop was told "busy";
  - the tablet's *Keep Hosting Here* question was hidden behind the pause screen.
- **Title case:** "TVs", "PINs" and "LG" are now written correctly.





📸 *Before and after (real screenshots: build 129 and build 130 side by side).*

Check	Result
`fix130a` (audience screens on the laptop: a TV opens the address and is listed; same picture as the Audience View; board, question, countdown; 720p + 1080p + 4K at once, each scaled, 16:9 kept; Wi-Fi drop → small *Reconnecting* line, last picture kept, recovers by itself, game never reset; reopening mid-question; the panel: rename (remembered), safe margin, Test Audience Screen 5/5, Refresh, Disconnect / Reconnect / Forget; a TV sending commands changes nothing and gets 403 for wagers, players, info and health; no Quizmaster controls in 816 pictures)	**41/41**
`fix130b` (privacy and a whole game: unplayed questions (57) and pictures (129), unrevealed answers, the Daily Double wager, Final wagers and typed Final answers, team codes and the PIN never reach a TV; every round to the Final reveal and winner; HDMI window and TVs together; a slow TV (slow network, 4× slower processor) keeps up)	**20/20**
`fix130t` (the Galaxy Tab hosting: two TVs show its screen, rename / test / disconnect from the tablet, remembered in the tablet's storage; Test Hosting on the tablet incl. background hosting and battery; battery chip at 9%, not on the TV)	**13/13**
`fix130x` (your success test: live quiz on the tablet with phones, a remote and a TV; Tablet Quiz Found; approve; same scores and question; countdown resumes; phones keep their team; the tablet becomes the remote; the TV follows with its name; the tablet refuses late writes; no double scoring; laptop restart; transfer before the quiz starts)	**32/32**
`fix130y` (during a Daily Double, with a Triple Gamble armed, during the Final with a wager still to come; Reject; repeated and competing requests; a laptop on another build is refused; the laptop's old game kept as a Safety Backup)	**22/22**
`fix130z` (laptop cut off during the pause → the tablet resumes after 30 s; tablet cut off just after the commit → the laptop finishes when it's back; the tablet app stopping mid-pause → *Keep Hosting Here*)	**18/18**
`fix130m` (laptop killed before the commit → the tablet keeps hosting; killed after the commit → the restarted laptop finishes from its saved snapshot; nobody answers → expires after 2 minutes)	**15/15**
`fix130e` (Test Hosting on the laptop; battery chip below 20%, red below 10%, gone when charging, never on the TV; Lock on the tablet/iPhone remote and the Phone Remote — taps ignored, short press doesn't unlock, hold does; signal in Connected Devices)	**17/17**
`fix130c` (new: How to Play identical to build 129; every new style uses new names only) — and it was shown to fail on the page before the fix	**7/7**
Windows joiner under Wine (with a stand-in for Windows' `certutil`): no certutil → joins, says the check was skipped; right fingerprint (new and old certutil formats) → joins, "Fingerprint checked"; wrong fingerprint → no EXE, "a part is damaged"	**4/4**
Full regression	**112 suites, 2,495 / 2,495** (75 min; no suite needed a retry), run on the final page — after both fixes above

Check	Result
Layout audits	host screens at 5 sizes + tablet home: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues** (final page)
Windows EXE	69,038,907 bytes, SHA-256 `03cbf028…` ; tested page inside (md5 `970333f9…`); `main.js` = build 130; the three parts re-join to the same file
Android APK	versionCode 130, versionName 1.69, 1,701,658 bytes, SHA-256 `729633a4…` ; v2+v3, original key `5f4819e4…` ; hosting service type `connectedDevice` ; tested page inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page + `main.js` inside; zip 100,301,418 bytes, SHA-256 `c053a8e0…` ; 5 parts re-join identically; **not run on a real Mac**

The numbers at the start of each check line in the test logs are the numbers of your spec's tests. The transfer checks carry 1–19; the full regression covers the existing features.

Limits of this testing — please read

- 🟡 **No real Galaxy Tab.** The tablet's server ran on the build machine (the same Java code as the APK) with its page in Chromium. The tablet's background hosting and battery reading were simulated. On a real Wi-Fi the laptop finds the tablet through the tablet's network announcement, which wasn't tested; if *Tablet Quiz Found* doesn't appear, use 📲 Connect ▾ → Take Over Tablet Quiz and type the tablet's address.
- 🟡 **No real smart TV.** The "TVs" were Chromium pages at 720p, 1080p and 4K. Some older TV browsers can't open local addresses or run modern pages; then use a cheap streaming stick with a browser, a laptop, or HDMI. Sound plays from the host, not the TV.
- 🟡 **The joiner's fingerprint check** ran under Wine with a stand-in for `certutil` , not on real Windows. If certutil is missing, the joiner still joins and says the check was skipped.
- If the laptop app is killed outright, its last few seconds of browser-side saving can be lost. The transfer itself is written to a file first, so a laptop that crashes after the commit still finishes the transfer from that file (tested).
- **Rehearse at home** (Rehearsal Checklist rows 34–36): Test Hosting; a TV on `/tv` with Full Screen and Test; moving a tablet quiz to the laptop.

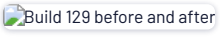
Live acceptance: **PENDING USER LIVE TEST**.

Build 129 addendum (16 October 2026)

Your choice: upgrades **1, 2, 4, 5, 6, 7**. Number 6 (smaller downloads) was dropped: the app's extra language files were already removed, so there was nothing left to trim — I told you at the time.

What changed

- **1 · The Galaxy Tab keeps hosting in the background.** While the tablet hosts, Android is told a quiz is running: a notification *Mind It! is hosting a quiz*, and the tablet keeps its Wi-Fi and server awake if the screen goes off or another app is opened. Swiping Mind It! away stops it. Android 13+ asks once to allow notifications.
- **2 · Other tablets find the tablet's game.** The hosting tablet announces its game on the Wi-Fi; another Galaxy Tab's start screen shows **Tablet Game Found** (and never lists its own game). Laptop games still say *Laptop Game Found*.
- **4 · Rename devices** in Connected Devices (laptop and tablet): the device in control and every trusted device. The name is used on request cards, *Currently in control* and the waiting remote, and is kept through reconnects and restarts. An empty name goes back to the device's own name.
- **5 · The webpage works offline.** The hosted webpage keeps a copy on the phone once opened; with no internet it opens from that copy (with its players and history). With internet it always loads the newest build first.
- **7 · 📱 Add Phone Remote is always there** on the remote in control: the host now sends it the PIN, so the button no longer disappears after a reconnect without the PIN.



📸 *Before and after (real screenshots; the offline "before" is a description, because the browser shows only its own "not connected" message).*

Check	Result
`fix129d` (laptop: Add Phone Remote after a reconnect without the PIN + its code decoded; Rename / Cancel / Enter / empty name; no redraw while typing; name in the request card and on the waiting iPhone; name saved in the trusted-devices file; kept after reconnect and laptop restart)	**21/21**
`fix129t` (the tablet's own server on the build machine: `kind: tablet` ; PIN sent to the controller; rename on the tablet, saved, kept after a server restart; another tablet's start screen shows *Tablet Game Found* + chip; the hosting tablet doesn't list itself; laptop games unchanged)	**16/16**
`fix129w` (the hosted webpage served as on GitHub Pages: offline helper installs; with the site switched off and the browser offline the page opens, with its saved data; a new tab opens offline; icon kept; other requests fail normally; back online a newer upload is used at once) — and a control run without the helper fails to open offline, as expected	**13/13**
Full regression	**103 suites, 2,316 / 2,316** (64 min). `fix127d` passed on retry: its wrong-PIN check reused a browser that has been trusted since build 128, so it sometimes connected without the PIN, which is correct. The check now uses a new device; re-run twice, **22/22** both times
Layout audits	host screens at 5 sizes + tablet home: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**

Check	Result
Android manifest	`foregroundServiceType=connectedDevice` on the hosting service, checked with `aapt dump xmltree` and with a second, independent reader (androguard); permissions FOREGROUND_SERVICE, FOREGROUND_SERVICE_CONNECTED_DEVICE, POST_NOTIFICATIONS; the Wi-Fi multicast permission Android 14 requires for this service type is declared
Windows EXE	69,016,865 bytes, SHA-256 `5d23dc2b…` ; tested page inside (md5 `ae10643d…`); `main.js` = build 129; parts re-join identically under Wine
Android APK	versionCode 129, SHA-256 `6b26e0b3…` , v2+v3, original key `5f4819e4…` ; tested page and new start screen inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page + `main.js` inside; zip 100,269,100 bytes (95.6 MiB), SHA-256 `1e78e68d…` ; 5 parts re-join identically; **not run on a real Mac**

Limits of this testing — please read

- 🟡 **The background hosting (1) and Tablet Game Found on a real network (2) were not tested at all.** They need a real Galaxy Tab running Android; the build machine has none. What was checked: the app is built as Android 14 and 15 require (manifest), and the code compiles. The tablet's server itself (rename, PIN) ran on the build machine.
- **Rehearse at home:** start a quiz on the tablet, join with a phone, turn the tablet's screen off for a minute and check the phone still works; look for the notification; on a second tablet, check *Tablet Game Found*.
- Offline webpage: tested in Chromium, not real iPhone Safari. An iPhone keeps the home-screen app's copy separately from Safari's. In Safari itself (not on the home screen), iOS can delete a site's saved data after 7 days without a visit, so use the home-screen icon.

Live acceptance: **PENDING USER LIVE TEST**.

Build 128 addendum (16 October 2026)

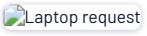
Your two specs ("Trusted Device Reconnection & Host-Controlled Device Management" and "Multi-Device Remote Control Hierarchy") with your decisions: the 📱 Phone Remote counts as a remote; a phone joins the laptop directly; requests wait 2 minutes; and "i need the tablet to be the server".

What changed

- **Trusted devices:** PIN once, then a device's own key; reconnects by itself after drops and restarts; Forget / New PIN revoke.
- **One controller:** only one remote (tablet, iPad/iPhone or Phone Remote) controls the quiz; it keeps its place through drops and restarts; only it receives the game (answers stay private).
- **Approval:** a second device waits; the host gets **Allow & Switch / Reject / Ignore** in the right-hand panel (never on the TV, not in Present mode); requests expire after 2 minutes.
- **Connected Devices** panel: in control · waiting · trusted (last connected) · Give Control / Disconnect / Forget.
- 📱 **Add Phone Remote** on a laptop-connected tablet sends a phone straight to the laptop.
- **Galaxy Tab as the server:** the tablet app runs the same server as the laptop (rewritten in Java) when it hosts a quiz alone — team phones, chips, buzzers, Final answers, Phone Remote and remotes.



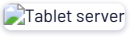
📸 *A second remote — before and after.*



📸 *The laptop when a second device asks — before and after.*



📸 *Connected Devices — before and after.*



📸 *The Galaxy Tab hosting on its own — before and after.*

Your 14 + 15 test scenarios

Scenario	Where	Result
New device connects with the PIN; reconnects without it	`fix128d` 1–2	✅
Trusted device reconnects after the laptop app restarts (data kept)	`fix128e` 3	✅
Second device while one is active; Allow & Switch; Reject	`fix128d` 4–6	✅ the first device keeps working until you decide
A pending request expires without interrupting the quiz	`fix128e` 7	✅ (2 min; scores unchanged)
Manual switching from Connected Devices; Disconnect; Forget	`fix128d` 8	✅ a disconnected device asks before it gets control back
Active device loses Wi-Fi and reconnects	`fix128e` 9	✅ same question still open
The old device reconnects after a switch → asks	`fix128d` 10	✅
Several devices ask at once	`fix128d` 11	✅ exactly one in control
Revoked / wrong-PIN device; commands from a device that lost control	`fix128d` 12	✅ refused
Active device drops during a question and during a Daily Double wager	`fix128e` 13	✅ state preserved
Team phones and the Audience View unaffected	`fix128d` 14, `fix128e`	✅ (request never on the TV or in Present mode)

Scenario	Where	Result
📱 Phone Remote counts as a controller	`fix128e`	✅ waits, Allow & Switch
Phone ↔ tablet: PIN, trusted reconnect, tablet hosts a whole quiz, phone controls it, switching, restart	`fix128t`	✅ on the build machine (JVM)
Tablet ↔ laptop, phone joins the laptop via the tablet's code	`fix128d` , `link`	✅

Check	Result
`fix128d` (laptop + iPad / iPhone / Android-sized pages)	**35/35**
`fix128e` (restart, Wi-Fi drop, expiry, Present mode, Phone Remote, drops mid-question / mid-wager)	**18/18**
`fix128t` (the tablet's own server — same Java source as the APK — on the build machine: phones join, DD wager incl. minimum, chips, buzzers, Final wager + answer, remotes, approval, restart, Check Connections, profiles)	**22/22**
Existing suites updated for the new rules	`link` (second tablet now waits; stale code with wrong PIN refused), `fix121` (Phone Remote approved first), `conn110`
Full regression	**100 suites, 2,269 / 2,269**, every suite first time (57 min)
Layout audits	host screens at 5 sizes + tablet home: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	69,015,668 bytes, SHA-256 `1f38f139…` ; tested page inside (md5 `59efc466…`); `main.js` = build 128; parts re-join identically under Wine
Android APK	versionCode 128, SHA-256 `20ed7e1d…` , v2+v3, original key `5f4819e4…` ; tested page inside; the server classes inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page + `main.js` inside; zip 100,268,686 bytes (95 MiB), SHA-256 `69f902de…` ; 5 parts re-join identically; **not run on a real Mac**

Limits of this testing — please read

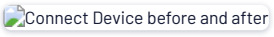
- 🟡 **The Galaxy Tab as the server has not run on a real Galaxy Tab.** The same Java code was tested on the build machine with the quiz page loaded as the app loads it and phones joining over the network. Android itself (the WebView, Wi-Fi, the app going to the background, power saving) was not part of it.
- iPhones and iPads were simulated in Chromium, not real Safari.
- "Encrypted communication" from your spec: connections stay on your local Wi-Fi over plain `http://` , as before. Pairing uses the PIN and a per-device key, and the laptop/tablet stores only a fingerprint (hash) of each key. Real encryption would need security certificates on every device, which can't be done offline without extra setup.
- "Google Cast / Android TV audience views" from your spec: Mind It! doesn't have these, so nothing could be affected. The Audience View window works as before.

Live acceptance: **PENDING USER LIVE TEST** — especially a rehearsal with the Galaxy Tab as the server.

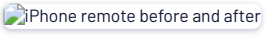
Build 127 addendum (16 October 2026)

Changes (your requests: the iPad/iPhone remote — "do it"; then "change connect tablet into connect device"):

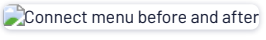
- 📲 **Connect Tablet is now Connect Device** — in the Connect ▾ menu, Setup → 6 · Advanced, the Devices window (*Connect device*, *Show Device QR / PIN*) and the window itself.
- The window shows **two codes**: **Galaxy Tab** (unchanged) and **iPad / iPhone** — a normal web address the Camera opens in Safari, giving the same full Quizmaster remote as the Galaxy Tab. The laptop app serves it at `/tablet` .
- A home-screen icon for that remote keeps working after the laptop app restarts (it picks up the new game number; the PIN is still required).
- On an iPhone the remote's board now fits the screen (all 6 columns).
- **Bug found and fixed:** a restored player could count as still deleted if the deletion and the restore happened in the same millisecond (only reachable by automation — `fix125` failed once on its first try and passed on the retry). A restore is now always dated after its deletion.

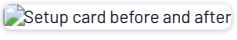


📸 *Connect Tablet → Connect Device, with the new iPad / iPhone code — before and after.*



📸 *The full remote on an iPhone — before (board runs off the screen) and after.*





📸 *Connect ▾ menu and the Setup card — before and after.*

Check	Result
`fix127d` (laptop app + iPhone- and iPad-sized browsers): the window is Connect Device with two equal codes; the right code is the laptop's `/tablet` address with game number and PIN (both codes decoded from a screenshot and checked); the Galaxy Tab code unchanged; the dialog fits the laptop screen; Setup and the Connect ▾ menu say Connect Device; the laptop serves `/tablet` to other devices; iPhone: Connected, board fits (6 columns, nothing off the edges, every value in full), tapping a value opens the question on the laptop, Correct / Wrong appear; an old game number in the address still connects; a wrong PIN can't; iPad: connects, tablet layout unchanged; no errors	**22/22**

Check	Result
Full regression	**97 suites, 2,188 / 2,188** — 96 first time; `fix125` once on retry (the bug above), then fixed; after the fix `fix125` 3× 27/27, `fix124` 19/19, `psync`, `libsync`, `fix125d`, `fix127d` 48/48
Layout audits	host screens at 5 sizes + tablet home at 3 sizes: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	69,009,786 bytes, SHA-256 `2470d7fa…`; tested page inside (md5 `c9db9f0f…`), `main.js` with the `/tablet` route; parts re-join identically under Wine
Android APK	versionCode 127, SHA-256 `a547d00e…`, v2+v3, original key `5f4819e4…`, tested page inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page and `main.js` inside; zip 100,261,994 bytes (95 MiB), SHA-256 `456f2c4e…`; 5 parts re-join identically; **not run on a real Mac**
Webpage	on GitHub Pages: https://coolrishi20.github.io/MindIt-Web/ (updated to build 127)

Limits of this testing: iPhone and iPad were simulated with Chromium (size, touch, user agent), not real Safari. The GitHub-hosted webpage can't be a remote for the laptop — a secure `https://` page isn't allowed to talk to the laptop's local `http://` address; use the **iPad / iPhone** code instead.

Live acceptance: **PENDING USER LIVE TEST**.

Build 126 addendum (13 October 2026)

Change (your request: "ok do both, include the webpage as a download option in github"): 📲 Mind It! on the home screen — on an iPhone or iPad, Safari's **Share → Add to Home Screen** now gives a **Mind It!** icon (the gold M logo) that opens the team-phone page or the 📱 Phone Remote full-screen. The page's `<head>` only: no game screen, rule or saved-data change. The webpage version `mindit.html` is now also on your GitHub download page.



📸 Home-screen icon and launch — before (build 125) and after (build 126).

Check	Result
`fix126` (the page itself): full-screen setting, name "Mind It!", browser tab title unchanged; 180×180 and 192×192 icons solid (no see-through corners); page starts normally at laptop and iPhone size, no errors	**8/8**
`fix126d` (laptop app + an iPhone-sized phone): the laptop's `/play` and `/remote` pages carry the full-screen setting, name and icon; the icon is the gold M logo; with the "opened from the home screen" flag set, the phone joins its team and shows "Connected"; nothing sticks out sideways; the Phone Remote still connects with its PIN; no errors	**13/13**
Webpage as a downloaded file	opened from disk in Chromium: starts, keeps a saved game and a new player after closing and reopening, Audience View opens; no errors
Full regression	**96 suites, 2,169 / 2,169**, every suite first time (53 min)
Layout audits	host screens at 5 sizes + tablet home at 3 sizes: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	69,008,995 bytes, SHA-256 `6cd8baec…`; tested page inside (md5 `d9d70e55…`), `main.js` unchanged; parts re-join identically under Wine
Android APK	versionCode 126, SHA-256 `c18ac5de…`, v2+v3, original key `5f4819e4…`, tested page inside
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page inside; zip 100,260,337 bytes (95 MiB — under GitHub's 100 MiB limit), SHA-256 `c04447c1…`; 5 parts re-join identically; **not run on a real Mac**
Web link	republished (alternate builds: 116, 118, 120, 122, 124, **126**); the published page carries the new head tags and the same app code

Limits of this testing: the phone checks used Chromium pretending to be an iPhone (size, touch, iPhone user agent). Real Safari wasn't available here, so how iOS shows the icon and opens it full-screen is **not verified on a real iPhone or iPad**. Android Chrome opens a plain `http://` address from the home screen as a normal tab (no full-screen).

Live acceptance: **PENDING USER LIVE TEST**.

Build 125 addendum (13 October 2026)

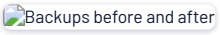
Changes (your request, all five): 🗑 Recently Deleted (restore a deleted player for 30 days) · 🛟 Safety backups (automatic, last 10) · ☑ Select several players · the tablet keeps what you type · 🖼 Results Picture for any Historic Game.



📸 *Player Profiles — before (build 124) and after (build 125, Select mode with two players ticked).*



📸 *Recently Deleted (new).*



📸 *Backups & Restore — before and after (🛟 Safety backups card).*



📸 *A Historic Game's bottom row — before and after.*

Check	Result
`fix125` (browser): Select / Select All / untick / Delete several with one question / Archive several; Recently Deleted list, Restore, picture and nickname back, 30-day expiry keeps the name, a sync can't re-delete a restored player; safety backups before deleting players, a game record and Start fresh, only 10 kept, listed in Backups & Restore, Save File gives a normal backup, Restore asks first; Results Picture saves a PNG for a past game	**27/27**
`fix125d` (laptop app + tablet): Adjust Score — amount, reason, chosen team and the cursor survive the laptop scoring meanwhile; Cancel starts empty next time; Apply works	**9/9**
Full regression	**94 suites, 2,127 / 2,127**, every suite first time
Layout audits	host screens at 5 sizes: **0 issues**; Audience View 234 screens at 9 sizes: **0 issues**
Windows EXE	68,963,836 bytes, SHA-256 `e995b25f…` ; tested page inside; parts re-join identically under Wine
Android APK	versionCode 125, SHA-256 `e39b61b4…` , v2+v3, original key
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page inside; 5 parts re-join to `5fcb4496…` ; **not run on a real Mac**

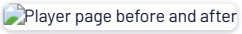
Note: safety backups are stored inside the app on each device (laptop app, or the tablet when it runs a game alone) — not on the linked tablet remote and not in a file you can see; **Back Up Now** files are still the backup to keep elsewhere.

Live acceptance: **PENDING USER LIVE TEST**.

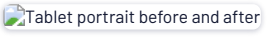
Build 124 addendum (13 October 2026)

Changes (your requests): 🗑 **delete a player** from Player Profiles; the Mac app is now **Apple Silicon only**.

Layout — before / after (the only layout change is one more button on a player's page; nothing else moved):



📸 *Laptop 1366×768 — a player's page, build 123 (left) and build 124 (right).*



📸 *Tablet portrait 800×1340 — the same row fits on one line.*



📸 *The question before deleting.*

Check	Result
New suite `fix124` : button on the same row as Edit/Archive/All Players/Merge; asks first; Cancel keeps; Delete removes from Player Profiles, the list and team lists; on Setup taken off their team; Quizmaster cleared; refused during a game; Historic Games keeps the name; other players' statistics unchanged; survives a reload; sync never brings the player back and passes deletions on; Returning Champion ignores deleted winners; restoring an older backup brings the player back	**19/19**
Full regression	**92 suites, 2,097 / 2,097**, every suite first time
Layout audits	laptop/tablet/phone host screens (incl. the player page) at 5 sizes: **0 issues**; Audience View 234 screens at 9 sizes (FHD, 4K, 16:10, 4:3, portrait): **0 issues**
Windows EXE	68,959,783 bytes, SHA-256 `bd485b3b…` ; tested page inside; parts re-join identically under Wine
Android APK	versionCode 124, SHA-256 `a0ca8c0a…` , v2+v3, original key `5f4819e4…`

Check	Result
Mac (Apple Silicon)	arm64, ad-hoc signed, tested page inside; zip 100,212,557 B (symlinks kept), 5 parts re-join to `fbdf9b69…` ; **not run on a real Mac**
Web link	republished (version 166), live page verified

Live acceptance: **PENDING USER LIVE TEST**.

Build 123 addendum (13 October 2026)

Changes (your decisions): restoring a backup or loading a game file **always asks first** (Restore Backup / Load File / Cancel), even with no game running — K4 settled; team photos stay in the save (K3, "leave it"); a **Mac version** of the laptop app (your request).

Check	Result
New suite `fix123` : asks with no game running; Cancel leaves the quiz untouched; Restore Backup restores game and players; with a game in progress only the usual "A game is paused…" question; game files ask "Load this game file?"; player-profile files merge without asking; Backups & Restore → Choose Backup File… asks; the in-app note is now true	**13/13**
Full regression	**91 suites, 2,081 / 2,081.** `qlib` and `crop` restore a backup as part of their checks; they first failed because the new question appeared, were updated to press Restore Backup, then passed (26/26, 27/27).
Windows EXE	68,958,930 bytes, SHA-256 `463b707e…` ; page md5 `feeb27bd7019…` inside; parts re-join identically under Wine
Android APK	versionCode 123, SHA-256 `b937f559…` , v2+v3, original key `5f4819e4…` ; page md5 `feeb27bd7019…`
Mac app — Apple Silicon	arm64, bundle `com.mindit.quiz` 1.69.0, macOS 11+, camera and local-network permission texts present, ad-hoc signed (app and frameworks), tested page and `main.js` inside; zip 100,211,536 B with the framework symlinks kept; 5 parts re-join to the same SHA-256 (`51b4421c…`)
Mac app — Intel	x86_64, same checks; zip 106,274,619 B (`18767abd…`)
Mac app on a real Mac	**Not run** — no Mac here. It isn't notarized by Apple, so macOS asks once (Open Anyway).
Guides	Complete Guide 4.1b (Mac install), 22.3/22.5 (restore asks), Quick Start Mac line; Cheat Sheet and Player's Guide still exactly 2 pages

Live acceptance (Windows, Galaxy Tab, phones, TV — and now the Mac): **PENDING USER LIVE TEST**.

Build 122 addendum (13 October 2026)

Change: your Returning Champion rule — the crown goes only to a team that has **every player of last game's winning team**; a one-player winner crowns the team that player is on; split winners → no crown (`app111.js`). This settles K1.

Check	Result
New suite `champ122` (split winners, full team, full team + new player, 1 of 4 / 2 of 4, one-player winner, deleted player, teams without players, joint winners, win streak and where it stops, rule switched off)	**15/15**
Existing `champ` (crown, TV intro, Title Defence, Dethroned, next champion)	9/9
Full regression	**90 suites, 2,061 / 2,061** (`desk` passed on retry: its first attempt couldn't connect to the freshly started laptop app)
Windows EXE	68,959,626 bytes, SHA-256 `2158c7c4…` ; page md5 `e2d075a926a6…` inside; parts re-join identically under Wine
Android APK	versionCode 122, SHA-256 `a4481b76…` , v2+v3, original key `5f4819e4…` ; page md5 `e2d075a926a6…`
Web link	republished (version 165); live page verified

Everything else in this report (build 121) still applies. Live acceptance: **PENDING USER LIVE TEST**.

A. Previous task completion

Interrupted task	Build 120 — "Interactive team swapping & Returning Champion logic" (your attachment 1): click-to-select / swap / + move / undo / reset / confirm in the Random Team Generator, with persistence, tests and layout testing; plus a review (not a change) of the Returning Champion logic.

Done before your QC message arrived	The team-swapping code (`app109.js`), its suites and the desktop/tablet layout check were finished; the Returning Champion review was written up.
Remaining when the QC message arrived	The build-120 EXE/APK, the full regression run, the guide update and the deliverables A–F.
Resumed and finished	Build 120 was built, verified and checkpointed (EXE `e93ffe65…` , APK `d65fd147…` vc120, web link v164); full run 2,019/2,019 (two suites passed on retry — the cause was found and fixed in this QC, see C2); the guide's §7.4 got the new team-swapping text and screenshot.
Verification of the build-120 requirements	`rtg120` **52/52** (mouse 1366×768 and portrait-tablet touch 800×1280): select, swap, same-team re-select, cancel, + move (refused when it would empty a team), undo (swaps and moves), reset (asks first), confirm saves the final teams, scroll position kept, 46 px+ touch targets. `fix120` 13/13. In build 121 both pass again (section B).
Returning Champion	Reviewed only — **not changed**, because you asked for approval first and no new rule has been approved. The current rule is now described exactly in the guide (§12.5). See K1.

B. QC results

Full automated regression, build 121 (final page): 89 suites (55 browser + 34 laptop-app), 2,055 / 2,055 checks passed, every suite first time (no retries), 49 minutes. (An earlier full run of build 121, before fix C7 was added, gave 2,025/2,025 in 88 suites + `fix121` 32/32. `qc2` plays random games, so its check count varies from run to run: 276–291.)

Area	Suites (examples)	Result
Game creation, Setup, rounds, Question Database	`qc2` , `zones` , `qlib` , `ext` , `l3` , `l4` , `fix120`	647/647 (7 suites)
Teams, players, Random Team Generator	`rtg` , `rtg120` , `g1` , `g3`	89/89 (4 suites)
Gameplay rules (scoring, Daily Double, Final, chips, buzzer, ties)	`rules35` , `tg` , `tg2` , `ud` , `bu` , `bz` , `bz3` , `r9` , `fix112` – `fix115`	250/250 (12 suites)
Historic Games, statistics, achievements, champion	`champ` , `g2` , `g6` , `g7` , `rh` , `v169`	74/74 (6 suites)
Phones, tablet, link, End Game	`link` , `pin` , `psync` , `hostreq` , `hostreq2` , `conn110` , `endgame110` , `fix116` , `fix117` , `fix119` , `fix121`	238/238 (11 suites)
Complete games end to end	`fullmatch` (whole match from the tablet), `onehand` , `desk`	64/64 (3 suites)
Audience View layout	`avaudit` — 234 screens × automatic checks at 9 sizes	**235/235**, 0 layout issues
Laptop host screens	`vaudit` — 18 screens × 5 sizes (90 screenshots)	**0 issues**
Live-game stress and soak	`stress110` phases	**24/24 + soak 2/2:** idle 2 × 10 min with the Quizmaster window hidden (117 samples each, TV + tablet + 10 phones: no false disconnects, no full-screen cover); network drop/return 6/6; 10 phones at once 4/4; phone lock 60 s 2/2; rapid/duplicate tablet taps 4/4; Check Connections 6/6; **60-minute real-time quiz:** 60 questions, 346 tablet actions, Quizmaster window hidden/shown 5×, an intermission — **0 false disconnects, 0 script errors**, laptop server memory 213 → 222 MB, page heap 12 → 13 MB
Rollback (build 121 data → build 119 app)	rollback test, section I	11/11

Skipped / not possible here: real hardware (Windows PC, Galaxy Tab A9+, phones, TV, real Wi-Fi, real sound); the Android rollback; Google Drive backup (needs internet and your script).

End Game (asked specifically): `endgame110` 19/19 — End Game is refused while the game is still being played; the game is recorded **once**; the button locks while saving (no double tap); a repeated End Game after a dropped connection is acknowledged without saving twice; an End Game for an old game is refused; rehearsal End Game records nothing; standalone tablet saves once.

C. Bugs found and fixed (build 121, `app110.js`)

Every item was reproduced or measured first, then fixed, then re-measured. No gameplay rule, layout or saved-data format was changed.

#	Defect (confirmed)	Effect on a quiz night	Fix	Evidence
C1	**Connection Health: phones flickered Responsive ↔ Unstable** although they checked in every 1.5 s. The laptop added the time since its own 3-second health refresh to each phone's age (1.5 s + 3 s > the 4 s limit).	False "Unstable" warnings; the tablet's 📶 chip changed colour for nothing.	Only time beyond the normal refresh counts. A phone that really stops still turns amber (after ~7 s) and red/stale (~15 s).	80/80 samples "ok" over 20 s for each phone (before: flapping every second); dropped phone → Unstable at 7 s, Stale at 19 s.

#	Defect (confirmed)	Effect on a quiz night	Fix	Evidence
C2	**Quizmaster tablet redrew its whole screen ~once a second.** The laptop re-sent the whole game because round-trip times inside Connection Health changed at every check-in.	Typed Adjust Score numbers could vanish; a tap could land on a button that was being replaced (this is why `hostreq2` and `fullmatch` needed a retry in the build-120 run).	The tablet's copy of Connection Health uses steady wording (round-trip bands, "just now"); the laptop keeps exact numbers.	Tablet state messages and full redraws: **1.07/s → 0/s** at idle; data sent to the tablet **~14.5 KB/s → 0**; `hostreq2` and `fullmatch` now pass first time.
C3	**Phone Buzzer Mode: BUZZ IN was rebuilt ~6.5 times a second**, so a tap whose press and release straddled a rebuild never arrived.	Lost buzzes — a fairness problem in a buzzer race.	The button is rebuilt only when something on it changes.	30 ordinary taps (120 ms): **14/30 → 30/30** reached the button; rebuilds **6.52/s → 0**. A real buzz still registers; the fast check pauses while the phone is locked.
C4	Phone screen rebuilt at **every check-in during a question** (the seconds left were part of the "has anything changed?" test).	Same lost-tap risk on chip/confirm buttons; wasted battery.	Seconds left no longer count as a change; the countdown runs on the phone and still follows +time / stop from the laptop.	Rebuilds **0.67/s → 0** in 6 s; +15 s reached the phone within 0.1 s of the laptop; stop cleared the phone timer.
C5	**Laptop Phones card (and wager and live-setup cards) rebuilt every 500 ms.**	"Join link & QR code" snapped shut; the network-address list closed while open; clicks during a rebuild were lost.	A card is replaced only when its content changes; open sections stay open; nothing is replaced while you type or choose in it.	Phones card rebuilt **2.0/s → 0/s**; a phone joining still updates it at once.
C6	📱 **Phone remote: every button press started another background check loop**; the page was redrawn every second while the timer ran.	After a few presses the phone hammered the laptop; presses could be lost.	One check loop; a press just makes the next check come sooner; the seconds update in place.	6 presses → 10 checks in 7 s (one loop); no redraw just for the seconds.
C7	**An unfinished game was lost if Mind It! was opened and closed again without choosing** Resume saved game / Start fresh. At start-up the app shows a fresh Setup while offering the saved game; closing saved that fresh Setup over it. **Also in build 119.**	A crash, reopen, close (e.g. to fix the TV), reopen → the game was gone (only a backup file could bring it back).	The unfinished game is kept aside until you choose; it is offered at every start. Save format unchanged; the extra copy (`mindit.resume.pending`) is ignored by older builds.	Restart test on the real laptop app: build 119 — offered once, then **never**; build 121 — offered at the 1st, 2nd and 3rd restart; Resume → right scores; Start fresh → no more offer.

Regression suite added: `fix121` (39 checks) covers C1–C7.

Found, not changed (reported): see K.

D. Connectivity results

Device	What was checked (simulated)	Result
Windows host (laptop app server)	Health endpoint, port fallback, saved-game and settings copies across a port change (`fix115`, `fix116`), Check Connections	Passed (`fix115` 22/22, `fix116` 26/26, stress `health` 6/6: host, tablet, TV and 8/8 teams green in 3.6 s)
Quizmaster tablet	Pairing/PIN, control hand-over, commands with acknowledgements, reconnect after Wi-Fi loss, End Game retry, Connection Health sheet (`link`, `pin`, `hostreq`, `hostreq2`, `conn110`, `endgame110`, `fix121`)	Passed (`link`, `pin`, `hostreq` 28/28, `hostreq2` 29/29 first time, `conn110` 35/35, `endgame110` 19/19, stress `net`: offline 20 s → small status line only, back in 0.3 s with the current state)
Audience View	No full-screen "Reconnecting" cover; heartbeat; the Quizmaster window hidden/minimised (build-110 fix) — `conn110`, stress `idle` / `soak`	Passed — no full-screen cover in 2 × 10 min idle and the 60-min soak with the Quizmaster window hidden half the time
Phones	Join, team codes, wagers, chips, buzzer, lock/unlock, network drop/return, Connection Health status (`psync`, `bz`, `conn110`, stress `lock` / `net`, `fix121`)	Passed (`psync`, `bz` 3 phones, stress `load` 10 phones at once, `lock` 60 s locked → "asleep", back on the same team with no re-pairing; `fix121` C1/C3/C4)
False "Reconnecting" (your live-game report)	The root cause found in build 110 (the Quizmaster window throttled when covered) stays fixed; build 121 removes the remaining false **Unstable** flicker (C1).	0 false disconnects in about 1 h 25 min of stress and soak sampling (2 × 10 min idle, the 60-min quiz and the short phases); Connection Health stays "Connected" for phones checking in normally (80/80 samples)

E. Performance results

Measured on the build machine with the laptop app, one phone and the tablet, 30 s per phase (`perfmeasure.py`), before and after build 121:

Phase	Measure	Before (build 120)	After (build 121)
Board idle	Tablet full redraws / s	1.07	0
Board idle	Game updates sent to the tablet / s · bytes / s	1.07 · 14,540	0 · 0
Board idle	Laptop Phones card rebuilt / s	2.0	0
Question, timer running	Tablet full redraws / s	1.07	0
Question, timer running	Phone screen rebuilds / s	0.67	0
Phone Buzzer Mode	BUZZ IN rebuilt / s · taps that arrived	6.52 · 14/30	0 · 30/30
Phone remote	Checks / s after 6 presses	grew with every press	~1.4 (one loop)

Requests per second are unchanged (phones check in every 1.5 s, 350 ms in Buzzer Mode); what changed is how often screens are thrown away and rebuilt.

Autosave: with four 300 KB team photos a save is ~1.2 MB and takes ~23 ms here; saves happen 0.7 s after a change (not on a timer — 0 saves in 20 s idle). Not changed (K3).

F. Visual QC

No layout was changed in build 121 (so there is no layout before/after). The only visible text change is in the **tablet's** Connection Health sheet — before/after:



📸 Tablet → 📶 Connection Health, build 120 (left) and build 121 (right), same devices.

Screen	Sizes tested	Result
Audience View (TV)	1920×1080 (FHD), 3840×2160 (4K), 2560×1600 and 1920×1200 (16:10), 1280×800, 1366×768, 1280×720, 1024×768 (4:3), 800×1280 (portrait) — 26 key screens each	234 screens, **0 issues** (text off screen, cut off, overlapping; board tiles equal and centred; smallest text)
Laptop host	1920×1080, 1366×768, 1340×800, 800×1340, 390×844 — 18 screens	**0 issues**
Quizmaster tablet (Galaxy Tab A9+ size, 1340×800 / 800×1340)	`fix119`, `rtg120`, `endgame110`, `butab` layout checks	passed (in the full run)
Phones (390×844 @2×)	`fix119`, `fix116`, `psync`	passed (in the full run)

Screenshots are from the sandbox; real-screen colour, brightness and viewing distance still need your rehearsal (Rehearsal Checklist).

G. Documentation update

Document	Updated for build 121	Check
Complete Idiot's Guide (PDF + Markdown)	"New in build 121"; §9.2 tablet Connection Health wording; §12.5 exact Returning Champion rule (current behaviour, 🔵 review pending); §22.3 restore wording corrected (asks only when a game is in progress; players/history merged); crash-recovery note (the offer now stays)	138 pages, rebuilt and text-checked
Quick Start	build stamp	3 pages
Quizmaster Cheat Sheet	build stamp	**exactly 2 A4 pages**, no overflow
Player's Guide	unchanged content (nothing changed for players)	**exactly 2 A4 pages**, no overflow
Rehearsal Checklist	"stay Connected for a minute, no flicker"; reopen twice without choosing; restore asks mid-game	2 pages
Project docs (status, changelog, rules, architecture, CLAUDE.md)	B31–B40, build 121 row, Build-121 module notes, spare-copy key	—
Documentation audit	new section for builds 116–121	—

Every build-121 change is described in the guide's "New in build 121" and in the changelog. Code vs documents: where they disagreed, the code was taken as the truth and the documents corrected (restore wording, Returning Champion detail); one in-app sentence is still too broad (K4).

H. Build verification

	Windows	Android
File	`Mind It!.exe` (portable, x64) — also delivered as `MindIt_v1.69.exe` (same bytes) and as 3 parts + `Join Mind It - build 121.bat`	`Mind It! for Android (Galaxy Tab).apk` — also as `MindIt_v1.69.apk` (same bytes)
Size	68,959,878 bytes; parts 3 × 22,986,626	1,574,609 bytes
SHA-256	`d8d14ca1a12187d44343eb52aa786c13a49aed15c71ee0ff6f0ac3c7e4a2f12e`	`67926597a774e18a8f84e6e43fe63c1ca2e35a1ec0a90a9348427ce68b274b62`
Version	1.69 (package 1.69.0)	`com.mindit.app`, versionName **1.69**, versionCode **121**
App page inside	`app.asar` → `app/mindit.html` md5 `a146a07d0bbc…` = the tested page; `main.js` identical to build 119	`assets/mindit.html` md5 `a146a07d0bbc…` = the tested page
Signing	not code-signed (as before — SmartScreen "More info → Run anyway")	v2 + v3 verified; one signer, certificate SHA-256 `5f4819e4b15e…` = your original key
Checks run	Built with electron-builder (exit 0); app.asar extracted and compared; parts re-joined with the `.bat` under Wine → identical SHA-256	`aapt dump badging`, `apksigner verify -v --print-certs`, page md5
Not run	The EXE on a real Windows PC (the same app shell and page ran all laptop tests here under Linux)	Install/run on a real tablet (no device or emulator here)

Source: `MindIt_Source_v1.69.zip` (7.0 MB, 438 files, extracts cleanly). Rebuilding the app page from the extracted zip gives a file identical to the shipped page except 3 characters — the timestamp inside the compressed HEIC helper (its unpacked content is identical). Git commit `c926b63`, tag `checkpoint-build121`.

I. Rollback verification

- **Preserved:** `MIND_IT_v1.69_PREVIOUS_WORKING_RELEASE.zip` (build 119 as delivered — not rebuilt): 86,577,817 bytes, SHA-256 `a745c41b…` ; 390 files, all checksums verified after unzipping; the EXE parts re-join to `49517ba1…` . Kept separate from the build-121 files and not overwritten. Git tags `checkpoint-build119` , `checkpoint-build120` , `checkpoint-build121` .
- **Rollback test (laptop app, sandbox):** build 121 wrote 5 players, a finished game, an unfinished game and a backup file; the app was closed, the build-119 page put in place and the app started on the **same profile**: players and Historic Games kept, the finished game's scores intact, the unfinished game offered and resumed with the right scores, and the build-121 backup file restored through ⋯ → Backups & Restore — **11/11**. (Run twice: with the build-121 page before and after fix C7.)
- **Android:** not tested (no tablet). Android refuses to install an older versionCode over a newer one; uninstalling deletes the tablet's Mind It! data. The guide gives three options, data-safe first.

J. Live acceptance status

PENDING USER LIVE TEST. Nothing here was run on your Windows laptop, Galaxy Tab A9+, phones or TV. Please play a 60–120-minute game with the real devices (the Rehearsal Checklist covers it) and watch for: false disconnects, full-screen Audience View interruptions, lost scores, duplicate actions, wrong game records, crashes, devices that don't reconnect. Keep the build-119 rollback until it passes.

K. Outstanding issues and risks

#	Issue	Severity	Status
~~K1~~	**Returning Champion (settled in build 122 — see the addendum):** a team with **any one** of last game's winners qualifies, and only the **first** qualifying team (team order) gets the crown — so a team with 1 of 4 winners can beat a team with 2 of 4. The proposed majority rule has open edge cases (tied winners, small rosters, deleted players, teams without players, the "N wins in a row" count).	—	**Done in build 122**
K2	Real-hardware behaviour (Wi-Fi quality, Android power saving, TV scaling, sound levels)	—	PENDING USER LIVE TEST
K3	Autosave includes team photos (~1.2 MB / ~23 ms per save with 4 photos).	Low	**Kept as is — your decision (13 Oct)**
~~K4~~	In-app Backups & Restore note said "You'll be asked before anything is replaced" but it asked only when a game was in progress.	—	**Fixed in build 123: it always asks now**
K5	The Android shell doesn't pause the page when the tablet app goes to the background (keeps the link alive — intended).	Info	Not changed
K6	The tablet still redraws fully when the game really changes (a score, a new question), so typing in Adjust Score while another device scores can still be interrupted.	Low	Not changed (rare; the once-a-second redraw is gone)
K7	Test scripts and `build.py` use the build machine's paths (`/home/claude` , `/tmp` , `/mnt`).	Info	Documented in the build instructions
K8	Build 119 (the rollback) has the C7 problem: press **Resume saved game** the first time it is offered.	Info	In the rollback guide

Release readiness: Ready for live acceptance testing. Not marked ready for unrestricted live use until your real-device game passes (J). No known open defect would stop a game; K1 is a rule decision waiting for you.